Into the Secret Garden

An Introduction to Schemes and Algebraic Geometry

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[...] *Oscar Zariski bewitched me. When he spoke the words "algebraic variety", there was a certain resonance in his voice that said distinctly that he was looking into a secret garden. I immediately wanted to be able to do this too. It led me to 25 years of struggling to make this world tangible and visible.*

David Mumford

Contents

0 Before we start **5**

1 Sheaves and $\mathcal{O}$-modules **9**

1.1 Key example: smooth functions 9
1.2 Presheaves, sheaves, morphisms 11
1.3 The sheaf condition via equalisers 16
1.4 Stalks, and what they (don't) tell us 18
1.5 Sheafification 24
1.6 Subsheaves, Quotient sheaves, and exactness 27
1.7 Supports 30
1.8 Sheaves = sheaves on a base 31
1.9 Pushforward, inverse image 36
1.9.1 Pushforward (or direct image) 36
1.9.2 Inverse image 40
1.10 Gluing sheaves 45
1.11 Locally ringed spaces 46
1.12 Ideal sheaves = Closed immersions up to isomorphism 48
1.13 $\mathcal{O}_X$-modules and (quasi)coherent sheaves 50
1.14 Pushforward and pullback of $\mathcal{O}$-modules 55

A Basic category theory **58**

A.1 Categories and functors 58
A.2 Final objects and universal properties 61
A.3 Equivalences 63
A.4 Adjunctions 65
A.5 Limits and colimits 66
A.6 Abelian categories 68
A.7 Yoneda Lemma 72

B Commutative algebra **76**

B.1 Frequently used theorems 76

B.1.1 Nakayama's Lemma(s) 76
B.1.2 Noether's Lemma 76
B.1.3 Zariski's Lemma 78
B.2 Tensor products 78
B.3 Localisation 80
B.3.1 General construction for modules 80
B.3.2 Universal property of localisation 82
B.3.3 Ring Localisations 83
B.3.4 Exactness of localisation 85
B.4 Length of modules 86
B.5 Associated points, and embedded components 88
B.5.1 Associated points: local case 88
B.5.2 Primary decomposition and embedded components 94
B.5.3 Associated points: global case 95
B.5.4 Links with finite length modules 96
B.5.5 The cycle associated to a coherent sheaf 99
B.6 Depth and Cohen–Macaulay schemes 101

0 | Before we start

About this course

These notes supplement the 60-hour course (3 cycles for SISSA students, 48 compulsory hours for UNITS master's students) *Algebraic Geometry* (533SM).

The exam consists of an oral presentation by the student about a topic mutually agreed upon by the student and the instructor, plus a few questions regarding the material covered in the course.

For UNITS students: the exam can only be scheduled within the official exam session, see the academic calendar.

Prerequisites

Familiarity with basic theory of commutative rings and modules is of great help, but not necessary. The relevant notions will be recalled as we need them. We have, however, included Appendix B to cover the basic commutative algebra constructions we will be using (and much more), and Appendix A to cover the basics of category theory as well. Category theory is, up to a certain point, used mainly as a tool, or rather a language, and is "invoked" in many proofs that would otherwise be too verbose or, as a matter of fact, obscure. The reader is, however, encouraged to *not* look at category theory as a black box, but rather embrace it as part of the language that shapes modern Algebraic Geometry.

Conventions

We list here a series of conventions that will be used throughout this text.

- The axiom of choice (or Zorn's Lemma) is assumed; so, for instance, every nonzero ring has a maximal ideal, and a poset $(P, \leqslant)$ in which every chain has an upper bound admits a maximal element.
- Given two sets A and B, the phrase '$A \subset B$' means that A is contained in B, *possibly equal* to B.

- We write '$A \simeq B$' to indicate a bijection of sets. We reserve '$A \cong B$' for isomorphisms involving objects with some structure, and '$A = B$' for straight equality or — occasionally — a canonical isomorphism that should be clear from the context.
- A *ring* is a commutative, unitary ring. The zero ring (the one where $1 = 0$) is allowed (and in fact needed), but we always assume our rings are nonzero unless we explicitly mention it. Ring homomorphisms preserve the identity.
- Some results will be valid over an arbitrary field, some over an algebraically closed one. Therefore, we will make the following notational distinction:

$$\mathbb{F} = \text{an arbitrary field}, \qquad \mathbf{k} = \text{an algebraically closed field}.$$

- An open cover of a topological space U is the datum of a set I, and an open subset $U_i \subset U$ for every $i \in I$, such that $U = \bigcup_{i \in I} U_i$. We set $U_{ij} = U_i \cap U_j$. If $I = \emptyset$, then $U = \emptyset$.
- To say that x is an object of a category $\mathscr{C}$ we simply write '$x \in \mathscr{C}$' instead of $x \in \mathrm{Ob}(\mathscr{C})$.

Proofs

This text is based on the following principle. Newcomers to algebraic geometers need to (1) learn the basics, and (2) learn the main techniques. For this reason, the first chapters, building up the basic material from scratch, include proofs for essentially *all* the statements presented. In later chapters, some proofs are replaced by appropriate references. This happens especially when the main argument was already given in a similar form for an earlier result, or when it is not harmful for pedagogical purposes to just refer the reader to one of the excellent references that are available.

Exercises

This text contains *lots* of exercises, of varying difficulty. They are marked with the symbol



and, as the reader will notice, they are integrated within the main body of the text, and not confined to the end of every chapter. This choice may seem disadvantageous, but the reader should not see the exercises as something "interrupting the flow" of the exposition or, more realistically, breaking the author's delirious stream of consciousness made of definitions and theorems coming non-stop one after the other. The exercises are "put there" to suggest what students *can already solve* on their own at any given stage of the journey, having learnt only what comes earlier in the text.

Finally, and this might seem confusing at first, the author would like to remind all students that the key point of the exercises is not quite to *solve* them (although this would be spectacular), but mainly *work on* them — on as many as possible — possibly in front of a blackboard or with a piece of paper and *together with other human beings* (not machines, please!).

Warnings

This text also contains *lots* of warning signs. They are marked with the symbol



and they mean to indicate the subtleties of the theory. Such a sign means: stop for a second and check your intuition! The reader is encouraged to think about these subtleties and form their own examples and counterexamples to what might have seemed, at first sight, something completely innocent and intuitive, but turned out false for some reason.

References

In tennis, the *Open Era* is the period beginning in 1968, and continuing to the present day, in which both amateurs and professionals are allowed to compete in all tournaments. Somewhat counterintuitively, it was the professionals that were "invited" to the amateurs' land, not the other way around.

Also the subject of Algebraic Geometry is now living its own *Open Era* which, remarkably, also started in the late 60s! That is the time when Grothendieck and Dieudonné published the various *tomes* of foundations of Algebraic Geometry, namely the much celebrated *Éléments de Géométrie Algébrique* (EGA), written in the span of 1960 through 1967. This is and always will be *the Bible* of Algebraic Geometry.

However, the first book on the subject with a clear intent of spreading algebraic geometry to a wide community in a self-contained, general-enough-but-not-overwhelming way, is Hartshorne's *Algebraic geometry,* appeared in 1977 [3]. And that was just the beginning.

The present text would not exist without the many books on Algebraic Geometry that have been written since the start of Algebraic Geometry's Open Era. It would not exist without the things I learnt from my teachers and from my collaborators.

We list below just four bibliographical references that supplement these notes. Each has influenced, in some way more or less visible, the author's view of the subject. But there are many more available nowadays, that we cite here and there in this text: the newcomer is encouraged to read everything!

- Q. Liu, *Algebraic geometry and arithmetic curves* [4],
- R. Hartshorne, *Algebraic geometry* [3],
- R. Vakil, *The rising sea* [8],
- D. Eisenbud, *Commutative Algebra: With a View Toward Algebraic Geometry* [1],

A final remark: the reader will notice that many times references, especially in later chapters, point to *The stacks project* [7]. We believe it is a wonderful, constantly evolving reference for the foundational material that "everybody uses" in their work of professional algebraic geometers; also, it has the advantage over Grothendieck's EGA of being more easily navigable, thanks to the efforts of its Creators. We encourage every student to get familiar with *The stacks project*, because that is now becoming more than a tool: it is becoming *The Bible 2.0*.

Caveat

These notes are written by a human. As such, they are imperfect, at best. You are kindly encouraged to let the author know if you find mistakes, inaccuracies, typos.

1 | Sheaves and $\mathscr{O}$-modules

The notion of a sheaf originates in work of Leray, developed in part during his years as a prisoner in Austria during World War II.

Sheaves are a key notion present in the toolbox of every mathematician keen to understand the "nature" of a *geometric space*. They incarnate one of the basic principles that will be unraveled throughout this course, which can be stated as the slogan

geometric spaces are determined by functions on them.

In algebraic geometry, a space is studied together with its functions on every open subset: the resulting sheaf of functions is part of the structure of the space itself.

1.1 Key example: smooth functions

Before diving into precise definitions, we explore a key, well-known example of a sheaf.

Let X be a smooth manifold. For each open subset $U \subset X$, we have a ring (actually, an $\mathbb{R}$-algebra)

$$C^\infty(U,\mathbb{R}) = \{\text{ smooth functions } U \to \mathbb{R}\}.$$

Indeed, smooth functions with the same source can naturally be added and multiplied pointwise using the ring structure on $\mathbb{R}$. If $V \hookrightarrow U$ is an open subset, we have a restriction map

$$C^\infty(U,\mathbb{R}) \xrightarrow{\rho_{UV}} C^\infty(V,\mathbb{R}), \quad f \mapsto f|_V,$$

which is an $\mathbb{R}$-algebra homomorphism, and if $W \hookrightarrow V \hookrightarrow U$ is a chain of open subsets of X, we have a commutative diagram

$$\begin{array}{ccccc} C^\infty(U,\mathbb{R}) & \xrightarrow{\rho_{UV}} & C^\infty(V,\mathbb{R}) & \xrightarrow{\rho_{VW}} & C^\infty(W,\mathbb{R}). \\ & \searrow^{\rho_{UW}} & & \nearrow & \end{array}$$

So far, we have just observed that the assignment $U \mapsto C^\infty(U,\mathbb{R})$ is (contravariantly) *functorial* in U, i.e. it defines a functor from the opposite category of the category of open subsets of X to the category of $\mathbb{R}$-algebras. The two distinguished features of the assignment $U \mapsto C^\infty(U,\mathbb{R})$, which make it into a *sheaf* of $\mathbb{R}$-algebras on X, are the following:

(i) Fix an open subset $U \subset X$ and an open cover $U = \bigcup_{i\in I} U_i$. If $f, g \in C^\infty(U, \mathbb{R})$ are smooth functions such that $f|_{U_i} = g|_{U_i}$ for every $i \in I$, then $f = g$. In other words, a smooth function is determined by its restriction to the open subsets forming a covering. This is the *locality axiom*.

(ii) Fix an open subset $U \subset X$ and an open cover $U = \bigcup_{i\in I} U_i$. Given a smooth function $f_i \in C^\infty(U_i, \mathbb{R})$ on each U_i, such that $f_i|_{U_i\cap U_j} = f_j|_{U_i\cap U_j}$ for every $(i, j) \in I \times I$, there is a smooth function $f \in C^\infty(U, \mathbb{R})$ such that $f_i = f|_{U_i}$ for every $i \in I$. In other words, functions glue along an open cover. This is the *gluing axiom*.

A sheaf is an abstract notion formalising this "ability to glue local data". The formal definition will be given in Important Definition 1.2.0.5. Note that the result of the gluing in Condition (ii) is *unique* by Condition (i).

Let us continue with our example. Let $x \in X$ be a point. Consider the ring

$$C^\infty_{X,x} = \left\{ (U, f) \,\middle|\, x \in U,\, f \in C^\infty(U, \mathbb{R}) \right\} \Big/ \sim$$

where $(U, f) \sim (V, g)$ whenever there exists an open subset $W \subset U \cap V$, containing x, such that $f|_W = g|_W$. Note that $C^\infty_{X,x}$ is indeed a ring, with addition and multiplication

$$[U, f] + [U', f'] = [U \cap U', f + f']$$
$$[U, f] \cdot [U', f'] = [U \cap U', f f'].$$

This ring, which is in fact an $\mathbb{R}$-algebra via $c \mapsto [X, c]$, for all $c \in \mathbb{R}$, is called the *stalk* of the sheaf $C^\infty(-, \mathbb{R})$ at x (cf. Important Definition 1.4.0.1), and it receives a natural $\mathbb{R}$-algebra homomorphism from $C^\infty(U, \mathbb{R})$ for every open subset U of X such that $x \in U$, sending $f \mapsto [U, f]$. The image of f along this map is called the *germ of f at x*. The subset

$$\mathfrak{m}_x = \left\{ [U, f] \in C^\infty_{X,x} \,\middle|\, f(x) = 0 \right\} \subset C^\infty_{X,x}$$

forms an ideal, which is a *maximal* ideal, being the kernel of the (surjective) evaluation map

$$C^\infty_{X,x} \longrightarrow\!\!\!\!\!\rightarrow \mathbb{R}$$
$$[U, f] \longmapsto f(x).$$

In fact, $\mathfrak{m}_x$ is the *unique* maximal ideal of $C^\infty_{X,x}$. To see this, it is enough to check that every element of $C^\infty_{X,x} \setminus \mathfrak{m}_x$ is invertible. But this is true, since a smooth function that is nonzero in a neighbourhood of x is invertible there.

The upshot is, then, that the pair $(C^\infty_{X,x}, \mathfrak{m}_x)$ defines a *local ring* with residue field $\mathbb{R}$. The geometric spaces X one deals with in algebraic geometry, namely *schemes*, have precisely this property: they come with a sheaf of rings $\mathscr{O}_X$ such that each stalk $\mathscr{O}_{X,x}$ is a local ring. These spaces $(X, \mathscr{O}_X)$ actually form a larger category, that of *locally ringed spaces* (cf. Section 1.11). Schemes are particular instances of locally ringed spaces.

1.2 Presheaves, sheaves, morphisms

Let $\mathscr{C}$ be a concrete category (Definition A.3.0.10) with a final object $0 \in \mathscr{C}$. The concreteness assumption means that part of the structure is the datum of a faithful functor $F\colon \mathscr{C} \to \mathsf{Sets}$. A further assumption we make is that F sends the final object 0 to a singleton (i.e. a final object) in Sets. This just means that the final object has a one-element underlying set. To fix ideas, $\mathscr{C}$ should be thought of as any of the following familiar categories:

- $\mathscr{C} = \mathsf{Sets}$,
- $\mathscr{C} = \mathsf{Rings}$,
- $\mathscr{C} = \mathsf{Ab} = \mathsf{Mod}_{\mathbb{Z}}$,
- $\mathscr{C} = \mathsf{Mod}_R$, where R is a ring.

If X is a topological space, we denote by τ_X the category of open subsets of X. The set $\mathrm{Hom}_{\tau_X}(V,U)$ between two open sets $V, U \subset X$ is just the empty set if $V \not\subset U$, or the singleton $\{\, V \hookrightarrow U \,\}$ in case V is contained in U. Thus the opposite category τ_X^{op} satisfies

$$\mathrm{Hom}_{\tau_X^{\mathrm{op}}}(U,V) = \begin{cases} \{\, V \hookrightarrow U \,\} & \text{if } V \subset U \\ \emptyset & \text{if } V \not\subset U \end{cases}$$

and a functor $\mathcal{F}\colon \tau_X^{\mathrm{op}} \to \mathscr{C}$ (i.e. a contravariant functor $\tau_X \to \mathscr{C}$) determines a map

$$\mathrm{Hom}_{\tau_X^{\mathrm{op}}}(U,V) \longrightarrow \mathrm{Hom}_{\mathscr{C}}(\mathcal{F}(U), \mathcal{F}(V)),$$

which is nothing but a choice of an element $\rho_{UV} \in \mathrm{Hom}_{\mathscr{C}}(\mathcal{F}(U), \mathcal{F}(V))$ for any inclusion of open subsets $V \subset U$.

Definition 1.2.0.1 (Presheaf, take I)**.** A *presheaf* on a topological space X, with values in $\mathscr{C}$, is a contravariant functor $\mathcal{F}$ from τ_X to $\mathscr{C}$, i.e. an object of the functor category $\mathsf{Fun}(\tau_X^{\mathrm{op}}, \mathscr{C})$.

For those who do not like the categorical definition, here is an equivalent definition, which just unravels the definition of a functor (cf. Definition A.1.0.9).

Definition 1.2.0.2 (Presheaf, take II)**.** A *presheaf* on a topological space X, with values in $\mathscr{C}$, is the assignment $U \mapsto \mathcal{F}(U)$ of an object $\mathcal{F}(U) \in \mathscr{C}$ for each open subset $U \subset X$, and of a morphism $\rho_{UV}\colon \mathcal{F}(U) \to \mathcal{F}(V)$ in $\mathscr{C}$ for each inclusion $V \hookrightarrow U$, such that

(1) $\rho_{UU} = \mathrm{id}_{\mathcal{F}(U)}$ for every $U \in \tau_X$, and

(2) $\rho_{UW} = \rho_{VW} \circ \rho_{UV}$ for every chain of inclusions $W \hookrightarrow V \hookrightarrow U$.

Terminology 1.2.0.3. Elements of $\mathcal{F}(U)$ are often called 'sections of $\mathcal{F}$ over U', or (somewhat more vaguely) 'local sections' when $U \subsetneq X$. Elements of $\mathcal{F}(X)$ are called 'global sections', or just 'sections'. Possible alternative notations for $\mathcal{F}(U)$ are $\Gamma(U,\mathcal{F})$ and $\mathrm{H}^0(U,\mathcal{F})$. The maps ρ_{UV} are often called 'restriction maps' (from U to V, the larger set being U).

Notation 1.2.0.4. Motivated by Terminology 1.2.0.3, we shall often write $s|_V$ for the image of a section $s \in \mathcal{F}(U)$ along the restriction map ρ_{UV}.

Important Definition 1.2.0.5 (Sheaf, take I)**.** A *sheaf* on a topological space X, with values in $\mathscr{C}$, is a presheaf $\mathcal{F}$ such that the following two conditions hold:

(loc) Fix an open subset $U \subset X$, an open cover $U = \bigcup_{i\in I} U_i$, and two sections $s, t \in \mathcal{F}(U)$ satisfying $s|_{U_i} = t|_{U_i}$ for all $i \in I$. Then $s = t$.

(glue) Fix an open subset $U \subset X$, an open cover $U = \bigcup_{i\in I} U_i$ and a tuple $(s_i)_{i\in I}$ of sections $s_i \in \mathcal{F}(U_i)$ such that $s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$ for all $(i,j) \in I\times I$. Then there exists a section $s \in \mathcal{F}(U)$ such that $s_i = s|_{U_i}$.

Conditions (loc) and (glue) generalise the conditions (i) and (ii), respectively, anticipated with the example $\mathcal{F} = C^\infty(-,\mathbb{R})$ in Section 1.1.

Terminology 1.2.0.6. A presheaf $\mathcal{F}$ is called *separated* if Condition (loc) holds. Sometimes this condition is called *locality axiom*. Condition (glue), on the other hand, is called the *gluing condition* (or *gluing axiom*).

Remark 1.2.0.7. Let $\mathcal{F}$ be a sheaf. Then, the section $s \in \mathcal{F}(U)$ in the gluing condition (glue) is necessarily unique because $\mathcal{F}$ is separated. In fact, the two sheaf conditions could be replaced by a single condition, identical to (glue), but imposing uniqueness of s.

Example 1.2.0.8 (Trivial sheaf)**.** Recall that we assumed $\mathscr{C}$ to admit a final object $0 \in \mathscr{C}$ and the faithful functor $F\colon \mathscr{C} \to \mathsf{Sets}$ to send 0 to a singleton. Under these assumptions, the presheaf defined by $U \mapsto 0$ for every U is a sheaf and is called the *trivial sheaf* (or sometimes the *zero sheaf*). It is simply denoted by '0'.

Example 1.2.0.9 (Restriction to an open)**.** Let $U \subset X$ be an open subset, $\mathcal{F}$ a presheaf on X. Then, setting $\mathcal{F}|_U(V) = \mathcal{F}(V)$ for V an open subset of U, defines a presheaf $\mathcal{F}|_U$ on U, which is a sheaf as soon as $\mathcal{F}$ is. It is called the *restriction of $\mathcal{F}$ to U*.

Definition 1.2.0.10 (Morphism of (pre)sheaves)**.** Let X be a topological space. A *morphism* between two presheaves $\mathcal{F},\mathcal{G}$ on X is a natural transformation $\eta\colon \mathcal{F} \Rightarrow \mathcal{G}$, i.e. a morphism in the functor category $\mathsf{Fun}(\tau_X^{\mathrm{op}}, \mathscr{C})$. A morphism of sheaves is just a morphism between the underlying presheaves.

Let us unravel the definition of natural transformation (cf. Definition A.3.0.1), to translate Definition 1.2.0.10 in naive terms.

To give a morphism of (pre)sheaves, one has to assign a homomorphism

$$\eta_U : \mathcal{F}(U) \longrightarrow \mathcal{G}(U)$$

in $\mathscr{C}$ to each $U \in \tau_X$, such that for every inclusion $V \hookrightarrow U$ of open subsets of X, the diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\eta_U} & \mathcal{G}(U) \\ {\scriptstyle \rho^{\mathcal{F}}_{UV}}\downarrow & & \downarrow{\scriptstyle \rho^{\mathcal{G}}_{UV}} \\ \mathcal{F}(V) & \xrightarrow{\eta_V} & \mathcal{G}(V) \end{array} \tag{1.2.1}$$

commutes. For the sake of clarity, we have emphasised the relevant sheaf in the restriction maps notation, but we will not be doing that systematically.

Notation 1.2.0.11. It is clear that presheaves on X with values in $\mathscr{C}$ form a category $\mathsf{pSh}(X, \mathscr{C})$, tautologically defined as

$$\mathsf{pSh}(X, \mathscr{C}) \coloneqq \mathsf{Fun}(\tau_X^{\mathrm{op}}, \mathscr{C}).$$

By Definition 1.2.0.10, sheaves form a full subcategory, denoted $\mathsf{Sh}(X, \mathscr{C})$. We denote by

$$j_{X,\mathscr{C}} : \mathsf{Sh}(X, \mathscr{C}) \hookrightarrow \mathsf{pSh}(X, \mathscr{C}) \tag{1.2.2}$$

the (fully faithful) inclusion functor.

An isomorphism of (pre)sheaves is an isomorphism in $\mathsf{pSh}(X, \mathscr{C})$, i.e. a *natural isomorphism*, i.e. a natural transformation $\eta : \mathcal{F} \Rightarrow \mathcal{G}$ such that η_U is an isomorphism in $\mathscr{C}$ for every $U \in \tau_X$ (cf. Definition A.3.0.2).

Notation 1.2.0.12. Since (pre)sheaves form a genuine category, from now on we shall use the classical arrow notation '$\mathcal{F} \to \mathcal{G}$' (instead of $\mathcal{F} \Rightarrow \mathcal{G}$) to denote a morphism of (pre)sheaves.

The following definition makes sense, because $\mathscr{C}$ is assumed to be a concrete category.

Definition 1.2.0.13 (Injective map of presheaves)**.** A morphism of (pre)sheaves $\eta : \mathcal{F} \to \mathcal{G}$ is *injective* if η_U is injective for every U. We denote this by writing η as '$\mathcal{F} \hookrightarrow \mathcal{G}$'.

We close this section with a few examples and exercises.

Example 1.2.0.14 (Smooth functions)**.** Let X be a smooth manifold. Then, sending $U \subset X$ to the set $C^\infty(U, \mathbb{R})$ of smooth functions $U \to \mathbb{R}$, defines a sheaf $C^\infty(-, \mathbb{R})$ with values in the category of $\mathbb{R}$-algebras.

Example 1.2.0.15 (Holomorphic functions). Let X be a complex manifold. Then, sending $U \subset X$ to the set $\mathscr{O}_X^{\mathrm{h}}(U)$ of holomorphic functions on U, defines a sheaf $\mathscr{O}_X^{\mathrm{h}}$ with values in the category of $\mathbb{C}$-algebras. Sending U to the set $\mathscr{O}_X^{\mathrm{h},\times}(U)$ of nowhere-zero holomorphic functions defines a sheaf of abelian groups on X (the group structure being given by pointwise multiplication of functions).

Example 1.2.0.16 (Continuous functions are a sheaf). Let X, Y be topological spaces. For $U \subset X$ open, define

$$\mathcal{F}(U) = \{\text{ continuous functions } U \to Y\}.$$

Then $\mathcal{F}$ is a sheaf of sets on X.

Example 1.2.0.17 (Separated presheaf, not a sheaf, take I). Set $X = \mathbb{C}$. Then, sending $U \subset X$ to the subset

$$\mathcal{F}(U) = \left\{ f \in \mathscr{O}_X^{\mathrm{h}}(U) \,\middle|\, f = g^2 \text{ for some } g \in \mathscr{O}_X^{\mathrm{h}}(U) \right\}$$

defines a (separated) presheaf. However, $\mathcal{F}$ is not a sheaf: the function $f(z) = z$ on the annulus

$$U_\varepsilon = \{ z \in \mathbb{C} \mid 1 - \varepsilon < |z| < 1 + \varepsilon \} \subset \mathbb{C},$$

where $0 < \varepsilon < 1$, has a square root in any neighbourhood of any point $x \in U_\varepsilon$, but there is no global $\sqrt{z}$ defined on the whole of U_ε, so Condition (glue) fails.



Exercise 1.2.0.18 (Separated presheaf, not a sheaf, take II). Let $X = \mathbb{R}$, with the standard topology. Show that

$$U \longmapsto \mathrm{B}(U) = \{\text{ bounded continuous functions } U \to \mathbb{R}\}$$

is a separated presheaf on X, but not a sheaf (i.e. Condition (glue) fails).

Example 1.2.0.19 (Constant presheaf). Work with $\mathscr{C} = \mathsf{Ab} = \mathsf{Mod}_{\mathbb{Z}}$, the category of abelian groups, and fix $G \neq 0$ in this category. Fix a topological space X, and define

$$\underline{G}_X^{\mathrm{pre}}(U) = \begin{cases} G & \text{if } U \neq \emptyset, \\ 0 & \text{if } U = \emptyset. \end{cases}$$

As for the restriction maps, set $\rho_{UV} = \mathrm{id}_G$ if both U and V are nonempty. This is a presheaf, which happens to be a sheaf only in precise circumstances (cf. Exercise 1.2.0.21). For instance, suppose $X = U_1 \amalg U_2$ is a disjoint union of two nonempty open subsets. Then $\underline{G}_X^{\mathrm{pre}}(X) = G = \underline{G}_X^{\mathrm{pre}}(U_i)$ for $i = 1, 2$. Now, $X = U_1 \amalg U_2$ is an open cover. Pick two *distinct* sections $s_i \in G = \underline{G}_X^{\mathrm{pre}}(U_i)$ for $i = 1, 2$. Then, $s_1|_{U_1 \cap U_2} = s_1|_\emptyset = 0 = s_2|_\emptyset = s_2|_{U_1 \cap U_2}$, but there is no section $s \in \underline{G}_X^{\mathrm{pre}}(X) = G$ such that $s|_{U_i} = s_i$ since $\rho_{XU_i} = \mathrm{id}_G$ for $i = 1, 2$ and $s_1 \neq s_2$ by assumption. Hence Condition (glue), i.e. the gluing axiom, fails (whereas Condition (loc) is trivially satisfied). We will see in Example 1.5.0.3 that $\underline{G}_X^{\mathrm{pre}}$ can be "transformed" into a sheaf by a canonical procedure.



Exercise 1.2.0.20. Provide examples of presheaves which satisfy the gluing axiom but not the separation axiom. (**Hint**: Try a presheaf which has two global sections that become equal on every proper open subset).



Exercise 1.2.0.21. Show that the constant presheaf $\underline{G}_X^{\mathrm{pre}}$ of Example 1.2.0.19 is a sheaf if and only if every nonempty open subset $U \subset X$ is connected.



Exercise 1.2.0.22 (Presheaves kernel and cokernel). Let $\mathscr{C} = \mathsf{Mod}_A$, so that every morphism has a kernel and a cokernel. Let $\eta\colon \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves with values in $\mathscr{C}$. Consider the assignments

$$U \longmapsto (\ker_{\mathrm{pre}} \eta)(U) = \ker(\eta_U),$$

$$U \longmapsto (\mathrm{coker}_{\mathrm{pre}} \eta)(U) = \mathrm{coker}(\eta_U).$$

Prove the following assertions.

(i) Both $\ker_{\mathrm{pre}} \eta$ and $\mathrm{coker}_{\mathrm{pre}} \eta$ are presheaves.

(ii) There is a canonical morphism $\ker_{\mathrm{pre}} \eta \to \mathcal{F}$ (resp. $\mathcal{G} \to \mathrm{coker}_{\mathrm{pre}} \eta$), and it satisfies the universal property of the kernel (resp. the cokernel) in $\mathsf{pSh}(X, \mathscr{C})$.

(iii) The presheaf $\ker_{\mathrm{pre}} \eta$ is in fact a sheaf, denoted $\ker(\eta)$, as soon as η is a morphism of *sheaves*.

(iv) If η is a morphism of sheaves, then $\ker(\eta) \hookrightarrow \mathcal{F}$ satisfies the universal property of the kernel in $\mathsf{Sh}(X, \mathscr{C})$, and η is injective if and only if $\ker(\eta) = 0$.[1]

Example 1.2.0.23 ($\mathrm{coker}_{\mathrm{pre}} \eta$ may not be a sheaf). Let $X = \mathbb{C}$ and $\mathscr{C} = \mathsf{Ab}$. Consider the morphism of sheaves

$$\mathscr{O}_X^{\mathrm{h}} \xrightarrow{\exp} \mathscr{O}_X^{\mathrm{h},\times}, \quad f \mapsto \exp(f),$$

where $\mathscr{O}_X^{\mathrm{h},\times}$ is the sheaf of nowhere-zero holomorphic functions on X (cf. Example 1.2.0.15). We have that the open subset $U = X \setminus \{0\} \subset X$ is covered by the two open subsets

$$U_1 = X \setminus [0, +\infty) \subset X, \quad U_2 = X \setminus (-\infty, 0] \subset X.$$

The function $g(z) = z$ viewed in $\mathscr{O}_X^{\mathrm{h},\times}(U)$ is not of the form $\exp(f)$ for any $f \in \mathscr{O}_X^{\mathrm{h}}(U)$. Thus the image $\widetilde{g}$ of g along

$$\mathscr{O}_X^{\mathrm{h},\times}(U) \longrightarrow \mathrm{coker}_{\mathrm{pre}}(\exp)(U)$$

is nonzero. However, U_1 and U_2 are simply connected, thus every function $h_i \in \mathscr{O}_X^{\mathrm{h},\times}(U_i)$ is of the form $\exp(f_i)$ for some $f_i \in \mathscr{O}_X^{\mathrm{h}}(U_i)$. Thus $\mathrm{coker}_{\mathrm{pre}}(\exp)(U_i) = 0$ for $i = 1, 2$. In particular, the restrictions $g|_{U_i}$ have this property, namely they go (necessarily) to 0 in $\mathrm{coker}_{\mathrm{pre}}(\exp)(U_i)$. If $\mathrm{coker}_{\mathrm{pre}}(\exp)$ were a sheaf, the gluing axiom would force $\widetilde{g} = 0$, which is not true.

[1] See Exercise 1.6.0.6 for the corresponding property of cokernels.

1.3 The sheaf condition via equalisers

We now present an alternative way to define sheaves. We will repeatedly use this reinterpretation throughout these notes.

Let $\mathscr{C}$ be a category with limits (cf. Definition A.5.0.1). In particular, $\mathscr{C}$ has products, equalisers, and a final object (cf. Appendix A.5 for full details). The reader may imagine $\mathscr{C}$ to be, for instance, any of the following categories: sets, groups, rings, algebras over a fixed ring, modules over a fixed ring. Fix a presheaf $\mathcal{F}$ with values in $\mathscr{C}$ on a topological space X. Let $\{U_i\}_{i\in I}$ be a family of open subsets of X, and set $U = \bigcup_{i\in I} U_i$. Then, by our assumption on $\mathscr{C}$, one can consider the map

$$\mathcal{F}(U) \xrightarrow{\rho} \prod_{i\in I} \mathcal{F}(U_i), \quad s \mapsto (s|_{U_i})_{i\in I},$$

as well as the family of maps

$$\mu_{ij}\colon \prod_{i\in I} \mathcal{F}(U_i) \to \mathcal{F}(U_i) \to \mathcal{F}(U_i \cap U_j), \qquad (s_i)_{i\in I} \mapsto s_i|_{U_i\cap U_j}$$
$$\nu_{ij}\colon \prod_{i\in I} \mathcal{F}(U_i) \to \mathcal{F}(U_j) \to \mathcal{F}(U_i \cap U_j), \qquad (s_i)_{i\in I} \mapsto s_j|_{U_i\cap U_j}$$

which, taking products over $(i,j) \in I \times I$, can be assembled into two maps

$$\prod_{i\in I} \mathcal{F}(U_i) \underset{\nu}{\overset{\mu}{\rightrightarrows}} \prod_{(i,j)\in I\times I} \mathcal{F}(U_i \cap U_j).$$

Definition 1.3.0.1 (Sheaf, take II)**.** Let $\mathscr{C}$ be a category with limits, and let X be a topological space. A presheaf $\mathcal{F} \in \mathsf{pSh}(X, \mathscr{C})$ is a *sheaf* if for every family of open subsets $\{U_i\}_{i\in I}$, with $U = \bigcup_{i\in I} U_i$, the diagram

$$\mathcal{F}(U) \xrightarrow{\rho} \prod_{i\in I} \mathcal{F}(U_i) \underset{\nu}{\overset{\mu}{\rightrightarrows}} \prod_{(i,j)\in I\times I} \mathcal{F}(U_i \cap U_j)$$

is an equaliser diagram in $\mathscr{C}$.

Informally, being an equaliser means that ρ is injective and its image agrees with the set of tuples $(s_i)_{i\in I}$ such that $s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$ for all pairs (i,j).

Note that Definition 1.3.0.1 is *element-free*. However, let us check that it agrees with Important Definition 1.2.0.5 when $\mathscr{C}$ is concrete: in this case the injectivity of ρ, implied by the equaliser condition, coincides with separatedness; the fact that the set-theoretic image of ρ coincides with the collection of tuples of sections $(s_i)_{i\in I}$ such that $s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j}$ is precisely the gluing condition.

Remark 1.3.0.2. Let $\mathcal{F}$ be a sheaf. Then, one has $\mathcal{F}(\emptyset) \cong 0$ canonically, where 0 is the final object in $\mathscr{C}$. Thus from now on we will simply write $\mathcal{F}(\emptyset) = 0$ for a sheaf $\mathcal{F}$. Note that this is sometimes listed as an axiom defining a (pre)sheaf, but it does in fact follow from our assumptions (cf. Exercise A.5.0.2).

Example 1.3.0.3. Let $\mathcal{F}$ be a sheaf on X. If $U = \coprod_{i\in I} U_i$ is a *disjoint* union of open subsets $U_i \subset U$, then ρ is an isomorphism, i.e. $s \mapsto (s|_{U_i})_{i\in I}$ defines an isomorphism

$$\rho : \mathcal{F}(U) \xrightarrow{\sim} \prod_{i\in I} \mathcal{F}(U_i).$$

Example 1.3.0.4. Let $\mathscr{C} = \mathsf{Mod}_R$.[2] Then a presheaf $\mathcal{F} \in \mathsf{pSh}(X, \mathscr{C})$ is a sheaf if and only if for every family of open subsets $\{U_i\}_{i\in I}$, with $U = \bigcup_{i\in I} U_i$, the sequence

$$0 \longrightarrow \mathcal{F}(U) \xrightarrow{\rho} \prod_{i\in I} \mathcal{F}(U_i) \xrightarrow{\mu-\nu} \prod_{(i,j)\in I\times I} \mathcal{F}(U_i \cap U_j)$$

is exact, where the map denoted $\mu - \nu$ sends $(s_i)_{i\in I} \mapsto (s_i|_{U_i\cap U_j} - s_j|_{U_i\cap U_j})_{i,j}$.

The following lemma applies, for instance, to categories of groups, rings, algebras over a ring, and modules over a ring. It allows one to check the sheaf condition of Definition 1.3.0.1 in the category of sets.

LEMMA 1.3.0.5 ([7, Tag 0073]). *Let $\mathscr{C}$ be a category, $F : \mathscr{C} \to \mathsf{Sets}$ a faithful functor such that $\mathscr{C}$ has limits and F commutes with them. Assume that F reflects isomorphisms. Then a presheaf $\mathcal{F} \in \mathsf{pSh}(X, \mathscr{C})$ is a sheaf if and only if the underlying presheaf of sets $F \circ \mathcal{F} : \tau_X^{\mathrm{op}} \to \mathsf{Sets}$ is a sheaf.*

At the beginning of this chapter we have defined (pre)sheaves of objects in an arbitrary concrete category $\mathscr{C}$. We still have to define a few things, though, e.g. stalks and sheafification. In order for everything to be well-defined and work well (but still be compatible with all we have discussed so far, including Definition 1.3.0.1), we need to add a few initial data. This is provided by the following definition.

Definition 1.3.0.6 ([7, Tag 007L])**.** A *type of algebraic structure* is a pair $(\mathscr{C}, F)$, where $\mathscr{C}$ is a category, $F : \mathscr{C} \to \mathsf{Sets}$ is a faithful functor, such that

(1) $\mathscr{C}$ has limits and F commutes with them,

(2) $\mathscr{C}$ has filtered colimits and F commutes with them,

(3) F reflects isomorphisms (i.e. F is *conservative*).

A few remarks are in order, before we go on.

- Equipping a category $\mathscr{C}$ with a faithful functor $F : \mathscr{C} \to \mathsf{Sets}$ is like saying that $\mathscr{C}$ is a *concrete category*, which we had already assumed in Section 1.2. Our earlier assumption that F preserves the final object is now implied by Condition (1).

[2] The example remains valid for *complete* abelian categories, i.e. abelian categories closed under arbitrary products.

- If we have a type of algebraic structure $(\mathscr{C}, F)$, then we can verify whether a presheaf is a sheaf in the category of sets, by Lemma 1.3.0.5.
- The condition that F be conservative implies that a bijective morphism in $\mathscr{C}$ is an isomorphism. This, as already mentioned, excludes categories such as the category of topological spaces.
- For every type of algebraic structure $(\mathscr{C}, F)$, one has the following properties:
  (i) $\mathscr{C}$ has a final object 0, and $F(0)$ is a final object in Sets (i.e. a singleton).
  (ii) $\mathscr{C}$ has products, fibre products, and equalisers — this follows from the examples in Appendix A.5. Moreover, F commutes with all of them.
- Examples of categories $\mathscr{C}$ having the additional structure of Definition 1.3.0.6 are:
  – monoids,
  – groups,
  – abelian groups,
  – rings,
  – modules over a ring.

  In all these cases, we take as the functor F the obvious forgetful functor. The reader is encouraged to just think of $\mathscr{C}$ as one of these familiar categories, and not bother too much about Definition 1.3.0.6. As a counterexample, however, consider the category Top of topological spaces: the forgetful functor to Sets exists but does not reflect isomorphisms (a continuous bijection need not be a homeomorphism).

1.4 Stalks, and what they (don't) tell us

Fix a type of algebraic structure $(\mathscr{C}, F\colon \mathscr{C} \to \mathsf{Sets})$ as in Definition 1.3.0.6. Let X be a topological space, $x \in X$ a point. The collection of open subsets $U \subset X$ containing x forms a directed system (the partial order $\succeq$ being the inclusion relation, i.e. $V \succeq U$ if and only if $V \subset U$). Indeed, given two open neighbourhoods U and V of x, there is always a third open neighbourhood of x contained in both U and V, namely $U \cap V$ or any smaller open subset containing x. In fancier language, the subcategory

$$\mathsf{Ngb}_x = \{\, U \in \tau_X \mid x \in U \,\}^{\mathrm{op}} \xhookrightarrow{\iota_x} \tau_X^{\mathrm{op}}$$

is a filtered category (see Definition A.5.0.10).

Important Definition 1.4.0.1 (Stalks)**.** Let $x \in X$ be a point, $\mathcal{F}$ a presheaf. The *stalk of* $\mathcal{F}$ *at* x is the filtered colimit

$$\mathcal{F}_x = \varinjlim_{\mathsf{Ngb}_x} \mathcal{F} \circ \iota_x = \varinjlim_{U \ni x} \mathcal{F}(U) \in \mathscr{C}.$$

Because F commutes with filtered colimits, the underlying set $F(\mathcal{F}_x)$, still denoted $\mathcal{F}_x$, is

$$\mathcal{F}_x = \{(U,s) \mid x \in U,\, s \in \mathcal{F}(U)\} / \sim$$

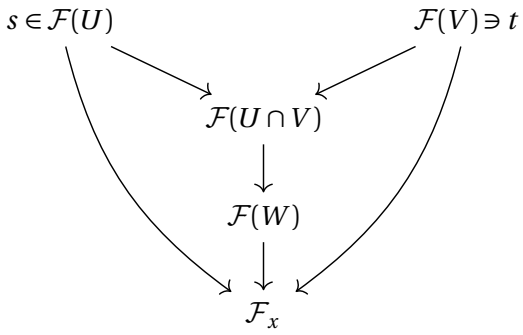
where $(U,s) \sim (V,t)$ whenever there is an open neighbourhood $W \subset U \cap V$ of x such that $s|_W = t|_W$. We denote by

$$s_x = [U,s] \in \mathcal{F}_x$$

the equivalence class of the pair (U,s). It is called the *germ of s at x*. By definition of direct limit, there are natural homomorphisms

$$\mathcal{F}(U) \longrightarrow \mathcal{F}_x, \quad s \mapsto s_x,$$

in $\mathscr{C}$, for every open neighbourhood U of x. The diagram



illustrates the fact that two sections $s \in \mathcal{F}(U)$ and $t \in \mathcal{F}(V)$ define the same element in the stalk $\mathcal{F}_x$ if and only if there is an intermediate open subset $W \subset U \cap V$ over which they agree.



Figure 1.1: A bunch of sheaves sitting in their natural habitat. The little tops of each leaf of corn are the stalks.

LEMMA 1.4.0.2. *If $\mathcal{F}$ is a separated presheaf of sets (e.g. a sheaf), then the natural map*

$$\sigma_U^{\mathcal{F}}\colon \mathcal{F}(U) \longrightarrow \prod_{x\in U} \mathcal{F}_x, \quad s \mapsto (s_x)_{x\in U} \tag{1.4.1}$$

is injective for every open subset U of X.

The lemma means, at an informal level, that sections are determined by their germs. In fact, one can prove that a presheaf $\mathcal{F}$ is separated if and only if the map $\sigma_U^{\mathcal{F}}$ is injective for all open subsets $U \subset X$.

Proof. If s and t are sections in $\mathcal{F}(U)$ such that $s_x = t_x$ in $\mathcal{F}_x$ for every $x \in U$, then for every $x \in U$ there is an open neighbourhood $U_x \subset U$ such that $s|_{U_x} = t|_{U_x}$. But this holds for every $x \in U$, and $U = \bigcup_{x\in U} U_x$ is an open covering, thus by the separation axiom we deduce $s = t$, i.e. $\sigma_U^{\mathcal{F}}$ is injective. □

Consider the following property of a tuple $(s_x)_{x\in U} \in \prod_{x\in U} \mathcal{F}_x$, for $U \subset X$ an open subset:

$$\begin{gathered} \text{for every } x \in U \text{ there exists a pair } (V_x, t^x), \\ \text{with } x \in V_x \subset U \text{ and } t^x \in \mathcal{F}(V_x), \\ \text{such that } t^x_y = s_y \text{ for all } y \in V_x. \end{gathered} \tag{1.4.2}$$

Definition 1.4.0.3 (Compatible germs)**.** Let $\mathcal{F}$ be a presheaf on X, and let $U \subset X$ be an open subset. We say that $(s_x)_{x\in U} \in \prod_{x\in U} \mathcal{F}_x$ is a *tuple of compatible germs* if Condition (1.4.2) is fulfilled.

We always have inclusions

$$\operatorname{im}(\sigma_U^{\mathcal{F}}) \subset \left\{ \text{tuples } (s_x)_{x\in U} \text{ of compatible germs} \right\} \subset \prod_{x\in U} \mathcal{F}_x \tag{1.4.3}$$

where the first inclusion is justified by taking $V_x = U$ and $t^x = s$ for every $x \in U$ as soon as $\sigma_U^{\mathcal{F}}(s) = (s_x)_{x\in U}$. If $\mathcal{F}$ is a sheaf, then tuples of compatible germs form precisely the image of the map (1.4.1), i.e. the first inclusion in (1.4.3) is an equality. Indeed, assume $(s_x)_{x\in U}$ consists of compatible germs. Let $\{(V_x, t^x) \mid x \in U\}$ be as in the displayed condition (1.4.2). By the compatibility condition, for every pair $(x, x') \in U \times U$ we have

$$t^x_y = t^{x'}_y, \quad y \in V_x \cap V_{x'}.$$

It follows from Lemma 1.4.0.2 that

$$t^x\big|_{V_x\cap V_{x'}} = t^{x'}\big|_{V_x\cap V_{x'}}. \tag{1.4.4}$$

Now, we have an open cover $U = \bigcup_{x\in U} V_x$, so by the gluing axiom, applicable by (1.4.4), the sections $t^x \in \mathcal{F}(V_x)$ glue to a (unique) section $t \in \mathcal{F}(U)$ such that $t|_{V_x} = t^x$. But $t^x_y = s_y$ for $y \in V_x$, and this holds for every $x \in U$, so $\sigma_U^{\mathcal{F}}(t) = (s_x)_{x\in U}$.

Summing up, when $\mathcal{F}$ is a sheaf, we have a *bijection*

$$\sigma_U^{\mathcal{F}}\colon \mathcal{F}(U) \xrightarrow{\sim} \{\text{tuples } (s_x)_{x\in U} \text{ of compatible germs}\}.$$

This also shows that *sections of a sheaf can always be identified with genuine functions!* Indeed, tuples $(s_x)_{x\in U}$ correspond to particular functions $U \to \coprod_{x\in U} \mathcal{F}_x$, sending $x \in U$ inside the corresponding stalk, and doing so in a compatible way.

LEMMA 1.4.0.4. *Let $s, t \in \mathcal{F}(X)$ be two global sections of a separated presheaf $\mathcal{F}$, such that $s_x = t_x \in \mathcal{F}_x$ for every $x \in X$. Then $s = t$.*

Proof. This is just a special case of Lemma 1.4.0.2. □



Exercise 1.4.0.5. Let $\mathcal{F}$ be a sheaf on X, and let $s, t \in \mathcal{F}(X)$ be two global sections. Show that

$$\{\, x \in X \mid s_x = t_x \,\} \subset X$$

is an open subset of X.

A morphism of presheaves $\eta\colon \mathcal{F} \to \mathcal{G}$ induces a morphism $\eta_x\colon \mathcal{F}_x \to \mathcal{G}_x$ at the level of stalks for every $x \in X$, defined by

$$s_x = [U, s] \longmapsto [U, \eta_U(s)] = (\eta_U(s))_x. \tag{1.4.5}$$



Exercise 1.4.0.6. Check that (1.4.5) is well-defined.

If $U \subset X$ is an open subset containing a point $x \in X$, then the diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\eta_U} & \mathcal{G}(U) \\ \downarrow & & \downarrow \\ \mathcal{F}_x & \xrightarrow{\eta_x} & \mathcal{G}_x \end{array} \qquad \begin{array}{ccc} s & \xmapsto{\eta_U} & \eta_U(s) \\ \downarrow & & \downarrow \\ s_x & \xmapsto{\eta_x} & (\eta_U(s))_x \end{array}$$

commutes. What we have just said can be rephrased by saying that the association $\mathcal{F} \mapsto \mathcal{F}_x$ defines a functor

$$\mathsf{stalk}_x\colon \mathsf{pSh}(X, \mathscr{C}) \longrightarrow \mathscr{C}. \tag{1.4.6}$$

We will see that in reasonable circumstances the restriction of this functor to the category of sheaves is *exact* (cf. Proposition 1.6.0.9).

Definition 1.4.0.7. A morphism of (pre)sheaves $\eta\colon \mathcal{F} \to \mathcal{G}$ is *surjective* if η_x is surjective for every $x \in X$.



Warning 1.4.0.8. You may have noticed that surjectivity of a map of sheaves (cf. Definition 1.4.0.7) is defined differently than injectivity (cf. Definition 1.2.0.13)!

Let $\eta\colon \mathcal{F}\to\mathcal{G}$ be a morphism of sheaves. Then

$$\eta_x \text{ is surjective} \iff \begin{array}{c}\text{for every } t_x\in\mathcal{G}_x \text{ there exists an open neighbourhood}\\ U \text{ of } x \text{ and a section } s\in\mathcal{F}(U) \text{ such that } (\eta_U(s))_x = t_x.\end{array}$$

$$\eta \text{ is surjective} \iff \begin{array}{c}\text{for every open subset } U\subset X \text{ and for every}\\ t\in\mathcal{G}(U), \text{ there exists a covering } U=\bigcup_{i\in I}U_i\\ \text{such that } t|_{U_i} \text{ is in the image of } \eta_{U_i} \text{ for every } i.\end{array}$$

The second equivalence is obtained as follows.

Proof of '$\Rightarrow$'. Assume η is surjective, i.e. η_x is surjective for every $x\in X$. Fix $U\subset X$ open and a local section $t\in\mathcal{G}(U)$. For every $x\in U$, we have a commutative diagram

$$\begin{array}{ccc}\mathcal{F}(U) & \xrightarrow{\eta_U} & \mathcal{G}(U)\\ \downarrow & & \downarrow\\ \mathcal{F}_x & \xrightarrow{\eta_x} & \mathcal{G}_x\end{array}\qquad\qquad\begin{array}{c} t\\ \downarrow\\ t_x\end{array}$$

where $t_x\in\mathcal{G}_x$ can be lifted along η_x to an element $s_x\in\mathcal{F}_x$. Let (V_x,s) be a representative for s_x, so that in particular $s\in\mathcal{F}(V_x)$. The identity $\eta_x(s_x)=t_x$ implies that there is an open neighbourhood $x\in U_x\subset V_x\cap U$ such that the diagram

$$\begin{array}{ccccc}s\in\mathcal{F}(V_x) & \xrightarrow{\rho^{\mathcal{F}}_{V_xU_x}} & \mathcal{F}(U_x) & \xrightarrow{\eta_{U_x}} & \mathcal{G}(U_x)\\ & & \downarrow & & \downarrow\\ & & \mathcal{F}_x & \xrightarrow{\eta_x} & \mathcal{G}_x\end{array}\qquad\qquad\begin{array}{ccccc}s & \longmapsto & s|_{U_x} & \longmapsto & t|_{U_x}\\ & & \downarrow & & \downarrow\\ & & s_x & \longmapsto & t_x\end{array}$$

commutes, in particular $\eta_{U_x}(s|_{U_x}) = t|_{U_x}$. Now this holds for every $x\in U$, and the elements of $\{U_x \mid x\in U\}$ form a covering of U, thus we have proved the condition, using the cover $U=\bigcup_{x\in U}U_x$ along with the sections $s|_{U_x}\in\mathcal{F}(U_x)$.

Proof of '$\Leftarrow$'. Conversely, assuming the condition, let us prove surjectivity of η. Fix $x\in X$ along with a germ $t_x\in\mathcal{G}_x$. We need to prove that t_x has a preimage in $\mathcal{F}_x$. Let (U,t) be a representative of t_x, so that $t\in\mathcal{G}(U)$. By the condition we are assuming, there exists a covering $U=\bigcup_{i\in I}U_i$ such that $t|_{U_i}=\eta_{U_i}(s_i)$ for some $s_i\in\mathcal{F}(U_i)$, for every $i\in I$. If $x\in U_i$, we have a commutative diagram

$$\begin{array}{ccc}\mathcal{F}(U_i) & \xrightarrow{\eta_{U_i}} & \mathcal{G}(U_i)\\ \downarrow & & \downarrow\\ \mathcal{F}_x & \xrightarrow{\eta_x} & \mathcal{G}_x\end{array}\qquad\qquad\begin{array}{ccc}s_i & \longmapsto & t|_{U_i}\\ \downarrow & & \downarrow\\ \star & \longmapsto & t_x\end{array}$$

so the element $\star\in\mathcal{F}_x$ is a preimage of t_x. The equivalence is proved.

The next result incarnates the local nature of sheaves.

LEMMA 1.4.0.9. *Let $\eta\colon\mathcal{F}\to\mathcal{G}$ be a morphism of sheaves. The following are equivalent:*

(i) *η is an isomorphism,*

(ii) *η_x is an isomorphism for every $x \in X$,*

(iii) *η is injective and surjective.*

Proof. Recall that η is an isomorphism if and only if η_U is an isomorphism for every U.

Proof of (i) $\Rightarrow$ (ii). By functoriality of $\mathcal{F} \mapsto \mathcal{F}_x$, we have that if η is an isomorphism, then so is η_x for every $x \in X$.

Proof of (ii) $\Rightarrow$ (i). Suppose η_x is an isomorphism for every x. Let $U \subset X$ be an open subset: we need to show that η_U is an isomorphism. In doing so, we implicitly use Lemma 1.3.0.5 (indeed we just show η_U is bijective).

To see that η_U is injective, pick $s, t \in \mathcal{F}(U)$ such that $\eta_U(s) = \eta_U(t) \in \mathcal{G}(U)$. Then, for any $x \in U$, one has

$$\eta_x(s_x) = (\eta_U(s))_x = (\eta_U(t))_x = \eta_x(t_x),$$

which implies $s_x = t_x$ by injectivity of η_x. This holds for every $x \in U$ by assumption, thus $s = t$ by Lemma 1.4.0.4. Therefore, η_U is injective for every U (i.e. η is injective).

To see that η_U is surjective, pick $t \in \mathcal{G}(U)$. By surjectivity of η (which we have since η_x is surjective for every $x \in X$), we can find an open cover $U = \bigcup_{i \in I} U_i$ along with a collection of sections $s_i \in \mathcal{F}(U_i)$ such that $\eta_{U_i}(s_i) = t|_{U_i}$. But η is injective, so s_i and s_j agree on $U_i \cap U_j$. Therefore, since $\mathcal{F}$ is a sheaf, they glue to a section $s \in \mathcal{F}(U)$ such that $s_i = s|_{U_i}$. By construction, $\eta_U(s)|_{U_i} = \eta_{U_i}(s_i) = t|_{U_i}$, which implies $\eta_U(s) = t$ since $\mathcal{G}$ is a sheaf. Thus η_U is surjective.

Proof of (ii) $\Rightarrow$ (iii). The first paragraph of '(ii) $\Rightarrow$ (i)' already shows that if η_x is an isomorphism for every $x \in X$, then η_U is injective for all U, i.e. η is injective. Surjectivity follows from the definition.

Proof of (iii) $\Rightarrow$ (ii). We only need to show that if η_U is injective for every U, then η_x is injective for every $x \in X$. Consider $s_x = [U, s]$ and $s'_x = [U', s']$ two germs in $\mathcal{F}_x$ such that $\eta_x(s_x) = \eta_x(s'_x)$ in $\mathcal{G}_x$. Then there is an open subset $W \subset U \cap U'$ such that $\eta_U(s)|_W = \eta_{U'}(s')|_W$. But by compatibility of η_W with restrictions, this is equivalent to the identity $\eta_W(s|_W) = \eta_W(s'|_W)$, which by our assumption implies $s|_W = s'|_W$. But then $s_x = s'_x$. □



Warning 1.4.0.10. It is not true that two sheaves with isomorphic stalks are isomorphic: there may be no map between them! For instance, consider a topological space X consisting of two points x_0, x_1 where only x_0 is a closed point. Thus X and $U = X \setminus \{x_0\}$ are the only nonempty open subsets of X. Fix an abelian group $G \neq 0$ and define $\mathcal{F}(X) = G = \mathcal{F}(U)$. Then choose either $\rho_{XU} = \mathrm{id}_G$ or $\rho_{XU} = 0$ to define two distinct sheaves on X. They have the same stalks but they are not isomorphic.



Exercise 1.4.0.11. Show that Lemma 1.4.0.9 fails for presheaves. (**Hint**: Construct a nonzero presheaf all of whose stalks vanish).

Example 1.4.0.12 (Surjectivity is subtle). Let $\mathcal{F} = \mathcal{O}_X^{\mathrm{h}}$ be the sheaf of holomorphic functions on $X = \mathbb{C}\setminus\{0\}$, and let $\mathcal{G} = \mathcal{F}^\times$ be the sheaf of invertible holomorphic functions on X. The map $\exp\colon \mathcal{F} \to \mathcal{G}$ is surjective, but $\exp_X\colon \mathcal{F}(X) \to \mathcal{G}(X)$ is not surjective, e.g. the function $f(z) = z$ in $\mathcal{G}(X)$ is not the exponential of a holomorphic function (cf. Example 1.2.0.23).

Example 1.4.0.13 (Skyscraper sheaf). Let X be a topological space, G a nontrivial abelian group, $x \in X$ a point. The assignment

$$U \mapsto G_x(U) = \begin{cases} G & \text{if } x \in U \\ 0 & \text{if } x \notin U \end{cases}$$

defines a sheaf, choosing as restriction maps the identity of G or the zero map in the obvious way. This sheaf is called the *skyscraper sheaf* attached to (X, x, G). At the level of stalks, one has

$$(G_x)_y = \begin{cases} G & \text{if } y \in \overline{\{x\}} \\ 0 & \text{if } y \notin \overline{\{x\}}, \end{cases}$$

because if y is in the closure of x then every neighbourhood of y also contains x, whereas if y is *not* in the closure of x, then the neighbourhood $U = X \setminus \overline{\{x\}}$ of y does not contain x, and thus $(G_x)_y = 0$.

Observe that G_x has only one nonzero stalk (at x) if and only if x is a closed point: this is the case where the name 'skyscraper sheaf' for G_x fits best.



Exercise 1.4.0.14. Let $\mathcal{F}$ be a presheaf, $\mathcal{G}$ a sheaf, and let $\eta_1, \eta_2\colon \mathcal{F} \to \mathcal{G}$ be two morphisms of presheaves of sets such that $\eta_{1,x} = \eta_{2,x}$ for every $x \in X$. Show that $\eta_1 = \eta_2$. Show that it is in fact necessary to assume $\mathcal{G}$ to be a sheaf (**Hint**: Use a nonzero presheaf all of whose stalks vanish, as in Exercise 1.4.0.11). This exercise will be needed in **??**.

1.5 Sheafification

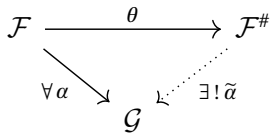
Fix a type of algebraic structure $(\mathscr{C}, F\colon \mathscr{C} \to \mathsf{Sets})$. Friendly translation: fix $\mathscr{C}$ to be either of the following categories:

$$\mathscr{C} \in \left\{ \mathsf{Sets}, \mathsf{Monoids}, \mathsf{Grp}, \mathsf{Ab}, \mathsf{Rings}, \mathsf{Mod}_A \right\}.$$

Let X be a topological space. Let $\mathcal{F}\colon \tau_X^{\mathrm{op}} \to \mathscr{C}$ be a presheaf. We next define a *sheaf* $\mathcal{F}^\#$, called the sheafification of $\mathcal{F}$, via an explicit universal property, and having precisely the same stalks as the initial presheaf $\mathcal{F}$.

Definition 1.5.0.1 (Sheafification of a presheaf). Let $\mathcal{F} \in \mathsf{pSh}(X, \mathscr{C})$ be a presheaf. A *sheafification* of $\mathcal{F}$ is a pair $(\mathcal{F}^\#, \theta)$, where $\mathcal{F}^\# \in \mathsf{Sh}(X, \mathscr{C})$ is a sheaf and $\theta\colon \mathcal{F} \to \mathcal{F}^\#$ is a morphism of presheaves, such that for every other pair $(\mathcal{G}, \alpha)$ where $\mathcal{G}$ is a sheaf and

$\alpha\colon \mathcal{F} \to \mathcal{G}$ is a morphism of presheaves, there exists a unique morphism of sheaves $\widetilde{\alpha}\colon \mathcal{F}^{\#} \to \mathcal{G}$ such that $\alpha = \widetilde{\alpha} \circ \theta$.



PROPOSITION 1.5.0.2. *Let $\mathcal{F} \in \mathsf{pSh}(X, \mathscr{C})$ be a presheaf. Then a sheafification $(\mathcal{F}^{\#}, \theta)$ exists, and the map $\theta_x\colon \mathcal{F}_x \to \mathcal{F}^{\#}_x$ is an isomorphism for every $x \in X$.*

What follows immediately from Proposition 1.5.0.2 is that $\mathcal{F}^{\#}$ is unique up to a unique isomorphism, and moreover the canonical map $\theta\colon \mathcal{F} \to \mathcal{F}^{\#}$ is an isomorphism precisely when $\mathcal{F}$ is already a sheaf.

Proof. Let $U \subset X$ be an open subset. Define

$$\mathcal{F}^{\#}(U) = \left\{ \text{functions } U \xrightarrow{f} \coprod_{x \in U} \mathcal{F}_x \;\middle|\; \begin{array}{c} \text{for every } x \in U,\ f(x) \in \mathcal{F}_x \text{ and there exist an} \\ \text{open neighbourhood } V \subset U \text{ of } x \text{ and } s \in \mathcal{F}(V) \\ \text{such that } f(y) = s_y \text{ for every } y \in V \end{array} \right\}.$$

For a thorough proof that $\mathcal{F}^{\#}(U)$ defines a genuine object of $\mathscr{C}$, we refer to [7, Tag 0086].[3] Note that, since $\mathscr{C}$ has products, we can view a function f as above as a tuple

$$(f(x))_{x \in U} \in \prod_{x \in U} \mathcal{F}_x$$

and we can rephrase the definition of $\mathcal{F}^{\#}(U)$ by saying that

$$\mathcal{F}^{\#}(U) = \left\{ \text{tuples } (s_x)_{x \in U} \text{ of compatible germs} \right\}.$$

See Definition 1.4.0.3 for the definition of compatible germs. Functoriality of the assignment $U \mapsto \mathcal{F}^{\#}(U)$ is clear (functions restrict!), thus $\mathcal{F}^{\#}$ is a presheaf. The morphism $\theta_U\colon \mathcal{F}(U) \to \mathcal{F}^{\#}(U)$ defined by sending $s \in \mathcal{F}(U)$ to the function

$$U \xrightarrow{f_s} \coprod_{x \in U} \mathcal{F}_x, \qquad x \mapsto s_x = [U, s] \in \mathcal{F}_x$$

determines a morphism of presheaves, being compatible with restrictions. It is just the function $\sigma^{\mathcal{F}}_U$ introduced in (1.4.1)!

The presheaf $\mathcal{F}^{\#}$ is a sheaf: By Lemma 1.3.0.5, the sheaf condition can be verified in the category of sets. Fix an open cover $U = \bigcup_{i \in I} U_i$ of some open subset $U \subset X$ and a collection of sections $f_i \in \mathcal{F}^{\#}(U_i)$ such that $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for every i and j. We need to find a unique $f \in \mathcal{F}^{\#}(U)$ such that $f|_{U_i} = f_i$. Define

$$f(x) = f_i(x) \in \mathcal{F}_x, \quad x \in U_i \subset U.$$

[3] The reader may easily verify this by hand for the familiar types of algebraic structure listed at the beginning of this section, which is all that really matters to us.

This is well-defined since, even though x can lie in more than one open U_i, by assumption we have $f_i(x) = f_j(x)$ as soon as $x \in U_i \cap U_j$. We need to check that f defines an element of $\mathcal{F}^\#(U)$, not just of the full product $\prod_{x\in U} \mathcal{F}_x$. But for every $i \in I$ we know the following: for every $x \in U_i$ there exist an open neighbourhood $x \in V_i \subset U_i$ and a section $s_i \in \mathcal{F}(V_i)$ such that $f(y) = f_i(y) = (s_i)_y$ for all $y \in V_i$. But V_i is also open in U, so the condition defining $\mathcal{F}^\#(U)$ also holds for f. Thus $f \in \mathcal{F}^\#(U)$ satisfies $f|_{U_i} = f_i$, and is clearly unique with this property.

The pair $(\mathcal{F}^\#, \theta)$ is the sheafification. Assume we have a sheaf $\mathcal{G}$ and a morphism of presheaves $\alpha\colon \mathcal{F} \to \mathcal{G}$. With a slight abuse of notation, we shall identify $\mathcal{G}$ and $\mathcal{G}^\#$ via the canonical isomorphism $\mathcal{G} \xrightarrow{\sim} \mathcal{G}^\#$. We need to define a morphism $\widetilde{\alpha}\colon \mathcal{F}^\# \to \mathcal{G}$ of presheaves such that $\alpha = \widetilde{\alpha} \circ \theta$. For every U open in X, we need to define a morphism $\widetilde{\alpha}_U : \mathcal{F}^\#(U) \to \mathcal{G}(U)$ in such a way that $\alpha_U = \widetilde{\alpha}_U \circ \theta_U$. Fix $s = (s_x)_{x\in U} \in \mathcal{F}^\#(U)$. The composition

$$\begin{array}{ccc} U & \xrightarrow{\ s\ } & \coprod_{x\in U} \mathcal{F}_x \\ & \searrow^{\widetilde{\alpha}_U(s)} & \downarrow{\scriptstyle \coprod_{x\in U} \alpha_x} \\ & & \coprod_{x\in U} \mathcal{G}_x \end{array}$$

defines a tuple of compatible germs for $\mathcal{G}$ over U, hence an element $\widetilde{\alpha}_U(s) \in \mathcal{G}^\#(U) = \mathcal{G}(U)$, using that $\mathcal{G}$ is a sheaf for this identity. This is the required morphism $\widetilde{\alpha}\colon \mathcal{F}^\# \to \mathcal{G}$.

The map θ is an isomorphism on stalks. The map θ, at the level of stalks, is defined by

$$\theta_x[U, s] = [U, f_s].$$

Injectivity: Suppose $\theta_x[U, s] = \theta_x[V, t]$ for two classes $[U, s], [V, t] \in \mathcal{F}_x$, i.e. assume $[U, f_s] = [V, f_t]$ in $\mathcal{F}^\#_x$. Then, by definition of germ, there exists an open neighbourhood $W \subset U \cap V$ of x such that $f_s|_W = f_t|_W$. But this means, by definition of f_s and f_t, that $s_y = t_y$ for all $y \in W$. Thus, in particular, $s_x = t_x$. But this is just the equality $[U, s] = [V, t]$ we were after.

Surjectivity: Pick a class $[U, f] \in \mathcal{F}^\#_x$ for some $f \in \mathcal{F}^\#(U)$ and open neighbourhood U of x. Then, for every $z \in U$, there exist an open neighbourhood $V \subset U$ of z and a section $s \in \mathcal{F}(V)$ such that $f(y) = s_y$ in $\mathcal{F}_y$ for every $y \in V$. Apply now this germ compatibility to the point $z = x$: we claim that $[U, f] = \theta_x(s_x)$, where $s_x = [V, s]$. Indeed, $\theta_x(s_x) \in \mathcal{F}^\#_x$ is the equivalence class of the map

$$V \xrightarrow{\ f_s\ } \coprod_{y\in V} \mathcal{F}_y, \quad y \mapsto s_y.$$

But this map agrees with the restriction of f to $V \subset U$ (by the condition $f(y) = s_y$ recalled above), i.e. $f_s = f|_V \in \mathcal{F}^\#(V)$. Since V is also an open neighbourhood of x, it follows that $(f|_V)_x = (f_s)_x = [V, f_s] = \theta_x(s_x) \in \mathcal{F}^\#_x$, but of course $(f|_V)_x = [U, f]$. Thus θ_x is surjective. □

Example 1.5.0.3 (Constant sheaf). Let G be a nontrivial abelian group. The *constant sheaf* on a topological space X, with values in G, is the sheafification $\underline{G}_X$ of the presheaf $\underline{G}_X^{\mathrm{pre}}$ defined in Example 1.2.0.19. This sheaf agrees with the sheaf whose sections over U are the locally constant functions $U \to G$. This, in turn, agrees with the following: endow G with the discrete topology and consider the assignment

$$U \longmapsto \{\text{continuous maps } U \to G\},$$

which we know is a sheaf by Example 1.2.0.16. If $U \subset X$ is a nonempty connected open subset, then $\underline{G}_X(U) = G$. By Proposition 1.5.0.2, at the level of stalks we have $\underline{G}_{X,x} = G$ for every $x \in X$, since the stalks of the constant presheaf are manifestly all equal to G.



Exercise 1.5.0.4. What is the sheafification of the presheaf of bounded continuous functions (cf. Exercise 1.2.0.18)?



Exercise 1.5.0.5. Let X be a topological space, x a point, G a nontrivial abelian group. Can you find a necessary and sufficient condition for the constant sheaf $\underline{G}_X$ being equal to the skyscraper sheaf G_x (cf. Example 1.4.0.13)?



Exercise 1.5.0.6. Show that $\mathcal{F} \mapsto \mathcal{F}^{\#}$ defines a functor $(-)^{\#}\colon \mathsf{pSh}(X,\mathscr{C}) \to \mathsf{Sh}(X,\mathscr{C})$, and that the forgetful functor $j_{X,\mathscr{C}}\colon \mathsf{Sh}(X,\mathscr{C}) \hookrightarrow \mathsf{pSh}(X,\mathscr{C})$ is a right adjoint. In other words (cf. Definition A.4.0.1), prove that there are bifunctorial bijections

$$\psi_{\mathcal{F},\mathcal{G}}\colon \mathrm{Hom}_{\mathsf{Sh}(X,\mathscr{C})}(\mathcal{F}^{\#},\mathcal{G}) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{pSh}(X,\mathscr{C})}(\mathcal{F},\mathcal{G}), \quad \widetilde{\alpha} \mapsto \widetilde{\alpha} \circ \theta$$

for any presheaf $\mathcal{F}$ and any sheaf $\mathcal{G}$. (**Hint**: the universal property of the sheafification!).

1.6 Subsheaves, Quotient sheaves, and exactness

We have essentially already proved the following general result.

PROPOSITION 1.6.0.1 ([7, Tag 007S]). *Let X be a topological space. Let $\mathcal{F},\mathcal{G} \in \mathsf{Sh}(X,\mathsf{Sets})$ be sheaves of sets, $\eta\colon \mathcal{F} \to \mathcal{G}$ a morphism. Then, the following are equivalent:*

(a) *η is a monomorphism,*

(b) *$\eta_x\colon \mathcal{F}_x \to \mathcal{G}_x$ is injective for all $x \in X$,*

(c) *$\eta_U\colon \mathcal{F}(U) \to \mathcal{G}(U)$ is injective for all open subsets $U \subset X$ (i.e. η is injective).*

Furthermore, the conditions

(i) *η is an epimorphism,*

(ii) *$\eta_x\colon \mathcal{F}_x \to \mathcal{G}_x$ is surjective for all $x \in X$ (i.e. η is surjective),*

are equivalent, and are implied (but not equivalent to, cf. Example 1.4.0.12!) by the condition

(iii) $\eta_U : \mathcal{F}(U) \to \mathcal{G}(U)$ *is surjective for all open subsets* $U \subset X$.

If $(\mathscr{C}, F)$ is a type of algebraic structure where $\mathscr{C}$ is abelian and has exact filtered colimits (e.g. Mod_A for a fixed ring A), then Proposition 1.6.0.1 holds replacing Sets with $\mathscr{C}$. In this setup, the condition that η_x be injective, resp, surjective, can safely be replaced by η_x being a monomorphism, resp. an epimorphism in $\mathscr{C}$.

Definition 1.6.0.2 (Subsheaf, quotient sheaf)**.** If there exists a morphism of sheaves $\eta : \mathcal{F} \to \mathcal{G}$ such that either of the equivalent conditions (a), (b) or (c) holds, we say that $\mathcal{F}$ is a *subsheaf* of $\mathcal{G}$ (and we may denote this by '$\mathcal{F} \subset \mathcal{G}$'). If either of the equivalent conditions (i) or (ii) holds, we say that $\mathcal{G}$ is a *quotient sheaf* of $\mathcal{F}$.

Example 1.6.0.3 (Quotient sheaf)**.** Let $(\mathscr{C}, F)$ be a type of algebraic structure such that $\mathscr{C}$ is an abelian category. If $\mathcal{F} \subset \mathcal{G}$ is a subsheaf (with values in $\mathscr{C}$), then sending

$$U \longmapsto \mathcal{G}(U)/\mathcal{F}(U) \tag{1.6.1}$$

is a presheaf on X, because the restriction maps respect the inclusions $\mathcal{F}(U) \hookrightarrow \mathcal{G}(U)$, and thus pass to the quotients. Its sheafification $\mathcal{G}/\mathcal{F}$ is called the *quotient sheaf of* $\mathcal{G}$ *by* $\mathcal{F}$. There is a natural morphism of sheaves $\mathcal{G} \to \mathcal{G}/\mathcal{F}$.

Definition 1.6.0.4 (Sheaf image, sheaf cokernel)**.** Let $(\mathscr{C}, F)$ be a type of algebraic structure with $\mathscr{C}$ abelian, $\eta : \mathcal{F} \to \mathcal{G}$ a morphism of sheaves (with values in $\mathscr{C}$), so that $\ker(\eta) \hookrightarrow \mathcal{F}$ is a subsheaf by Exercise 1.2.0.22. The sheafification $\mathrm{im}(\eta)$ of the presheaf

$$U \longmapsto \mathrm{im}_{\mathrm{pre}}(U) = \mathrm{im}(\eta_U) = \mathcal{F}(U)/\ker(\eta_U)$$

is called the *image of* η. It is a special case of Example 1.6.0.3 and defines a subsheaf

$$\mathrm{im}(\eta) = \mathcal{F}/\ker(\eta) \hookrightarrow \mathcal{G}.$$

The quotient sheaf

$$\mathrm{coker}(\eta) = \mathcal{G}/\mathrm{im}(\eta),$$

again a special case of Example 1.6.0.3, is called the *sheaf cokernel*.

Remark 1.6.0.5. Set $\mathscr{C} = \mathsf{Mod}_A$, or any Grothendieck abelian category supporting a type of algebraic structure. By definition, $\mathscr{C}$ has axiom (AB5), i.e. filtered colimits exist and are (also *left*) exact. Let $\mathcal{F} \subset \mathcal{G}$ be a subsheaf, $x \in X$ a point. Then

$$(\mathcal{G}/\mathcal{F})_x = \mathcal{G}_x/\mathcal{F}_x \tag{1.6.2}$$

in Mod_A. This follows from the fact that $(\mathcal{G}/\mathcal{F})_x$ agrees with the stalk of the *presheaf* (1.6.1), and from right exactness of filtered colimits. Moreover, if $\eta : \mathcal{F} \to \mathcal{G}$ is a morphism of sheaves and $x \in X$ is a point, then

$$\begin{aligned} \ker(\eta)_x &= \ker(\eta_x) \\ \mathrm{im}(\eta)_x &= \mathrm{im}(\eta_x) \\ \mathrm{coker}(\eta)_x &= \mathrm{coker}(\eta_x). \end{aligned} \tag{1.6.3}$$

The first identity in (1.6.3) follows from the fact that filtered colimits are *also left exact* in Mod_A, thus

$$\begin{aligned}\ker\left(\mathcal{F}_x \xrightarrow{\eta_x} \mathcal{G}_x\right) &= \ker\left(\varinjlim_{U\ni x} \mathcal{F}(U) \to \varinjlim_{U\ni x} \mathcal{G}(U)\right)\\ &= \varinjlim_{U\ni x} \ker(\mathcal{F}(U)\to\mathcal{G}(U))\\ &= \ker(\eta)_x.\end{aligned}$$

The last two identities in (1.6.3) are a special case of (1.6.2).

Exercise 1.6.0.6. Set $\mathscr{C} = \mathsf{Mod}_A$, or any Grothendieck abelian category supporting a type of algebraic structure. By definition, $\mathscr{C}$ has axiom (AB5), i.e. filtered colimits exist and are (also *left*) exact. Let $\eta\colon \mathcal{F}\to\mathcal{G}$ be a morphism of sheaves with values in $\mathscr{C}$. Show that the composition

$$\mathcal{G} \longrightarrow \operatorname{coker}_{\mathrm{pre}}\eta \longrightarrow \operatorname{coker}(\eta),$$

where the first morphism is given by the natural maps $\mathcal{G}(U)\twoheadrightarrow \mathcal{G}(U)/\operatorname{im}(\eta_U)$ and the last morphism is the sheafification, is a cokernel in the category $\mathsf{Sh}(X,\mathscr{C})$. Finally, show that η is an epimorphism if and only if $\operatorname{coker}(\eta)=0$, and if $\mathscr{C}=\mathsf{Mod}_A$ this is also equivalent to stalkwise surjectivity.

THEOREM 1.6.0.7 ([2, §10]). *If $\mathscr{C}$ is a Grothendieck abelian category, then $\mathsf{Sh}(X,\mathscr{C})$ is a Grothendieck abelian category.*

Definition 1.6.0.8. A *short exact sequence of sheaves* with values in a Grothendieck abelian category $\mathscr{C}$ is a short exact sequence

$$0 \longrightarrow \mathcal{F} \xrightarrow{\iota} \mathcal{G} \xrightarrow{\theta} \mathcal{H} \longrightarrow 0$$

of objects in the abelian category $\mathsf{Sh}(X,\mathscr{C})$. Explicitly, exactness means that ι is a monomorphism, θ is an epimorphism and $\operatorname{im}(\iota)=\ker(\theta)$.

PROPOSITION 1.6.0.9. *Let $\mathscr{C}$ be a Grothendieck abelian category. A sequence*

$$0 \longrightarrow \mathcal{F} \xrightarrow{\iota} \mathcal{G} \xrightarrow{\theta} \mathcal{H} \longrightarrow 0$$

of objects in $\mathsf{Sh}(X,\mathscr{C})$ is a short exact sequence if and only if

$$0 \longrightarrow \mathcal{F}_x \xrightarrow{\iota_x} \mathcal{G}_x \xrightarrow{\theta_x} \mathcal{H}_x \longrightarrow 0$$

is a short exact sequence in $\mathscr{C}$ for every $x\in X$.

Proof. Combine Remark 1.6.0.5 and Lemma 1.4.0.9 with one another. □



Exercise 1.6.0.10. Let $\eta\colon \mathcal{F}\to\mathcal{G}$ be a morphism of sheaves of A-modules, where A is a ring. Prove that there is an exact sequence of sheaves

$$0 \longrightarrow \ker(\eta) \longrightarrow \mathcal{F} \xrightarrow{\eta} \mathcal{G} \longrightarrow \operatorname{coker}(\eta) \longrightarrow 0.$$

In particular, if η is injective, this reduces to

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{G}/\mathcal{F} \longrightarrow 0.$$



Exercise 1.6.0.11. Let A be a ring. For a nonempty open subset U of a topological space X, consider the functor $\Gamma(U,-)\colon \mathsf{Sh}(X,\mathsf{Mod}_A)\to\mathsf{Mod}_A$ sending $\mathcal{F}\mapsto\mathcal{F}(U)$. Show that it is left exact. That is, it transforms an exact sequence of sheaves $0\to\mathcal{F}\to\mathcal{G}\to\mathcal{H}$ into an exact sequence of A-modules

$$0 \longrightarrow \mathcal{F}(U) \longrightarrow \mathcal{G}(U) \longrightarrow \mathcal{H}(U).$$

When $U=X$, this functor takes $\mathcal{F}\mapsto\mathcal{F}(X)$ and is thus called the *global section functor*. Another notation used for it in the literature is $\mathrm{H}^0(X,-)$, cf. Terminology 1.2.0.3.

1.7 Supports

Let A be a ring. Let $\mathcal{F}\in\mathsf{Sh}(X,\mathsf{Mod}_A)$ be a sheaf of A-modules on a topological space X. Let $U\subset X$ be an open subset, and fix a section $s\in\mathcal{F}(U)$. We have two notions of support: the support of $\mathcal{F}$, and the support of s, defined respectively as

$$\begin{aligned}\mathsf{Supp}(\mathcal{F}) &= \{\, x\in X \mid \mathcal{F}_x\neq 0\,\},\\ \mathsf{Supp}(s) &= \{\, x\in U \mid s_x\neq 0 \text{ in } \mathcal{F}_x\,\}.\end{aligned} \tag{1.7.1}$$

If $s_x=0$, then there is an open neighbourhood $x\in V\subset U$ such that $s|_V=0\in\mathcal{F}(V)$. Thus $V\subset U\setminus\mathsf{Supp}(s)$ and hence $\mathsf{Supp}(s)\subset U$ is closed. In fact, this follows from (or solves) Exercise 1.4.0.5. In general, however, $\mathsf{Supp}(\mathcal{F})\subset X$ is *not* closed, as the two next examples show.

Example 1.7.0.1 ($\mathsf{Supp}(\mathcal{F})$ need not be closed, take I)**.** Let X be an irreducible topological space. This means that any two nonempty open subsets of X intersect. Fix a nontrivial abelian group G, a point $x\in X$, and for $U\in\tau_X$ define

$$\mathcal{F}(U)=\begin{cases}0 & \text{if } U=\emptyset \text{ or } x\in U\\ G & \text{otherwise.}\end{cases}$$

Let $\rho_{UV}\in\{\mathrm{id}_G,0\}$ be chosen in the obvious way for all $U,V\in\tau_X$. Then $\mathcal{F}$ is a sheaf of abelian groups on X, with stalks

$$\mathcal{F}_y=\begin{cases}0 & \text{if } y\in\overline{\{x\}}\\ G & \text{otherwise.}\end{cases}$$

It follows that

$$\mathsf{Supp}(\mathcal{F}) = X \setminus \overline{\{x\}},$$

which is not closed in X as soon as $\overline{\{x\}} \hookrightarrow X$ is not open.

Example 1.7.0.2 ($\mathsf{Supp}(\mathcal{F})$ need not be closed, take II)**.** Let $j : U \hookrightarrow X$ be the inclusion of an open subset U of a topological space X. Let $\mathcal{F} \in \mathsf{Sh}(U, \mathsf{Ab})$ be a sheaf. Define $j_!\mathcal{F} \in \mathsf{Sh}(X, \mathsf{Ab})$ to be the sheafification of the presheaf $j_!^{\mathrm{pre}}\mathcal{F}$ defined by

$$j_!^{\mathrm{pre}}\mathcal{F}(V) = \begin{cases} \mathcal{F}(V) & \text{if } V \subset U \\ 0 & \text{otherwise} \end{cases}$$

on X. One has

$$(j_!\mathcal{F})_x = \begin{cases} \mathcal{F}_x & \text{if } x \in U \\ 0 & \text{otherwise,} \end{cases}$$

so that $\mathsf{Supp}(j_!\mathcal{F}) = \mathsf{Supp}(\mathcal{F})$. Fix $G \neq 0$ in Ab and consider the constant sheaf on U (cf. Example 1.5.0.3). We have $\mathsf{Supp}(j_!\underline{G}_U) = \mathsf{Supp}(\underline{G}_U) = U$. In particular, $\mathsf{Supp}(j_!\underline{G}_U) \subset X$ is not closed as soon as U is not closed in X.

If $\mathcal{F}$ is a sheaf of rings, the notions of support defined in (1.7.1) still make sense, and one has $\mathsf{Supp}(\mathcal{F}) = \mathsf{Supp}(1)$, where $1 \in \mathcal{F}(X)$ is the ring identity (recall that the '0 ring' is the one where $1 = 0$). Thus $\mathsf{Supp}(\mathcal{F})$ *is* in fact closed in this case.

1.8 Sheaves = sheaves on a base

Fix a type of algebraic structure $(\mathscr{C}, F : \mathscr{C} \to \mathsf{Sets})$.

Definition 1.8.0.1 ($\mathsf{Base\ of\ open\ sets}$)**.** Let X be a topological space. A *base of open sets* for X is a collection of open subsets $\mathcal{B} \subset \tau_X$ satisfying the following requirements:

(a) $\mathcal{B}$ is stable under finite intersections,

(b) every $U \in \tau_X$ can be written as a union of open sets belonging to $\mathcal{B}$.

The terminology just introduced is slightly nonstandard, in the sense that condition (a) is usually omitted. For convenience, we use the term *base* only for bases stable under finite intersections: this stronger convention will be particularly convenient for the basis of distinguished open subsets of an affine scheme.

Definition 1.8.0.2 ($\mathcal{B}$-sheaf)**.** A *$\mathcal{B}$-presheaf* (resp. *$\mathcal{B}$-sheaf*) is an assignment

$$U \xmapsto{\mathcal{F}_{\mathcal{B}}} \mathcal{F}_{\mathcal{B}}(U) \in \mathscr{C}, \text{ for each } U \in \mathcal{B},$$

such that the presheaf conditions (1)–(2) of Definition 1.2.0.2 (resp. those conditions and the sheaf conditions (loc)–(glue) of Important Definition 1.2.0.5) hold, considering only open sets belonging to $\mathcal{B}$.

Note that restriction maps

$$\rho_{UV}\colon \mathcal{F}_{\mathcal{B}}(U) \longrightarrow \mathcal{F}_{\mathcal{B}}(V)$$

are part of the data of a $\mathcal{B}$-(pre)sheaf whenever $V \subset U$ is an inclusion of open sets both belonging to $\mathcal{B}$. Note, also, that condition (a) in Definition 1.8.0.1 ensures that open subsets of the form $U \cap V$ still belong to $\mathcal{B}$ if $U, V \in \mathcal{B}$. In particular, as in Section 1.3, a $\mathcal{B}$-presheaf $\mathcal{F}_{\mathcal{B}}$ is a sheaf precisely when the following condition is fulfilled: for every open subset $U \in \mathcal{B}$ and for any open cover $U = \bigcup_{i\in I} U_i$ with all $U_i \in \mathcal{B}$, the diagram

$$\mathcal{F}_{\mathcal{B}}(U) \xrightarrow{\rho} \prod_{i\in I} \mathcal{F}_{\mathcal{B}}(U_i) \underset{\nu}{\overset{\mu}{\rightrightarrows}} \prod_{(i,j)\in I\times I} \mathcal{F}_{\mathcal{B}}(U_i \cap U_j)$$

is an equaliser diagram in $\mathscr{C}$.

Remark 1.8.0.3. Let $x \in X$ be a point. The collection of open neighbourhoods

$$\mathcal{B}_x = \{\, B \in \mathcal{B} \mid x \in B \,\}^{\mathrm{op}} \subset \tau_X^{\mathrm{op}}$$

is a fundamental system of open neighbourhoods of x, also called a local basis at x (i.e. for any $W \in \mathsf{Ngb}_x$ there exists $B \in \mathcal{B}_x$ such that $B \subset W$). In more technical terms, one may say that the filtered categories Ngb_x and $\mathcal{B}_x$ are *cofinal*, i.e. the inclusion $\mathcal{B}_x \hookrightarrow \mathsf{Ngb}_x$ is a cofinal functor. We will not use this terminology.

The stalk

$$\mathcal{F}_{\mathcal{B},x} = \varinjlim_{B\in\mathcal{B}_x} \mathcal{F}_{\mathcal{B}}(B) \in \mathscr{C} \tag{1.8.1}$$

of a $\mathcal{B}$-(pre)sheaf $\mathcal{F}_{\mathcal{B}}$ at a point $x \in X$ is well-defined as an object of $\mathscr{C}$. It receives, by definition of direct limit, canonical morphisms

$$\mathcal{F}_{\mathcal{B}}(B) \longrightarrow \mathcal{F}_{\mathcal{B},x}, \qquad B \in \mathcal{B}_x.$$

We denote by $s_x \in \mathcal{F}_{\mathcal{B},x}$, as ever, the image of $s \in \mathcal{F}_{\mathcal{B}}(B)$ under this morphism.

Moreover, if $B \in \mathcal{B}$ and $\mathcal{F}_{\mathcal{B}}$ is a $\mathcal{B}$-sheaf, the natural map

$$\sigma_B^{\mathcal{F}_{\mathcal{B}}}\colon \mathcal{F}_{\mathcal{B}}(B) \longrightarrow \prod_{x\in B} \mathcal{F}_{\mathcal{B},x}, \qquad s \mapsto (s_x)_{x\in B}, \tag{1.8.2}$$

is injective (as in Lemma 1.4.0.2), and its image agrees with the collections of compatible germs; to be more precise, we should now call them '$\mathcal{B}$-compatible', for they are, by definition, those tuples

$$(s_x)_{x\in B} \in \prod_{x\in B} \mathcal{F}_{\mathcal{B},x}$$

such that for every $x \in B$ there is a pair (V_x, t^x), where $V_x \in \mathcal{B}_x$ satisfies $V_x \subset B$ and $t^x \in \mathcal{F}_{\mathcal{B}}(V_x)$ satisfies $t^x_y = s_y$ for every $y \in V_x$. Note that, because $\mathcal{B}_x$ is a local basis at x, the notion of a tuple of $\mathcal{B}$-compatible germs makes sense for an arbitrary open subset $U \subset X$!

Definition 1.8.0.4 (Morphism of $\mathcal{B}$-sheaves)**.** A *morphism* of $\mathcal{B}$-(pre)sheaves

$$\eta_{\mathcal{B}}\colon \mathcal{F}_{\mathcal{B}} \longrightarrow \mathcal{G}_{\mathcal{B}} \tag{1.8.3}$$

is the datum of a collection of maps $\eta_B\colon \mathcal{F}_{\mathcal{B}}(B) \to \mathcal{G}_{\mathcal{B}}(B)$, one for each $B \in \mathcal{B}$, such that Diagram (1.2.1) commutes for all $B, C \in \mathcal{B}$ such that $C \subset B$.

With this definition, $\mathcal{B}$-sheaves form a category, denoted $\mathsf{Sh}_{\mathcal{B}}(X, \mathscr{C})$.[4]

Remark 1.8.0.5. Let X be a topological space, $\mathcal{B}$ a base of open subsets of X. A (pre)sheaf $\mathcal{F}$ on X is a $\mathcal{B}$-(pre)sheaf in a natural way. More precisely, there is (say, at the level of sheaves) a *restriction functor*

$$\mathsf{res}_{\mathcal{B}}\colon \mathsf{Sh}(X, \mathscr{C}) \longrightarrow \mathsf{Sh}_{\mathcal{B}}(X, \mathscr{C}), \tag{1.8.4}$$

defined on objects in the obvious way. Its functoriality is just a consequence of the definition of morphism of $\mathcal{B}$-sheaves, and is an easy routine check.

The next result says that the restriction functor (1.8.4) is essentially surjective.

LEMMA 1.8.0.6 (Extension from a base). *Let X be a topological space and let $\mathcal{B} \subset \tau_X$ be a base of open subsets. Let $\mathcal{F}_{\mathcal{B}} \in \mathsf{Sh}_{\mathcal{B}}(X, \mathscr{C})$ be a $\mathcal{B}$-sheaf. Then there exists a sheaf $\mathcal{F} \in \mathsf{Sh}(X, \mathscr{C})$ together with an isomorphism of $\mathcal{B}$-sheaves*

$$\alpha\colon \mathcal{F}_{\mathcal{B}} \xrightarrow{\sim} \mathsf{res}_{\mathcal{B}}(\mathcal{F}).$$

In particular, the restriction functor (1.8.4) *is essentially surjective.*

Proof. Recall from (1.8.1) the definition of the stalks $\mathcal{F}_{\mathcal{B},x}$ of a $\mathcal{B}$-sheaf. Let $U \in \tau_X$ be an arbitrary open set. Define

$$\mathcal{F}(U) = \left\{ \text{tuples } (s_x)_{x\in U} \text{ of } \mathcal{B}\text{-compatible germs} \right\} \subset \prod_{x\in U} \mathcal{F}_{\mathcal{B},x}.$$

As noted above, this makes sense for an arbitrary U, and confirms that $\mathcal{F}$ is a sheaf, for the same reason why the sheaf property is fulfilled by the sheafification of a presheaf (see the proof of Proposition 1.5.0.2). For every $B \in \mathcal{B}$, the natural map

$$\alpha_B\colon \mathcal{F}_{\mathcal{B}}(B) \xrightarrow{\sim} \mathcal{F}(B), \qquad s \mapsto (s_x)_{x\in B}$$

is an isomorphism (since the injective map $\sigma_B^{\mathcal{F}_{\mathcal{B}}}$ of (1.8.2) hits precisely the tuples of $\mathcal{B}$-compatible germs), and these maps are compatible with restriction. Hence they define an isomorphism of $\mathcal{B}$-sheaves $\alpha\colon \mathcal{F}_{\mathcal{B}} \xrightarrow{\sim} \mathsf{res}_{\mathcal{B}}(\mathcal{F})$ as required. □

We shall now prove that the restriction functor (1.8.4) is in fact an equivalence. A quasi-inverse is given precisely by Lemma 1.8.0.6 above for objects and by Theorem 1.8.0.8 below for morphisms.

[4] Also $\mathcal{B}$-presheaves form a category, but it is not as well-behaved as $\mathsf{Sh}_{\mathcal{B}}(X, \mathscr{C})$, and we do not need it, so we shall ignore it.

Remark 1.8.0.7. In the notation of Lemma 1.8.0.6, we have a canonical isomorphism $\mathcal{F}_{\mathcal{B},x} \xrightarrow{\sim} \mathcal{F}_x$ for all $x \in X$. This follows directly from Remark 1.8.0.3 (cofinal filtered categories have isomorphic colimits).

A (stronger) analogue of Lemma 1.8.0.6 for *morphisms* is the following.

THEOREM 1.8.0.8. *Let X be a topological space, $\mathcal{B} \subset \tau_X$ a base of open sets and let $\mathcal{F}$, $\mathcal{G} \in \mathsf{Sh}(X, \mathscr{C})$ be two sheaves on X. Suppose given a morphism*

$$\eta_{\mathcal{B}} \colon \mathsf{res}_{\mathcal{B}}(\mathcal{F}) \longrightarrow \mathsf{res}_{\mathcal{B}}(\mathcal{G})$$

between the underlying $\mathcal{B}$-sheaves. Then $\eta_{\mathcal{B}}$ extends uniquely to a sheaf homomorphism $\eta \colon \mathcal{F} \to \mathcal{G}$. Furthermore, if η_U is surjective (or injective, or an isomorphism) for every $U \in \mathcal{B}$, then so is η.

Proof. Before we start: By Lemma 1.3.0.5 we may perform the construction on the underlying sheaves of sets; locally the resulting maps agree with the morphisms $\eta_{\mathcal{B},B}$, hence they are morphisms in $\mathcal{C}$.

To ease notation, we set $\mathcal{F}_{\mathcal{B}} = \mathsf{res}_{\mathcal{B}}(\mathcal{F})$ and $\mathcal{G}_{\mathcal{B}} = \mathsf{res}_{\mathcal{B}}(\mathcal{G})$. Clearly we have $\mathcal{F}_{\mathcal{B}}(B) = \mathcal{F}(B)$ for $B \in \mathcal{B}$, and similarly for $\mathcal{G}$. Given $\eta_{\mathcal{B}}$, for every $B \in \mathcal{B}$ we already have a morphism

$$\eta_B \coloneqq \eta_{\mathcal{B},B} \colon \mathcal{F}_{\mathcal{B}}(B) \longrightarrow \mathcal{G}_{\mathcal{B}}(B),$$

compatible with restriction maps. We construct the extension $\eta \colon \mathcal{F} \to \mathcal{G}$ as follows. Let $U \subset X$ be an arbitrary open subset and let $s \in \mathcal{F}(U)$. Choose a covering

$$U = \bigcup_{i \in I} B_i, \qquad B_i \in \mathcal{B}.$$

For every i, consider the section

$$t_i = \eta_{B_i}(s|_{B_i}) \in \mathcal{G}(B_i).$$

We claim that the sections t_i agree on pairwise intersections. Indeed, since $\mathcal{B}$ is stable under finite intersections, $B_i \cap B_j \in \mathcal{B}$, and the compatibility of $\eta_{\mathcal{B}}$ with restriction maps gives

$$t_i|_{B_i \cap B_j} = \eta_{B_i \cap B_j}\big(s|_{B_i \cap B_j}\big) = t_j|_{B_i \cap B_j}.$$

Since $\mathcal{G}$ is a sheaf, there exists a unique section $t \in \mathcal{G}(U)$ such that $t|_{B_i} = t_i$ for every $i \in I$. We define

$$\eta_U(s) = t \in \mathcal{G}(U).$$

This definition is independent of the chosen covering. Indeed, if

$$U = \bigcup_{j \in J} B'_j$$

is another covering by elements of $\mathcal{B}$, then $B_i \cap B'_j \in \mathcal{B}$, and the two resulting sections of $\mathcal{G}(U)$ have the same restriction to every $B_i \cap B'_j$. Since these open subsets cover U, the locality axiom for $\mathcal{G}$ implies that the two sections coincide.

The maps η_U are compatible with restrictions. Indeed, let $V \subset U$ be an open subset and let $s \in \mathcal{F}(U)$. Cover V by elements $B_i \in \mathcal{B}$ satisfying $B_i \subset V$. Then

$$\eta_U(s)|_{B_i} = \eta_{B_i}(s|_{B_i}) = \eta_V(s|_V)|_{B_i} \in \mathcal{G}(B_i),$$

for every $i \in I$. Since the B_i cover V, the locality axiom for $\mathcal{G}$ gives

$$\eta_U(s)|_V = \eta_V(s|_V) \in \mathcal{G}(V).$$

Thus the collection $\{\eta_U\}_{U \in \tau_X}$ defines a morphism of sheaves

$$\eta \colon \mathcal{F} \longrightarrow \mathcal{G}.$$

By construction, $\eta_B = \eta_{\mathcal{B},B}$ as maps $\mathcal{F}(B) \to \mathcal{G}(B)$ for every $B \in \mathcal{B}$, so η extends $\eta_{\mathcal{B}}$.

Let us prove uniqueness. Suppose $\alpha \colon \mathcal{F} \to \mathcal{G}$ is another morphism extending $\eta_{\mathcal{B}}$. Fix an open $U \subset X$, a section $s \in \mathcal{F}(U)$, and a covering

$$U = \bigcup_{i \in I} B_i, \qquad B_i \in \mathcal{B}.$$

Then the section $\alpha_U(s) \in \mathcal{G}(U)$ satisfies

$$\alpha_U(s)|_{B_i} = \alpha_{B_i}(s|_{B_i}) = \eta_{B_i}(s|_{B_i}) = \eta_U(s)|_{B_i}$$

for every $i \in I$. Hence $\alpha_U(s) = \eta_U(s)$ by the locality axiom for $\mathcal{G}$. Thus $\alpha = \eta$, proving uniqueness.

We now prove the last assertions.

Suppose first that $\eta_B \colon \mathcal{F}(B) \to \mathcal{G}(B)$ is injective for every $B \in \mathcal{B}$. Let $U \subset X$ be open and let $s, t \in \mathcal{F}(U)$ satisfy

$$\eta_U(s) = \eta_U(t).$$

Choose a covering $U = \bigcup_i B_i$ by elements of $\mathcal{B}$. Then

$$\eta_{B_i}(s|_{B_i}) = \eta_U(s)|_{B_i} = \eta_U(t)|_{B_i} = \eta_{B_i}(t|_{B_i}).$$

Since η_{B_i} is injective, $s|_{B_i} = t|_{B_i}$ for every i, and therefore $s = t$. Thus η is injective.

Suppose now that every η_B is surjective. We claim that $\eta_x \colon \mathcal{F}_x \to \mathcal{G}_x$ is surjective for every $x \in X$. Let $t_x \in \mathcal{G}_x$ be a germ. Since the elements of $\mathcal{B}$ containing x form a local basis at x, we can represent t_x by a section $t \in \mathcal{G}(B)$ for some $B \in \mathcal{B}$ with $x \in B$. By surjectivity of η_B, there exists $s \in \mathcal{F}(B)$ such that $\eta_B(s) = t$. Passing to germs at x, we obtain

$$\eta_x(s_x) = t_x.$$

Hence η_x is surjective for every $x \in X$, so η is surjective.

Finally, if every η_B is an isomorphism, then η is both injective and surjective. Hence, by Lemma 1.4.0.9, we deduce that η is an isomorphism. □

COROLLARY 1.8.0.9. *The restriction functor of* (1.8.4),

$$\mathsf{res}_{\mathcal{B}}\colon \mathsf{Sh}(X,\mathscr{C}) \longrightarrow \mathsf{Sh}_{\mathcal{B}}(X,\mathscr{C}),$$

is an equivalence of categories.

Proof. Lemma 1.8.0.6 shows that $\mathsf{res}_{\mathcal{B}}$ is essentially surjective. On the other hand, Theorem 1.8.0.8 shows that, for every pair of sheaves $\mathcal{F},\mathcal{G}$, the map

$$\mathrm{Hom}_{\mathsf{Sh}(X,\mathscr{C})}(\mathcal{F},\mathcal{G}) \longrightarrow \mathrm{Hom}_{\mathsf{Sh}_{\mathcal{B}}(X,\mathscr{C})}\big(\mathsf{res}_{\mathcal{B}}(\mathcal{F}),\mathsf{res}_{\mathcal{B}}(\mathcal{G})\big)$$

is bijective. Thus $\mathsf{res}_{\mathcal{B}}$ is fully faithful and essentially surjective, hence an equivalence. □

1.9 Pushforward, inverse image

In this section we learn how to "move" sheaves from a topological space X to another topological space Y, in the presence of a continuous map between the two spaces.

$$\begin{array}{ccc} \mathcal{F} & \rightsquigarrow & f_*\mathcal{F} \\ \big| & & \big| \\ X & \xrightarrow{f} & Y \end{array} \qquad\qquad \begin{array}{ccc} f^{-1}\mathcal{G} & \leftsquigarrow & \mathcal{G} \\ \big| & & \big| \\ X & \xrightarrow{f} & Y \end{array}$$

1.9.1 Pushforward (or direct image)

Let $f\colon X\to Y$ be a continuous map of topological spaces, and let $\mathcal{F}$ be a presheaf on X. The assignment

$$V \longmapsto f_*\mathcal{F}(V)=\mathcal{F}(f^{-1}V)$$

defines a presheaf $f_*\mathcal{F}$ on Y, called the *pushforward* (or *direct image*) of $\mathcal{F}$ by f. It is a sheaf as soon as $\mathcal{F}$ is, because if $V=\bigcup_{i\in I}V_i$ is an open covering of an open subset $V\subset Y$, then $f^{-1}V=\bigcup_{i\in I}f^{-1}(V_i)$ is an open covering of $f^{-1}V\subset X$.

Example 1.9.1.1. If X is arbitrary and $Y=\mathsf{pt}$, then $f_*\mathcal{F}(\mathsf{pt})=\mathcal{F}(X)$, an object of $\mathscr{C}$. We will see in a minute that the direct image along any continuous map defines a functor. The direct image along the constant map $(X\to\mathsf{pt})_*\colon \mathsf{Sh}(X,\mathscr{C})\to\mathscr{C}$ is also called the *global section functor*. If $\mathscr{C}=\mathsf{Mod}_A$, it is a left exact functor (you proved a more general statement in Exercise 1.6.0.11).

Example 1.9.1.2. If $f\colon X\hookrightarrow Y$ is the inclusion of a subspace, then $f_*\mathcal{F}$ is defined, for any open subset $V\subset Y$, by

$$f_*\mathcal{F}(V)=\mathcal{F}(V\cap X).$$

Example 1.9.1.3 (Skyscraper sheaf as a pushforward)**.** Let $x\in X$ be a point, G a nontrivial abelian group. Consider the constant sheaf $\underline{G}_{\{x\}}$ on $\{x\}$. Let $i_x\colon\{x\}\hookrightarrow X$ be

the inclusion. Then the skyscraper sheaf $G_x \in \mathsf{Sh}(X, \mathsf{Mod}_{\mathbb{Z}})$ defined in Example 1.4.0.13 can be described as

$$G_x = i_{x,*}\underline{G}_{\{x\}}.$$

Next, we observe that pushforward of sheaves is functorial, i.e. sending $\mathcal{F} \mapsto f_*\mathcal{F}$ defines functors

$$\begin{array}{ccc} \mathsf{Sh}(X, \mathscr{C}) & \xrightarrow{f_*} & \mathsf{Sh}(Y, \mathscr{C}) \\ \downarrow & & \downarrow \\ \mathsf{pSh}(X, \mathscr{C}) & \xrightarrow{f_*} & \mathsf{pSh}(Y, \mathscr{C}) \end{array}$$

where the vertical maps are the natural inclusions (1.2.2). Indeed, given a morphism of (pre)sheaves $\eta\colon \mathcal{F} \to \mathcal{G}$, we can construct a morphism of (pre)sheaves

$$f_*\eta\colon f_*\mathcal{F} \longrightarrow f_*\mathcal{G}$$

simply by setting

$$(f_*\eta)_V = \eta_{f^{-1}V}\colon \mathcal{F}(f^{-1}V) \longrightarrow \mathcal{G}(f^{-1}V)$$

for an open subset $V \subset Y$. The compatibility with restriction maps follows from those of η (and the obvious observation that if $V' \subset V$ then $f^{-1}V' \subset f^{-1}V$).

Moreover, $(-)_*$ is compatible with compositions of continuous maps, in the following sense: if $f\colon X \to Y$ and $g\colon Y \to Z$ are continuous maps of topological spaces, then, as functors, we have an equality $(g \circ f)_* = g_* \circ f_*$ on the nose (both for presheaves and for sheaves). In other words, the diagram

$$\begin{array}{ccc} \mathsf{Sh}(X, \mathscr{C}) & \xrightarrow{f_*} & \mathsf{Sh}(Y, \mathscr{C}) \\ & \searrow^{(g\circ f)_*} & \downarrow{\scriptstyle g_*} \\ & & \mathsf{Sh}(Z, \mathscr{C}) \end{array} \tag{1.9.1}$$

commutes. Indeed, if $\mathcal{F}$ is a (pre)sheaf on X, then for every open $W \subset Z$ one has

$$\begin{aligned} (g \circ f)_*\mathcal{F}(W) &= \mathcal{F}((g \circ f)^{-1}(W)) \\ &= \mathcal{F}(f^{-1}g^{-1}(W)) \\ &= f_*\mathcal{F}(g^{-1}(W)) \\ &= (g_*f_*\mathcal{F})(W) \\ &= (g_* \circ f_*)\mathcal{F}(W). \end{aligned}$$

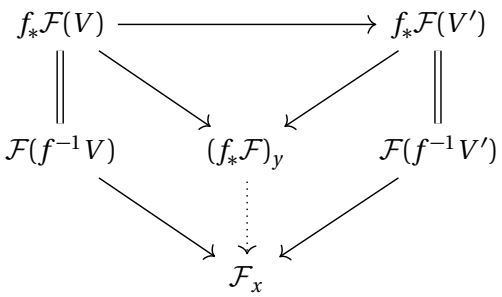
Note that no identifications are made here: all equalities are actual equalities!

LEMMA 1.9.1.4. *Let $f\colon X \to Y$ be a continuous map of topological spaces, and fix a sheaf $\mathcal{F} \in \mathsf{Sh}(X, \mathscr{C})$. Set $y = f(x)$. There is a canonical morphism*

$$(f_*\mathcal{F})_y \longrightarrow \mathcal{F}_x,$$

which is an isomorphism when f is the inclusion of a subspace $X \hookrightarrow Y$.

Proof. If $y \in V' \subset V \subset Y$, then $x \in f^{-1}V' \subset f^{-1}V \subset X$, and the commutative diagram



induces, via the universal property of the stalk

$$(f_*\mathcal{F})_y = \varinjlim_{V \ni y} \mathcal{F}(f^{-1}V),$$

a canonical morphism $(f_*\mathcal{F})_y \to \mathcal{F}_x$, as required.

Now, let us assume $f : X \hookrightarrow Y$ is the inclusion of a subspace, and let us take $y \in X$. Note that every neighbourhood $y \in U \subset X$ is of the form $U = V \cap X$ for some open neighbourhood $y \in V \subset Y$. Thus

$$(f_*\mathcal{F})_y = \varinjlim_{Y \supset V \ni y} \mathcal{F}(V \cap X) \xrightarrow{\sim} \varinjlim_{X \supset U \ni y} \mathcal{F}(U) = \mathcal{F}_y. \tag{1.9.2}$$

The proof is complete. □

Remark 1.9.1.5. We shall use Lemma 1.9.1.4 crucially with $\mathscr{C} = \mathrm{Rings}$, when defining morphisms of locally ringed spaces (cf. Definition 1.11.0.3 and the footnote there).



Caution 1.9.1.6. Even if $f : X \hookrightarrow Y$ is the inclusion of a subspace, it is not true that $(f_*\mathcal{F})_y = 0$ for all $y \in Y \setminus X$. This is nevertheless true when f is the inclusion of a *closed* subspace, cf. Remark 1.9.1.7.

Remark 1.9.1.7. If $f : X \hookrightarrow Y$ is the inclusion of a *closed* subspace, and $\mathcal{F}$ is a sheaf on X, then

$$(f_*\mathcal{F})_y = \begin{cases} \mathcal{F}_y & \text{if } y \in X \\ 0 & \text{if } y \notin X. \end{cases} \tag{1.9.3}$$

The case $y \in X$ is the computation (1.9.2). As for the case $y \notin X$, we use the definition

$$(f_*\mathcal{F})_y = \varinjlim_{Y \supset V \ni y} f_*\mathcal{F}(V) = \varinjlim_{Y \supset V \ni y} \mathcal{F}(V \cap X),$$

and the observation that, since X is closed in Y, there are arbitrarily small neighbourhoods V of y which are disjoint from X. For these, we have $\mathcal{F}(V \cap X) = \mathcal{F}(\emptyset) = 0$ since $\mathcal{F}$ is a sheaf (Remark 1.3.0.2). This causes the colimit to vanish.

Exactness of pushforward

We set $\mathscr{C} = \mathsf{Mod}_A$ in this subsection (for A a fixed ring), and we fix a continuous map $f: X \to Y$. We now say two (important) words on the exactness of the direct image functor

$$f_*\colon \mathsf{Sh}(X, \mathsf{Mod}_A) \longrightarrow \mathsf{Sh}(Y, \mathsf{Mod}_A).$$

It is important to remember that

f_* is always left exact, and it is exact if $f: X \hookrightarrow Y$ is a closed subspace.

Since f_* will turn out to be a right adjoint (Lemma 1.9.2.8), it is left exact by general category theory. However, we prove it directly here. Note that you have already proved the case $Y = \mathsf{pt}$ in Exercise 1.6.0.11. You will notice in the proof of the above slogan that this was essentially enough to handle the general case.

PROPOSITION 1.9.1.8. *Let A be a ring, $f: X \to Y$ a continuous map of topological spaces. The functor $f_*\colon \mathsf{Sh}(X, \mathsf{Mod}_A) \to \mathsf{Sh}(Y, \mathsf{Mod}_A)$ is left exact. If f is the inclusion of a closed subspace, then f_* is exact.*

Proof. Let us prove the first assertion. We need to show that an exact sequence

$$0 \longrightarrow \mathcal{F} \xrightarrow{\alpha} \mathcal{G} \xrightarrow{\beta} \mathcal{H}$$

in $\mathsf{Sh}(X, \mathsf{Mod}_A)$ induces an exact sequence

$$0 \longrightarrow f_*\mathcal{F} \xrightarrow{f_*\alpha} f_*\mathcal{G} \xrightarrow{f_*\beta} f_*\mathcal{H}$$

in $\mathsf{Sh}(Y, \mathsf{Mod}_A)$. We know by Exercise 1.6.0.11 that we have an exact sequence

$$0 \longrightarrow \mathcal{F}(f^{-1}V) \xrightarrow{\alpha_{f^{-1}V}} \mathcal{G}(f^{-1}V) \xrightarrow{\beta_{f^{-1}V}} \mathcal{H}(f^{-1}V) \tag{1.9.4}$$

for any open subset $V \subset Y$, by applying the functor $\Gamma(f^{-1}V, -)$ to the original sequence. In particular, $\alpha_{f^{-1}V} = (f_*\alpha)_V$ is injective for all V, which shows that $f_*\alpha$ is injective. There is an equality of presheaves

$$\mathrm{im}_{\mathrm{pre}}(f_*\alpha) = \ker(f_*\beta)$$

again thanks to exactness of (1.9.4) in the middle, ensuring precisely that $\mathrm{im}(\alpha_{f^{-1}V}) = \ker(\beta_{f^{-1}V})$. But $\ker(f_*\beta)$ is a sheaf, therefore we get exactness in the middle, i.e. $\mathrm{im}(f_*\alpha) = \ker(f_*\beta)$.

Let us show the second statement. Assume f is the inclusion of a closed subspace. By the first part of the proof, we only need to show that if $\eta: \mathcal{G} \twoheadrightarrow \mathcal{H}$ is surjective as a map of sheaves on X, then $f_*\mathcal{G} \twoheadrightarrow f_*\mathcal{H}$ is surjective as a map of sheaves on Y. If $y \in Y \setminus X$, then (using that X is closed, cf. Remark 1.9.1.7)

$$(f_*\mathcal{G})_y = 0 = (f_*\mathcal{H})_y. \tag{1.9.5}$$

Assume $y \in X$. Since $\mathcal{G}$ surjects onto $\mathcal{H}$, in the commutative diagram

$$\begin{array}{ccc} (f_*\mathcal{G})_y & \xrightarrow{(f_*\eta)_y} & (f_*\mathcal{H})_y \\ \| & & \| \\ \mathcal{G}_y & \xrightarrow{\eta_y} & \mathcal{H}_y \end{array}$$

the bottom map is surjective. The vertical equalities are given by Remark 1.9.1.7. Thus the top map is surjective as well. Hence $f_*\eta$ is surjective on all stalks, hence it is surjective. □

1.9.2 Inverse image

Let $f: X \to Y$ be a continuous map of topological spaces. Let $\mathcal{G}$ be a presheaf on Y. Given $U \subset X$, the collection of open subsets $V \subset Y$ containing $f(U)$ form a directed set via reverse inclusions. Sending

$$U \longmapsto (f_{\text{pre}}^{-1}\mathcal{G})(U) = \varinjlim_{V \supset f(U)} \mathcal{G}(V)$$

defines a presheaf on X. Indeed, assume $U' \subset U$ is an open subset. Then there is an inclusion $f(U') \subset f(U)$, inducing a map of directed systems

$$\{ V \in \tau_Y \mid V \supset f(U) \} \hookrightarrow \{ V \in \tau_Y \mid V \supset f(U') \},$$

which in turn induces a morphism

$$(f_{\text{pre}}^{-1}\mathcal{G})(U) = \varinjlim_{V \supset f(U)} \mathcal{G}(V) \longrightarrow \varinjlim_{V \supset f(U')} \mathcal{G}(V) = (f_{\text{pre}}^{-1}\mathcal{G})(U').$$

This is the restriction morphism for $f_{\text{pre}}^{-1}\mathcal{G}$.

Remark 1.9.2.1. If $f(U)$ is an open subset of Y, then $(f_{\text{pre}}^{-1}\mathcal{G})(U) = \mathcal{G}(f(U))$.

Now assume $\mathcal{G}$ is a sheaf. We define the *inverse image* of $\mathcal{G}$ by f to be the sheafification

$$f^{-1}\mathcal{G} = \left(f_{\text{pre}}^{-1}\mathcal{G}\right)^{\#}.$$

By Proposition 1.5.0.2, the canonical map $f_{\text{pre}}^{-1}\mathcal{G} \to f^{-1}\mathcal{G}$ of presheaves induces an isomorphism

$$(f_{\text{pre}}^{-1}\mathcal{G})_x \xrightarrow{\sim} (f^{-1}\mathcal{G})_x$$

on all the stalks.



Exercise 1.9.2.2. Both f_{pre}^{-1} and f^{-1} are functors.

Example 1.9.2.3. Let $\iota_y: \{y\} \hookrightarrow Y$ be the inclusion of a point $y \in Y$, and let $\mathcal{G}$ be a sheaf on Y. Then $\iota_y^{-1}\mathcal{G} = \mathcal{G}_y$, since $\iota_y^{-1}\mathcal{G}(\{y\}) = \varinjlim_{V \ni y} \mathcal{G}(V) = \mathcal{G}_y$. Thus ι_y^{-1} agrees with the stalk functor

$$\mathsf{stalk}_y: \mathsf{Sh}(Y, \mathscr{C}) \longrightarrow \mathscr{C}, \quad \mathcal{G} \mapsto \mathcal{G}_y.$$

Example 1.9.2.4. If $p: X \to \mathsf{pt}$ is the constant map, and $G \in \mathscr{C} \cong \mathsf{Sh}(\mathsf{pt}, \mathscr{C})$, then $p^{-1}G = \underline{G}_X$, the constant sheaf on X with values in the object G.

Example 1.9.2.5. Let $j: U \hookrightarrow Y$ be the inclusion of an open subset. Then $j_{\mathrm{pre}}^{-1}\mathcal{G} = \mathcal{G}|_U$ for any sheaf $\mathcal{G}$ on Y. The reason is that if U' is open in U, it is also open in Y, and thus

$$j_{\mathrm{pre}}^{-1}\mathcal{G}(U') = \varinjlim_{V \supset U'} \mathcal{G}(V) = \mathcal{G}(U') = \mathcal{G}|_U(U').$$

In particular, $j_{\mathrm{pre}}^{-1}\mathcal{G}$ is already a sheaf, and hence

$$\boxed{j^{-1}\mathcal{G} = \mathcal{G}|_U, \quad j: U \hookrightarrow Y \text{ open.}}$$

Remark 1.9.2.6. Despite Example 1.9.2.5, sheafifying f_{pre}^{-1} is in general necessary: consider a constant map $f: X = \{\star, \bullet\} \to \{*\} = Y$ from a discrete two point space, and fix a nontrivial abelian group G. The constant sheaf $\mathcal{G} = \underline{G}_Y$ has the property $f_{\mathrm{pre}}^{-1}\mathcal{G} = \underline{G}_X^{\mathrm{pre}}$, which is not a sheaf (cf. Example 1.2.0.19).

Functoriality (cf. Exercise 1.9.2.2) can be translated into a diagram of functors

$$\begin{array}{ccc} \mathsf{Sh}(Y, \mathscr{C}) & \xrightarrow{f^{-1}} & \mathsf{Sh}(X, \mathscr{C}) \\ {\scriptstyle j_{Y,\mathscr{C}}}\downarrow & & \downarrow\uparrow{\scriptstyle (-)^{\#}} \\ \mathsf{pSh}(Y, \mathscr{C}) & \xrightarrow{f_{\mathrm{pre}}^{-1}} & \mathsf{pSh}(X, \mathscr{C}) \end{array}$$

where $f^{-1} = (-)^{\#} \circ f_{\mathrm{pre}}^{-1} \circ j_{Y,\mathscr{C}}$.



Exercise 1.9.2.7. Let $f: X \to Y$ and $g: Y \to Z$ be continuous maps of topological spaces. Show that there is a canonical natural isomorphism[5]

$$f^{-1} \circ g^{-1} \xrightarrow{\sim} (g \circ f)^{-1}$$

of functors $\mathsf{Sh}(Z, \mathscr{C}) \to \mathsf{Sh}(X, \mathscr{C})$.

LEMMA 1.9.2.8 (Unit and counit maps). *For any pair of presheaves $\mathcal{F} \in \mathsf{pSh}(X, \mathscr{C})$ and $\mathcal{G} \in \mathsf{pSh}(Y, \mathscr{C})$ there are canonical presheaf homomorphisms*

$$\mathcal{G} \xrightarrow{\mathsf{unit}^{\mathcal{G}}} f_* f_{\mathrm{pre}}^{-1}\mathcal{G}, \qquad f_{\mathrm{pre}}^{-1} f_* \mathcal{F} \xrightarrow{\mathsf{counit}^{\mathcal{F}}} \mathcal{F}.$$

Proof. We start with the map $\mathsf{unit}^{\mathcal{G}}$. The observation here is that there is, for any open subset $V \subset Y$, a canonical inclusion $f(f^{-1}V) \hookrightarrow V$. Thus $\mathcal{G}(V)$ appears in the colimit

$$\varinjlim_{W \supset f(f^{-1}V)} \mathcal{G}(W).$$

[5] Note: unlike the pushforward, this is not an equality on the nose, just a natural isomorphism.

This induces a canonical morphism

$$\mathsf{unit}_V^{\mathcal{G}}\colon \mathcal{G}(V) \longrightarrow \varinjlim_{W\supset f(f^{-1}V)} \mathcal{G}(W) = f_{\mathrm{pre}}^{-1}\mathcal{G}(f^{-1}V) = f_*f_{\mathrm{pre}}^{-1}\mathcal{G}(V)$$

which does define a natural transformation $\mathcal{G}\to f_*f_{\mathrm{pre}}^{-1}\mathcal{G}$ because if $V'\subset V$, then any open $W\subset Y$ containing $f(f^{-1}V)$ also contains $f(f^{-1}V')$, simply because $f(f^{-1}V')\subset f(f^{-1}V)$. Thus there is a natural morphism

$$\varinjlim_{W\supset f(f^{-1}V)} \mathcal{G}(W) \longrightarrow \varinjlim_{W\supset f(f^{-1}V')} \mathcal{G}(W)$$

and the induced diagram

$$\begin{array}{ccccc}
\mathcal{G}(V) & \xrightarrow{\mathsf{unit}_V^{\mathcal{G}}} & \varinjlim_{W\supset f(f^{-1}V)} \mathcal{G}(W) & = & f_*f_{\mathrm{pre}}^{-1}\mathcal{G}(V) \\
\downarrow & & \downarrow & & \\
\mathcal{G}(V') & \xrightarrow{\mathsf{unit}_{V'}^{\mathcal{G}}} & \varinjlim_{W\supset f(f^{-1}V')} \mathcal{G}(W) & = & f_*f_{\mathrm{pre}}^{-1}\mathcal{G}(V')
\end{array}$$

commutes. This defines the map $\mathsf{unit}^{\mathcal{G}}\colon \mathcal{G}\to f_*f_{\mathrm{pre}}^{-1}\mathcal{G}$ of presheaves.

To construct the map $\mathsf{counit}^{\mathcal{F}}$, one observes that for any open subset $U\subset X$ there is a canonical map

$$f_{\mathrm{pre}}^{-1}f_*\mathcal{F}(U) = \varinjlim_{V\supset f(U)} f_*\mathcal{F}(V) = \varinjlim_{V\supset f(U)} \mathcal{F}(f^{-1}V) \longrightarrow \mathcal{F}(U),$$

since if $V\supset f(U)$ inside Y, then $U\subset f^{-1}f(U)\subset f^{-1}V$ inside X. This map is also functorial in $U'\subset U$, thus the map $\mathsf{counit}^{\mathcal{F}}\colon f_{\mathrm{pre}}^{-1}f_*\mathcal{F}\to\mathcal{F}$ is defined. □

The usefulness of the homomorphisms $\mathsf{unit}^{\mathcal{G}}$ and $\mathsf{counit}^{\mathcal{F}}$ is that they make $(f_{\mathrm{pre}}^{-1}, f_*)$ into an adjoint pair of functors. More precisely, there are bijections

$$\varphi_{\mathcal{F},\mathcal{G}}\colon \mathrm{Hom}_{\mathsf{pSh}(Y,\mathscr{C})}(\mathcal{G}, f_*\mathcal{F}) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{pSh}(X,\mathscr{C})}(f_{\mathrm{pre}}^{-1}\mathcal{G}, \mathcal{F}),$$

functorial in both $\mathcal{F}$ and $\mathcal{G}$. Specifically, $\varphi_{\mathcal{F},\mathcal{G}}$ sends $\eta\colon\mathcal{G}\to f_*\mathcal{F}$ to

$$f_{\mathrm{pre}}^{-1}\mathcal{G} \xrightarrow{f_{\mathrm{pre}}^{-1}\eta} f_{\mathrm{pre}}^{-1}f_*\mathcal{F} \xrightarrow{\mathsf{counit}^{\mathcal{F}}} \mathcal{F}$$

with inverse sending $\zeta\colon f_{\mathrm{pre}}^{-1}\mathcal{G}\to\mathcal{F}$ to

$$\mathcal{G} \xrightarrow{\mathsf{unit}^{\mathcal{G}}} f_*f_{\mathrm{pre}}^{-1}\mathcal{G} \xrightarrow{f_*\zeta} f_*\mathcal{F}.$$

Using the adjunction

$$\mathsf{pSh}(Y,\mathscr{C}) \underset{f_*}{\overset{f_{\mathrm{pre}}^{-1}}{\rightleftarrows}} \mathsf{pSh}(X,\mathscr{C}) \tag{1.9.6}$$

it is immediate to show that also

$$\mathsf{Sh}(Y,\mathscr{C}) \underset{f_*}{\overset{f^{-1}}{\rightleftarrows}} \mathsf{Sh}(X,\mathscr{C}) \tag{1.9.7}$$

is an adjoint pair of functors. Indeed, for any pair of sheaves $\mathcal{F} \in \mathsf{Sh}(X,\mathscr{C})$ and $\mathcal{G} \in \mathsf{Sh}(Y,\mathscr{C})$, we have

$$\begin{aligned}
\mathrm{Hom}_{\mathsf{Sh}(Y,\mathscr{C})}(\mathcal{G}, f_*\mathcal{F}) &= \mathrm{Hom}_{\mathsf{pSh}(Y,\mathscr{C})}(\mathcal{G}, f_*\mathcal{F}) && j_{Y,\mathscr{C}} \text{ is fully faithful} \\
&\xrightarrow{\sim} \mathrm{Hom}_{\mathsf{pSh}(X,\mathscr{C})}(f_{\mathrm{pre}}^{-1}\mathcal{G}, \mathcal{F}) && \text{adjunction (1.9.6)} \\
&\xrightarrow{\sim} \mathrm{Hom}_{\mathsf{Sh}(X,\mathscr{C})}(f^{-1}\mathcal{G}, \mathcal{F}) && \text{Exercise 1.5.0.6.}
\end{aligned}$$

Remark 1.9.2.9. Let us revisit Exercise 1.9.2.7. The canonical isomorphism claimed there is in fact easily obtained as a consequence of uniqueness of adjoints, applied to the observation that both $f^{-1} \circ g^{-1}$ and $(g \circ f)^{-1}$ serve as left adjoints to $(g \circ f)_* = g_* \circ f_*$.

Remark 1.9.2.10. Fix $\mathcal{G} \in \mathsf{Sh}(Y,\mathscr{C})$. Once more, the adjunction (1.9.7) gives a canonical morphism $\mathcal{G} \to f_* f^{-1}\mathcal{G}$, corresponding to $\mathrm{id}_{f^{-1}\mathcal{G}}$ under

$$\mathrm{Hom}_{\mathsf{Sh}(Y,\mathscr{C})}(\mathcal{G}, f_* f^{-1}\mathcal{G}) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{Sh}(X,\mathscr{C})}(f^{-1}\mathcal{G}, f^{-1}\mathcal{G}).$$

Sending $\mathcal{G} \mapsto f_* f^{-1}\mathcal{G}$ is a functor $f_* f^{-1} \colon \mathsf{Sh}(Y,\mathscr{C}) \to \mathsf{Sh}(Y,\mathscr{C})$, and the naturality of this operation yields a natural transformation

$$\mathsf{unit} \colon \mathrm{Id}_{\mathsf{Sh}(Y,\mathscr{C})} \Longrightarrow f_* f^{-1}$$

of functors $\mathsf{Sh}(Y,\mathscr{C}) \to \mathsf{Sh}(Y,\mathscr{C})$, which is called the *unit* of the adjunction (1.9.7). Similarly, if $\mathcal{F} \in \mathsf{Sh}(X,\mathscr{C})$, there is a canonical morphism $f^{-1} f_* \mathcal{F} \to \mathcal{F}$ corresponding to $\mathrm{id}_{f_*\mathcal{F}}$ under

$$\mathrm{Hom}_{\mathsf{Sh}(Y,\mathscr{C})}(f_*\mathcal{F}, f_*\mathcal{F}) \xrightarrow{\sim} \mathrm{Hom}_{\mathsf{Sh}(X,\mathscr{C})}(f^{-1} f_*\mathcal{F}, \mathcal{F}).$$

Sending $\mathcal{F} \mapsto f^{-1} f_* \mathcal{F}$ is a functor $f^{-1} f_* \colon \mathsf{Sh}(X,\mathscr{C}) \to \mathsf{Sh}(X,\mathscr{C})$, and the naturality of this operation yields a natural transformation

$$\mathsf{counit} \colon f^{-1} f_* \Longrightarrow \mathrm{Id}_{\mathsf{Sh}(X,\mathscr{C})}$$

of functors $\mathsf{Sh}(X,\mathscr{C}) \to \mathsf{Sh}(X,\mathscr{C})$, which is called the *counit* of the adjunction (1.9.7).

The next lemma says that the stalk of the inverse image is somewhat easy to compute (unlike for the pushforward).

LEMMA 1.9.2.11 (Stalk of inverse image). *Let $f \colon X \to Y$ be a continuous map of topological spaces, $\mathcal{G}$ a sheaf on Y, and $x \in X$ a point. There is a canonical isomorphism*

$$(f^{-1}\mathcal{G})_x \cong \mathcal{G}_{f(x)}.$$

Proof. We have

$$\begin{aligned}(f^{-1}\mathcal{G})_x &= (\{x\} \hookrightarrow X)^{-1}(f^{-1}\mathcal{G}) && \text{by Example 1.9.2.3}\\ &\cong (\{x\} \hookrightarrow X \to Y)^{-1}\mathcal{G} && \text{by Exercise 1.9.2.7}\\ &= (\{f(x)\} \hookrightarrow Y)^{-1}\mathcal{G}\\ &= \mathcal{G}_{f(x)}\end{aligned}$$

where we have used Example 1.9.2.3 once more for the last identity. □

PROPOSITION 1.9.2.12. *Let $\mathscr{C} = \mathsf{Mod}_A$, for a ring A. Then the inverse image functor*

$$f^{-1}\colon \mathsf{Sh}(Y, \mathsf{Mod}_A) \longrightarrow \mathsf{Sh}(X, \mathsf{Mod}_A)$$

is exact.

Proof. Indeed, let

$$0 \longrightarrow \mathcal{G} \longrightarrow \mathcal{H} \longrightarrow \mathcal{K} \longrightarrow 0$$

be an exact sequence in $\mathsf{Sh}(Y, \mathsf{Mod}_A)$. Then,

$$0 \longrightarrow \mathcal{G}_{f(x)} \longrightarrow \mathcal{H}_{f(x)} \longrightarrow \mathcal{K}_{f(x)} \longrightarrow 0$$

is exact in Mod_A by Proposition 1.6.0.9, for every $x \in X$. But this is, by Lemma 1.9.2.11, canonically isomorphic to the sequence

$$0 \longrightarrow (f^{-1}\mathcal{G})_x \longrightarrow (f^{-1}\mathcal{H})_x \longrightarrow (f^{-1}\mathcal{K})_x \longrightarrow 0.$$

Thus

$$0 \longrightarrow f^{-1}\mathcal{G} \longrightarrow f^{-1}\mathcal{H} \longrightarrow f^{-1}\mathcal{K} \longrightarrow 0$$

is exact, again by Proposition 1.6.0.9. □



Exercise 1.9.2.13. Show that if $f\colon X \hookrightarrow Y$ is the inclusion of a subspace, then the counit

$$f^{-1}f_*\mathcal{F} \xrightarrow{\sim} \mathcal{F}$$

is an isomorphism for every $\mathcal{F} \in \mathsf{Sh}(X, \mathscr{C})$. (**Hint**: check it on stalks).

PROPOSITION 1.9.2.14. *Let $f\colon X \hookrightarrow Y$ be the inclusion of a closed subspace. Let A be a ring.*

(1) *Let $\mathcal{G} \in \mathsf{Sh}(Y, \mathsf{Mod}_A)$ be a sheaf such that $\mathsf{Supp}(\mathcal{G}) \subset X$. Then the unit map*

$$\mathcal{G} \xrightarrow{\sim} f_*f^{-1}\mathcal{G}$$

is an isomorphism.

(2) *The functor f_* induces an equivalence of categories*

$$f_*\colon \mathsf{Sh}(X,\mathsf{Mod}_A) \xrightarrow{\sim} \mathsf{Sh}_X(Y,\mathsf{Mod}_A),$$

where $\mathsf{Sh}_X(Y,\mathsf{Mod}_A) \hookrightarrow \mathsf{Sh}(Y,\mathsf{Mod}_A)$ is the full subcategory of sheaves on Y with support contained in X.

Proof. To prove (1) it is enough to prove that the unit map is an isomorphism on all the stalks. If $y \in Y \setminus X$, we get $0 \xrightarrow{\sim} 0$, since $\mathcal{G}_y = 0$ by the assumption $\mathsf{Supp}(\mathcal{G}) \subset X$ and $(f_*f^{-1}\mathcal{G})_y = 0$ by (1.9.3). On the other hand, if $y \in X$, then $\mathcal{G}_y \to (f_*f^{-1}\mathcal{G})_y$ is nothing but the inverse of the isomorphism

$$(f_*f^{-1}\mathcal{G})_y \xrightarrow{\sim} (f^{-1}\mathcal{G})_y = \mathcal{G}_y$$

of Lemma 1.9.1.4.

To prove (2), observe first of all that f_* lands in the category $\mathsf{Sh}_X(Y,\mathsf{Mod}_A)$ by (1.9.3). Next, note that sending $\mathcal{G} \mapsto f^{-1}\mathcal{G}$ is an inverse to f_* by (1). In a little more detail, the equivalence (cf. Definition A.3.0.4) is set up by considering the pair of functors (f_*, f^{-1}) and exploiting the unit and counit *natural isomorphisms*

$$\mathsf{unit}\colon \mathrm{Id}_{\mathsf{Sh}_X(Y,\mathsf{Mod}_A)} \Longrightarrow f_*f^{-1} \qquad \mathsf{counit}\colon f^{-1}f_* \Longrightarrow \mathrm{Id}_{\mathsf{Sh}(X,\mathsf{Mod}_A)}$$

using (1) and Exercise 1.9.2.13. □



Exercise 1.9.2.15. Find examples of maps f and sheaves $\mathcal{G}$ such that $\mathcal{G} \to f_*f^{-1}\mathcal{G}$ is not an isomorphism. (**Hint**: Consider the inclusion of a suitable proper open subspace).

Remark 1.9.2.16. If $j\colon X \hookrightarrow Y$ is *open* and $\mathcal{G}$ is a sheaf on Y, then $j_*j^{-1}\mathcal{G}$ satisfies

$$j_*j^{-1}\mathcal{G}(V) = (j_*\mathcal{G}|_X)(V) = \mathcal{G}(V \cap X), \quad V \subset Y \text{ open.}$$

The natural map $\mathcal{G}(V) \to j_*j^{-1}\mathcal{G}(V)$ sends $s \mapsto s|_{V\cap X}$.

1.10 Gluing sheaves

The purpose of this section is to state the next theorem, which is of crucial importance.

THEOREM 1.10.0.1 ([7, Tag 00AK]). *Let X be a topological space, $X = \bigcup_{i\in I} U_i$ an open covering. Set $U_{ij} = U_i \cap U_j$ and similarly $U_{ijk} = U_{ij} \cap U_k$. Assume given a sheaf $\mathcal{F}_i$ on U_i for every $i \in I$, along with a collection of isomorphisms*

$$\varphi_{ij}\colon \mathcal{F}_i\big|_{U_{ij}} \xrightarrow{\sim} \mathcal{F}_j\big|_{U_{ij}}$$

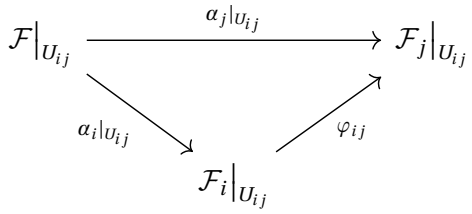
such that $\varphi_{ii} = \mathrm{id}_{\mathcal{F}_i}$ for every i, and such that

$$\begin{array}{ccc} \mathcal{F}_i\big|_{U_{ijk}} & \xrightarrow{\varphi_{ij}|_{U_{ijk}}} & \mathcal{F}_j\big|_{U_{ijk}} \\ & \searrow^{\varphi_{ik}|_{U_{ijk}}} & \downarrow{\scriptstyle \varphi_{jk}|_{U_{ijk}}} \\ & & \mathcal{F}_k\big|_{U_{ijk}} \end{array}$$

commutes for every triple intersection. Then there exists a sheaf $\mathcal{F}$ on X, unique up to unique isomorphism compatible with the isomorphisms

$$\alpha_i : \mathcal{F}\big|_{U_i} \xrightarrow{\sim} \mathcal{F}_i$$

such that the diagrams



commute for every $(i,j) \in I \times I$. The sheaf $\mathcal{F}$ is called the gluing of $(\mathcal{F}_i, \varphi_{ij})_{i,j}$ along the given covering.

1.11 Locally ringed spaces

Given our background on sheaves, we are ready for the definition of locally ringed space.

Definition 1.11.0.1 (Locally ringed space)**.** A *ringed space* is a pair $(X, \mathcal{O}_X)$ where X is a topological space and $\mathcal{O}_X$ is a sheaf of commutative rings on X. The sheaf $\mathcal{O}_X$ is called the *structure sheaf*. A *locally ringed space* is a ringed space such that the stalk $\mathcal{O}_{X,x}$ is a local ring for every $x \in X$.

Notation 1.11.0.2. Let $(X, \mathcal{O}_X)$ be a locally ringed space, $x \in X$ a point. We will write $\mathfrak{m}_x$ for the maximal ideal $\mathcal{O}_{X,x}$, and $\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$ for the corresponding *residue field*.

Recall that, given two local rings $(B, \mathfrak{m}_B)$ and $(A, \mathfrak{m}_A)$, a local homomorphism between them is a ring homomorphism $h : B \to A$ such that $h^{-1}(\mathfrak{m}_A) = \mathfrak{m}_B$, or, equivalently, $h(\mathfrak{m}_B) \subset \mathfrak{m}_A$.

Definition 1.11.0.3 (Morphism of locally ringed spaces)**.** A morphism of ringed spaces, denoted

$$(X, \mathcal{O}_X) \xrightarrow{(f, f^\#)} (Y, \mathcal{O}_Y), \tag{1.11.1}$$

is a pair $(f, f^\#)$ where $f : X \to Y$ is a continuous map between the underlying topological spaces and $f^\# : \mathcal{O}_Y \to f_* \mathcal{O}_X$ is a homomorphism of sheaves of rings on Y. If $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ are locally ringed spaces, a morphism between them is a morphism of ringed spaces such that the canonical map

$$f_x^\# : \mathcal{O}_{Y, f(x)} \longrightarrow (f_* \mathcal{O}_X)_{f(x)} \longrightarrow \mathcal{O}_{X,x}$$

is a local homomorphism of local rings for every $x \in X$.[6]

[6] The second map in this composition is the one provided in Lemma 1.9.1.4.

Example 1.11.0.4. Assume $X = \mathsf{pt}$. Then a (locally) ringed space $(\mathsf{pt}, \mathscr{O}_{\mathsf{pt}})$ is just the choice of a (local) ring A. A morphism of (locally) ringed spaces $(\mathsf{pt}, A) \to (\mathsf{pt}, B)$ is the same as a (local) ring homomorphism $B \to A$.

Notation 1.11.0.5. In what follows, when there is no confusion possible, we shall omit the sheaf of rings from the notation, and simply write X to denote the (locally) ringed space $(X, \mathscr{O}_X)$, or $f : X \to Y$ to denote a morphism $(f, f^\#)$ of (locally) ringed spaces as in (1.11.1). When we want to emphasise the underlying topological space of $(X, \mathscr{O}_X)$, we write $|X|$.

Remark 1.11.0.6. Note that any morphism of locally ringed spaces $X \to Y$ induces, at every x, a field extension

$$\kappa(y) \hookrightarrow \kappa(x), \quad y = f(x).$$

Example 1.11.0.7. Let $(X, \mathscr{O}_X)$ be a (locally) ringed space, $U \subset X$ an open subset. Then $(U, \mathscr{O}_X|_U)$ is a (locally) ringed space. We shall *always* take $\mathscr{O}_X|_U$ as the structure sheaf of an open subset $U \subset X$ of a locally ringed space X. We denote it by $\mathscr{O}_U$.

The composition of two morphisms of locally ringed spaces is defined in a straightforward way (but you need to know that pushforward commutes with composition, see (1.9.1)). In a little more detail, consider two morphisms

$$(X, \mathscr{O}_X) \xrightarrow{(f, f^\#)} (Y, \mathscr{O}_Y), \quad (Y, \mathscr{O}_Y) \xrightarrow{(g, g^\#)} (Z, \mathscr{O}_Z)$$

and define their composition to be the morphism

$$(X, \mathscr{O}_X) \xrightarrow{(g \circ f, (g \circ f)^\#)} (Z, \mathscr{O}_Z)$$

where the map on sheaves $(g \circ f)^\#$ is the composition

$$\mathscr{O}_Z \xrightarrow{g^\#} g_* \mathscr{O}_Y \xrightarrow{g_* f^\#} g_* f_* \mathscr{O}_X = (g \circ f)_* \mathscr{O}_X.$$

Locally ringed spaces thus form a (large) category, denoted

$$\mathsf{LRS},$$

where isomorphisms are simply the invertible morphisms (those that compose to the identity both ways).

Remark 1.11.0.8. A morphism of locally ringed spaces $(f, f^\#)$ as in (1.11.1) is an isomorphism if and only if

- $f : X \to Y$ is a topological homeomorphism, and
- $f^\# : \mathscr{O}_Y \to f_* \mathscr{O}_X$ is an isomorphism of sheaves.

Definition 1.11.0.9 (Immersions)**.** Let $(f, f^{\#})\colon (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a morphism of locally ringed spaces. It is called an *open immersion* (resp. a *closed immersion*) if $f\colon X \to Y$ is a topological open immersion (resp. closed immersion) and $f^{\#}_x$ is an isomorphism (resp. surjective) for every $x \in X$. It is called an *immersion* (or a *locally closed immersion*) if it factors as a closed immersion followed by an open immersion.

Notation 1.11.0.10. We sometimes will denote an immersion by '$X \hookrightarrow Y$'.

It is clear that a morphism of locally ringed spaces $(f, f^{\#})\colon (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is an open immersion if and only if there exists an open subset $V \subset Y$ such that $(f, f^{\#})$ induces an isomorphism $(X, \mathscr{O}_X) \xrightarrow{\sim} (V, \mathscr{O}_Y|_V)$. It is also clear that the composition of two open (resp. closed) immersions is an open immersion (resp. a closed immersion).

1.12 Ideal sheaves = Closed immersions up to isomorphism

Unlike open immersions, closed immersions are quite subtle. They are characterised by *ideal sheaves*, in a way that we now describe.

Definition 1.12.0.1 (Ideal sheaf)**.** Fix a locally ringed space $(X, \mathscr{O}_X)$. An *ideal sheaf* (or a *sheaf of ideals*) is a subsheaf $\mathscr{I} \subset \mathscr{O}_X$ (as abelian groups) such that $\mathscr{I}(U) \subset \mathscr{O}_X(U)$ is an ideal for every open subset $U \subset X$.

Given an ideal sheaf $\mathscr{I}$, the subset

$$\mathrm{V}(\mathscr{I}) = \left\{ x \in X \;\middle|\; \mathscr{I}_x \neq \mathscr{O}_{X,x} \right\} \xhookrightarrow{j} X \tag{1.12.1}$$

is a closed subset. Indeed, for any $x \in X \setminus \mathrm{V}(\mathscr{I})$, i.e. for any x such that $\mathscr{I}_x = \mathscr{O}_{X,x}$, there is a neighbourhood U of x and a section $f \in \mathscr{I}(U)$ such that $f_x = 1 \in \mathscr{O}_{X,x}$. But this means that $f|_V = 1 \in \mathscr{O}_X(V)$ for some open subset $V \subset U$. Thus $V \subset X \setminus \mathrm{V}(\mathscr{I})$, and thus $X \setminus \mathrm{V}(\mathscr{I})$ is open.

The quotient sheaf $\mathscr{O}_X/\mathscr{I}$ is a sheaf of rings (because, by definition, it is the sheafification of a presheaf of rings), not just abelian groups. The pair

$$(\mathrm{V}(\mathscr{I}), j^{-1}(\mathscr{O}_X/\mathscr{I}))$$

defines a locally ringed space (indeed, for any $x \in \mathrm{V}(\mathscr{I})$, the stalk $(j^{-1}(\mathscr{O}_X/\mathscr{I}))_x = (\mathscr{O}_X/\mathscr{I})_{j(x)} = \mathscr{O}_{X,j(x)}/\mathscr{I}_{j(x)}$ is a local ring: we have used Lemma 1.9.2.11 and Remark 1.6.0.5), and the canonical surjection

$$j^{\#}\colon \mathscr{O}_X \twoheadrightarrow \mathscr{O}_X/\mathscr{I} = j_* j^{-1}(\mathscr{O}_X/\mathscr{I}) \tag{1.12.2}$$

turns $(j, j^{\#})$ into a closed immersion

$$(\mathrm{V}(\mathscr{I}), j^{-1}(\mathscr{O}_X/\mathscr{I})) \xhookrightarrow{(j, j^{\#})} (X, \mathscr{O}_X). \tag{1.12.3}$$

Note that we have used Proposition 1.9.2.14(1) for the identification $\mathscr{O}_X/\mathscr{I} = j_* j^{-1}(\mathscr{O}_X/\mathscr{I})$ in (1.12.2). So we have defined an assignment

$$\mathscr{O}_X \supset \mathscr{I} \qquad \longmapsto \qquad \mathrm{V}(\mathscr{I}) \hookrightarrow X.$$

Conversely, to a closed immersion $\iota\colon Y \hookrightarrow X$ one can associate an ideal sheaf, namely

$$\mathscr{I}_Y = \ker(\mathscr{O}_X \xrightarrow{\iota^\#} \!\!\!\rightarrow\ \iota_*\mathscr{O}_Y) \subset \mathscr{O}_X.$$

These two operations are inverse to each other "up to isomorphism", as the next proposition clarifies.

PROPOSITION 1.12.0.2 (Closed immersions and ideal sheaves). *Let*

$$(\iota, \iota^\#)\colon (Y, \mathscr{O}_Y) \hookrightarrow (X, \mathscr{O}_X)$$

be a closed immersion of locally ringed spaces. Set $\mathscr{I}_Y = \ker \iota^\# \subset \mathscr{O}_X$ and consider the associated closed immersion

$$(\mathrm{V}(\mathscr{I}_Y), j^{-1}(\mathscr{O}_X/\mathscr{I}_Y)) \xhookrightarrow{(j, j^\#)} (X, \mathscr{O}_X).$$

as in (1.12.3)*. There is a unique isomorphism of locally ringed spaces*

$$(g, g^\#)\colon (Y, \mathscr{O}_Y) \xrightarrow{\sim} (\mathrm{V}(\mathscr{I}_Y), j^{-1}(\mathscr{O}_X/\mathscr{I}_Y))$$

such that $(\iota, \iota^\#) = (j, j^\#) \circ (g, g^\#)$.

Moreover, an inclusion of ideal sheaves $\mathscr{I}_2 \hookrightarrow \mathscr{I}_1 \hookrightarrow \mathscr{O}_X$ determines, and is determined by (in the above sense) a chain of closed immersions $\mathrm{V}(\mathscr{I}_1) \hookrightarrow \mathrm{V}(\mathscr{I}_2) \hookrightarrow X$.

Proof. The last statement is immediate. We thus only prove the first.

Let us use the shorthand notation

$$(Z, \mathscr{O}_Z) = (\mathrm{V}(\mathscr{I}_Y), j^{-1}(\mathscr{O}_X/\mathscr{I}_Y)).$$

where $j\colon Z \hookrightarrow X$ is the topological closed embedding that first appeared above in (1.12.1). We need to find the isomorphism g as in the statement. By Remark 1.9.1.7, we have

$$(\iota_*\mathscr{O}_Y)_x = \begin{cases} \mathscr{O}_{Y,y} & \text{if } x = \iota(y) \\ 0 & \text{if } x \notin \iota(Y). \end{cases} \tag{1.12.4}$$

Combine the exact sequence

$$0 \longrightarrow \mathscr{I}_Y \longrightarrow \mathscr{O}_X \longrightarrow \iota_*\mathscr{O}_Y \longrightarrow 0$$

with Proposition 1.6.0.9 and Equation (1.12.4) to deduce that

$$\mathscr{I}_{Y,x} = \mathscr{O}_{X,x} \qquad \Longleftrightarrow \qquad x \notin \iota(Y).$$

This shows $Z = \mathrm{V}(\mathscr{I}_Y) = \iota(Y)$. Let $g\colon Y \xrightarrow{\sim} Z$ be the induced homeomorphism. We have a factorisation

$$\iota\colon Y \xrightarrow{\ g\ } Z \xhookrightarrow{\ j\ } X$$

as topological maps, and

$$j_*\mathscr{O}_Z = j_* j^{-1}(\mathscr{O}_X/\mathscr{I}_Y) \cong \mathscr{O}_X/\mathscr{I}_Y \cong \iota_*\mathscr{O}_Y = j_* g_* \mathscr{O}_Y$$

as sheaves of rings, the last identity being a consequence of Diagram (1.9.1). Therefore, we have

$$\mathscr{O}_Z \cong j^{-1} j_* \mathscr{O}_Z = j^{-1} j_* g_* \mathscr{O}_Y \cong g_* \mathscr{O}_Y.$$

This extends g to the desired isomorphism $(g, g^\#)$. It is straightforward to verify the identity

$$(\iota, \iota^\#) = (j, j^\#) \circ (g, g^\#)$$

as morphisms of locally ringed spaces.

It remains to prove uniqueness. The underlying topological map g is forced by the factorisation $\iota = j \circ g$, since j is an inclusion (cf. Exercise 1.9.2.13). Once g is fixed, the identity

$$\iota^\# = j_* g^\# \circ j^\#$$

determines $j_* g^\#$ uniquely, because $j^\#$ is surjective. Finally, j_* is fully faithful for the inclusion of a subspace, since $j^{-1} j_* \cong \mathrm{id}$. Hence $g^\#$ is uniquely determined. □

Proposition 1.12.0.2 will be used to characterise closed subschemes of a scheme X (cf. **??**) later in **??**.

1.13 $\mathscr{O}_X$-modules and (quasi)coherent sheaves

Let $(X, \mathscr{O}_X)$ be a ringed space.

Definition 1.13.0.1 ($\mathscr{O}_X$-module)**.** An *$\mathscr{O}_X$-module* on the ringed space $(X, \mathscr{O}_X)$ is a sheaf of abelian groups $\mathcal{F}$ such that $\mathcal{F}(U)$ is an $\mathscr{O}_X(U)$-module for every open subset $U \subset X$, and such that for every inclusion $V \subset U$ of open subsets the diagram

$$\begin{array}{ccc} \mathscr{O}_X(U) \times \mathcal{F}(U) & \longrightarrow & \mathcal{F}(U) \\ \downarrow & & \downarrow \\ \mathscr{O}_X(V) \times \mathcal{F}(V) & \longrightarrow & \mathcal{F}(V) \end{array} \tag{1.13.1}$$

commutes. This means, explicitly, that $(a \cdot s)|_V = a|_V \cdot s|_V$ for every $a \in \mathscr{O}_X(U)$ and $s \in \mathcal{F}(U)$.

A morphism of $\mathscr{O}_X$-modules is a morphism of sheaves that preserves the linear structures. With this notion of morphisms, $\mathscr{O}_X$-modules form a category, denoted $\mathsf{Mod}(\mathscr{O}_X)$. We write $\mathrm{Hom}_{\mathscr{O}_X}(-,-)$ for the hom-sets in this category.

Example 1.13.0.2. Consider the following examples.

(1) An ideal sheaf $\mathscr{I} \subset \mathscr{O}_X$ is an $\mathscr{O}_X$-module, as well as the quotient $\mathscr{O}_X/\mathscr{I}$.

(2) The sheaf of 1-forms Ω^1_X on a smooth manifold $(X, \mathscr{C}^\infty_X)$ is a $\mathscr{C}^\infty_X$-module.

(3) The constant sheaf (cf. Example 1.5.0.3) $\mathcal{F} = \underline{G}_X$ is not an $\mathscr{O}_X$-module in general: for instance consider a smooth manifold $(X, \mathscr{C}^\infty_X)$ and set $G = \mathbb{Z}$. There is no natural action of $\mathscr{C}^\infty_X(U)$ on the set of continuous functions $U \to \mathbb{Z}$. The same issue occurs when $\mathcal{F} = \mathbb{Z}_x$ is a skyscraper sheaf (cf. Example 1.4.0.13) for a point $x \in X$. Again, say $x \in U \subset X$. How would one define an action of $\mathscr{C}^\infty_X(U)$ on $\mathbb{Z}$?

Example 1.13.0.3. Let $(X, \mathscr{O}_X)$ be a locally ringed space and $x \in X$. If $i_x \colon \{x\} \hookrightarrow X$ is the inclusion, then

$$i_{x*}\kappa(x)$$

is naturally an $\mathscr{O}_X$-module. Indeed, for every open U containing x, the action is induced by

$$\mathscr{O}_X(U) \longrightarrow \mathscr{O}_{X,x} \twoheadrightarrow \kappa(x).$$

More generally, every $\mathscr{O}_{X,x}$-module gives in the same way a skyscraper $\mathscr{O}_X$-module supported on $\overline{\{x\}}$.

Remark 1.13.0.4. Let $\mathcal{F}$ be an $\mathscr{O}_X$-module. There is a canonical identification

$$\mathcal{F}(X) \xrightarrow{\sim} \mathrm{Hom}_{\mathscr{O}_X}(\mathscr{O}_X, \mathcal{F}).$$

It sends $s \in \mathcal{F}(X)$ to the $\mathscr{O}_X$-linear map $\mathscr{O}_X \to \mathcal{F}$ defined on an open $U \subset X$ as $f \mapsto f \cdot s|_U$. Conversely, an $\mathscr{O}_X$-linear map $\eta \colon \mathscr{O}_X \to \mathcal{F}$ determines the section $s = \eta_X(1) \in \mathcal{F}(X)$.

It is easy to check that the category $\mathsf{Mod}(\mathscr{O}_X)$ has arbitrary direct sums (categorical coproducts), kernels and cokernels. One can prove (cf. [7, Tag 01AG]) that $\mathsf{Mod}(\mathscr{O}_X)$ is in fact an abelian category, in which a complex

$$\mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H}$$

is exact in the middle if and only if

$$\mathcal{F}_x \longrightarrow \mathcal{G}_x \longrightarrow \mathcal{H}_x$$

is exact in the middle for every $x \in X$. In particular, a complex of $\mathscr{O}_X$-modules is exact if and only if it is exact as a complex of sheaves of abelian groups.

In $\mathsf{Mod}(\mathscr{O}_X)$, the *tensor product* $\mathcal{F} \otimes_{\mathscr{O}_X} \mathcal{G}$ of two objects $\mathcal{F}$ and $\mathcal{G}$ is a new $\mathscr{O}_X$-module, defined to be the sheafification of the presheaf $U \mapsto \mathcal{F}(U) \otimes_{\mathscr{O}_X(U)} \mathcal{G}(U)$. It satisfies

$$\begin{aligned}(\mathcal{F} \otimes_{\mathscr{O}_X} \mathcal{G})_x &\cong \varinjlim_{U \ni x} \mathcal{F}(U) \otimes_{\mathscr{O}_X(U)} \mathcal{G}(U) \\ &\cong \varinjlim_{U \ni x} \mathcal{F}(U) \otimes_{\varinjlim_{U \ni x} \mathscr{O}_X(U)} \varinjlim_{U \ni x} \mathcal{G}(U) \\ &= \mathcal{F}_x \otimes_{\mathscr{O}_{X,x}} \mathcal{G}_x\end{aligned} \tag{1.13.2}$$

for every $x \in X$. The first isomorphism is justified since stalks can be computed before sheafification, the second isomorphism follows since filtered colimits commute with tensor products (even when the base ring varies compatibly).

Moreover, the functor $-\otimes_{\mathscr{O}_X} \mathcal{G}$ is right exact for all $\mathcal{G} \in \mathsf{Mod}(\mathscr{O}_X)$.

The restriction $\mathcal{F}|_U$ of an $\mathscr{O}_X$-module $\mathcal{F}$ to an open subset $U \subset X$ is naturally an $\mathscr{O}_U$-module, where we set $\mathscr{O}_U = \mathscr{O}_X|_U$. Given $\mathcal{F}, \mathcal{G} \in \mathsf{Mod}(\mathscr{O}_X)$, one also has a new $\mathscr{O}_X$-module

$$\mathscr{H}om_{\mathscr{O}_X}(\mathcal{F}, \mathcal{G})$$

called the *Hom sheaf*. Its sections over an open $U \subset X$ are given by

$$\mathscr{H}om_{\mathscr{O}_X}(\mathcal{F}, \mathcal{G})(U) = \mathrm{Hom}_{\mathscr{O}_U}(\mathcal{F}|_U, \mathcal{G}|_U).$$



Exercise 1.13.0.5. Show the following:

(i) $\mathscr{H}om_{\mathscr{O}_X}(\mathcal{F}, \mathcal{G})$ is a sheaf of $\mathscr{O}_X$-modules.

(ii) As functors $\mathsf{Mod}(\mathscr{O}_X) \to \mathsf{Mod}(\mathscr{O}_X)$, we have that $\mathscr{H}om_{\mathscr{O}_X}(\mathcal{F}, -)$ is (covariant and) left exact, and $\mathscr{H}om_{\mathscr{O}_X}(-, \mathcal{G})$ is (contravariant and) left exact. (This is a shadow of a more general fact about Homs in abelian categories).

(iii) Prove the next lemma.

LEMMA 1.13.0.6 (Adjunction $(\otimes, \mathscr{H}om)$). *Given $\mathcal{F}, \mathcal{G}, \mathcal{H} \in \mathsf{Mod}(\mathscr{O}_X)$, there is an isomorphism of $\mathscr{O}_X$-modules*

$$\mathscr{H}om_{\mathscr{O}_X}(\mathcal{F}, \mathscr{H}om_{\mathscr{O}_X}(\mathcal{G}, \mathcal{H})) \cong \mathscr{H}om_{\mathscr{O}_X}(\mathcal{F} \otimes_{\mathscr{O}_X} \mathcal{G}, \mathcal{H}).$$



Exercise 1.13.0.7. Show that both Lemma 1.8.0.6 and Theorem 1.8.0.8 on recovering sheaves (or morphisms thereof) from sheaves on a base extend to the setup of $\mathscr{O}_X$-modules.

Definition 1.13.0.8 (Free, locally free, globally generated)**.** Let I be a set. An $\mathscr{O}_X$-module $\mathcal{F}$ isomorphic to $\mathscr{O}_X^{\oplus I}$ is called *free*. If I is finite and $|I| = r$, we say that $\mathcal{F}$ is *free of rank r*. We say that $\mathcal{F}$ is *locally free of rank r* if for every $x \in X$ there is an open neighbourhood U such that $\mathcal{F}|_U \cong \mathscr{O}_U^{\oplus r}$. A locally free $\mathscr{O}_X$-module of rank 1 is called an *invertible sheaf* (or a *line bundle*). Finally, we say that $\mathcal{F}$ is *globally generated* if it receives a surjection $\mathscr{O}_X^{\oplus I} \twoheadrightarrow \mathcal{F}$ from a free sheaf, for some set I.

Definition 1.13.0.9 (Quasicoherent sheaf)**.** An $\mathscr{O}_X$-module $\mathcal{F}$ is *quasicoherent* if it is locally the cokernel of a morphism of free sheaves, i.e. if every point $x \in X$ has an open neighbourhood $U \subset X$ such that there is an exact sequence

$$\mathscr{O}_X^{\oplus I}\big|_U \longrightarrow \mathscr{O}_X^{\oplus J}\big|_U \longrightarrow \mathcal{F}\big|_U \longrightarrow 0$$

for some sets I and J. We denote by $\mathsf{QCoh}(X) \subset \mathsf{Mod}(\mathscr{O}_X)$ the full subcategory of quasicoherent $\mathscr{O}_X$-modules on the ringed space $(X, \mathscr{O}_X)$.

Definition 1.13.0.10 (Finitely presented sheaf)**.** An $\mathcal{O}_X$-module $\mathcal{F}$ is *finitely presented* if it is locally the cokernel of a morphism of free sheaves of finite rank, in the sense of Definition 1.13.0.9.

Definition 1.13.0.11 (Finitely generated, coherent)**.** Let $(X, \mathcal{O}_X)$ be a ringed space. An $\mathcal{O}_X$-module $\mathcal{F}$ is said to be

- *finitely generated* (or *of finite type*) if every $x \in X$ has an open neighbourhood $U \subset X$ admitting a surjection $\mathcal{O}_X^{\oplus n}|_U \twoheadrightarrow \mathcal{F}|_U$ for some $n \geqslant 1$,
- *coherent* if it is finitely generated and for every morphism $\theta\colon \mathcal{O}_X^{\oplus n}|_U \to \mathcal{F}|_U$ on an open $U \subset X$ one has that $\ker\theta$ is also finitely generated.

We denote by $\mathsf{Coh}(X) \subset \mathsf{Mod}(\mathcal{O}_X)$ the full subcategory of coherent sheaves on $(X, \mathcal{O}_X)$.

Example 1.13.0.12. The structure sheaf $\mathcal{O}_X$ is quasicoherent. More generally, any finitely presented sheaf is quasicoherent.

Example 1.13.0.13. Consider the following examples.

(i) Take $(X, \mathcal{O}_X) = (\mathrm{pt}, A)$ as ringed space, where $A = \mathbb{F}[x_i \mid i \geqslant 1]/(x_i x_j \mid i, j \geqslant 1)$. Then $\mathcal{O}_X = A$ is not coherent, as the kernel of the homomorphism $A \to A$ sending $a \mapsto a x_1$, namely the ideal $\mathsf{Ann}_A(x_1) = (\overline{x}_i \mid i \geqslant 1)$, is not finitely generated. In particular, notice that locally free of finite rank does not imply coherent over an arbitrary ringed space.

(ii) Let $\alpha\colon \mathcal{O}_X^{\oplus m} \to \mathcal{O}_X^{\oplus n}$ be a morphism. Then $\mathcal{F} = \mathrm{coker}(\alpha)$ is finitely presented, hence quasicoherent. If $\mathcal{O}_X$ is coherent, then so is $\mathcal{F}$.

(iii) Take $(X, \mathcal{O}_X) = (\mathrm{pt}, \mathbb{Z})$ as ringed space, and consider the $\mathcal{O}_X$-module $\mathbb{Q}$. Then $\mathbb{Q}$ is quasicoherent, but not coherent since it is not finitely generated over $\mathbb{Z}$. In fact, on a one-point ringed space, every module is quasicoherent. Coherence reduces to the usual algebraic notion of a coherent module.

(iv) Take $(X, \mathcal{O}_X) = (\mathbb{R}, \underline{\mathbb{Z}}_{\mathbb{R}})$ as ringed space and set $\mathcal{F} = \underline{\mathbb{Z}/2\mathbb{Z}}_X$. Then $\mathcal{F}$ has a global presentation

$$\underline{\mathbb{Z}}_{\mathbb{R}} \xrightarrow{\cdot 2} \underline{\mathbb{Z}}_{\mathbb{R}} \longrightarrow \mathcal{F} \longrightarrow 0,$$

and indeed $\mathcal{F}$ is coherent: locally, the kernels appearing in the definition of coherence correspond to kernels of homomorphisms between finitely generated $\mathbb{Z}$-modules, and these are finitely generated.

(v) Let $\mathbb{F}$ be a field, and consider the ringed space $(X, \mathcal{O}_X) = (\mathbb{R}, \underline{\mathbb{F}}_{\mathbb{R}})$, where $\underline{\mathbb{F}}_{\mathbb{R}}$ is the constant sheaf associated with $\mathbb{F}$. Let $\mathcal{F} = \mathbb{F}_0$ be the skyscraper sheaf supported at $0 \in \mathbb{R}$. There is a natural surjection of $\mathcal{O}_X$-modules

$$\underline{\mathbb{F}}_{\mathbb{R}} \twoheadrightarrow \mathcal{F},$$

which sends a locally constant function to its value at 0 (on open sets containing 0), and to zero otherwise. Thus $\mathcal{F}$ is of finite type.

We claim that $\mathcal{F}$ is not quasicoherent. Indeed, suppose that it were. Then there would exist an open interval U containing 0 and an exact sequence

$$\mathscr{O}_U^{\oplus I} \longrightarrow \mathscr{O}_U^{\oplus J} \longrightarrow \mathcal{F}|_U \longrightarrow 0.$$

Since U is connected, every morphism between free $\underline{\mathbb{F}}_{\mathbb{R}}|_U$-modules is induced by a homomorphism of $\mathbb{F}$-vector spaces

$$\mathbb{F}^{\oplus I} \longrightarrow \mathbb{F}^{\oplus J}.$$

Consequently, its cokernel is a constant sheaf on U. In particular, all stalks of $\mathcal{F}|_U$ must be isomorphic as $\mathbb{F}$-vector spaces. But

$$\mathcal{F}_x = \begin{cases} \mathbb{F} & \text{if } x = 0, \\ 0 & \text{if } x \neq 0, \end{cases}$$

a contradiction. Therefore $\mathcal{F}$ is of finite type but not quasicoherent.

(vi) Let $\mathbb{F}$ be a field, and consider again the ringed space $(X, \mathscr{O}_X) = (\mathbb{R}, \underline{\mathbb{F}}_{\mathbb{R}})$. Define

$$\mathcal{F} = \bigoplus_{n \geqslant 1} \mathscr{O}_X.$$

Then $\mathcal{F}$ is quasicoherent, since it is a free $\mathscr{O}_X$-module (of infinite rank).

However, $\mathcal{F}$ is not coherent. Indeed, for every $x \in X$, taking stalks gives

$$\mathcal{F}_x \cong \bigoplus_{n \geqslant 1} \mathscr{O}_{X,x} \cong \bigoplus_{n \geqslant 1} \mathbb{F},$$

which is not finitely generated as a module over $\mathscr{O}_{X,x} \cong \mathbb{F}$.

Since a coherent sheaf is necessarily of finite type, and hence has finitely generated stalks, we conclude that $\mathcal{F}$ is not coherent.

Recall the notion of support, introduced in (1.7.1). We have the following fundamental lemma on its closedness.

LEMMA 1.13.0.14. *Let $(X, \mathscr{O}_X)$ be a ringed space and let $\mathcal{F}$ be an $\mathscr{O}_X$-module of finite type. Then $\mathsf{Supp}(\mathcal{F})$ is closed in X.*

Proof. Let $x \notin \mathsf{Supp}(\mathcal{F})$, so $\mathcal{F}_x = 0$. Shrinking around x, choose generators $s_1, \ldots, s_n$ of $\mathcal{F}|_U$. Since $(s_i)_x = 0$ for every i, after shrinking U further we may assume $s_i = 0$ for all i. Hence $\mathcal{F}|_U = 0$, so $U \cap \mathsf{Supp}(\mathcal{F}) = \emptyset$. Thus the complement of $\mathsf{Supp}(\mathcal{F})$ is open. □

We will revisit (quasi)coherent sheaves and the categories $\mathsf{Mod}(\mathscr{O}_X)$, $\mathsf{QCoh}(X)$ and $\mathsf{Coh}(X)$ in greater detail in the context of schemes in **??**.

1.14 Pushforward and pullback of $\mathcal{O}$-modules

Let $f: X \to Y$ be a morphism of ringed spaces, and consider the situation

$$\begin{array}{ccc} \mathcal{F} & & \mathcal{G} \\ | & & | \\ X & \xrightarrow{f} & Y \end{array}$$

where $\mathcal{F} \in \mathsf{Mod}(\mathcal{O}_X)$ and $\mathcal{G} \in \mathsf{Mod}(\mathcal{O}_Y)$. If these were sheaves with no additional $\mathcal{O}$-module structure, we would already know, in such a situation, how to go "back and forth" using f_* and f^{-1}. Going forward is, even for $\mathcal{O}$-modules, simpler than going back. To go back, we will first confirm that $f^{-1}(-)$ preserves $\mathcal{O}$-linearity along the morphism $f^{-1}\mathcal{O}_Y \to \mathcal{O}_X$ adjoint to $\mathcal{O}_Y \to f_*\mathcal{O}_X$, and then observe that there is a "better" operation than f^{-1}, called the *pullback*, that is denoted f^*. It should be seen as a global version of *extension of scalars*, whereas pushforward in the context of $\mathcal{O}$-modules should be seen as a *restriction of scalars*. Let us briefly review what this means in the local case.

Consider a ring homomorphism $\phi: A \to B$. Then a B-module N is naturally an A-module via $a \cdot n = \phi(a)n$. This A-module structure, called restriction of scalars along ϕ, is simply denoted by N again, or sometimes $\phi_* N$. On the other hand, if M is an A-module, then $M \otimes_A B$ is a B-module, via the rule

$$b \cdot z = \big(M \otimes_A B \xrightarrow{\mathrm{id}_M \otimes t_b} M \otimes_A B\big)(z), \quad z \in M \otimes_A B,$$

where $t_b: B \to B$ is multiplication by $b \in B$. This structure is called *extension of scalars*. Of course, one can make the same construction replacing B by an arbitrary B-module N. Then using $\mathrm{id}_M \otimes t_b$, with $t_b: N \to N$ being multiplication $b \in B$, yields a natural B-module structure on $M \otimes_A N$.



Exercise 1.14.0.1. Let $\phi: A \to B$ be a ring homomorphism. Show the following assertions.

(i) Both $M \mapsto M \otimes_A B$ and $N \mapsto \phi_* N$ are functors.

(ii) There is an adjunction

$$\mathsf{Mod}_A \underset{\phi_*}{\overset{-\otimes_A B}{\rightleftarrows}} \mathsf{Mod}_B .$$

That is, there are bifunctorial bijections

$$\mathrm{Hom}_B(M \otimes_A B, N) \cong \mathrm{Hom}_A(M, \phi_* N).$$

In the 'global' case, consider a morphism $f: X \to Y$ of ringed spaces, and let $f^{\#}: \mathcal{O}_Y \to f_*\mathcal{O}_X$ be the map on sheaves of rings on Y. Let $V \subset Y$ be an open set, so that

$$A = \mathcal{O}_Y(V) \xrightarrow{\phi} f_*\mathcal{O}_X(V) = \mathcal{O}_X(f^{-1}(V)) = B$$

is a ring homomorphism. For the pushforward: assume $\mathcal{F}$ is an $\mathcal{O}_X$-module, so that by definition $f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}(V))$ is a module over $\mathcal{O}_X(f^{-1}(V)) = f_*\mathcal{O}_X(V)$. This says that $f_*\mathcal{F}$ is an $f_*\mathcal{O}_X$-module. But then, $f_*\mathcal{F}(V)$ is also an $\mathcal{O}_Y(V)$-module via restriction of scalars along the map $\phi = f_V^{\#}$, so

$$\boxed{f_*\mathcal{F} \text{ has a natural structure of } \mathcal{O}_Y\text{-module.}}$$

For the pullback: assume $\mathcal{G}$ is an $\mathcal{O}_Y$-module. At the presheaf level we have an action of

$$(f_{\text{pre}}^{-1}\mathcal{O}_Y)(U) = \varinjlim_{V \supset f(U)} \mathcal{O}_Y(V)$$

on

$$(f_{\text{pre}}^{-1}\mathcal{G})(U) = \varinjlim_{V \supset f(U)} \mathcal{G}(V)$$

defined as follows: if $a \in \mathcal{O}_Y(V)$, $s \in \mathcal{G}(W)$ and $U \subset X$ is an open such that $f(U) \subset V \cap W$, we define

$$[a] \cdot [s] = [a|_{V\cap W} \cdot s|_{V\cap W}].$$

This is well-defined and makes $f_{\text{pre}}^{-1}\mathcal{G}$ into a module over $f_{\text{pre}}^{-1}\mathcal{O}_Y$. After sheafification we obtain an $f^{-1}\mathcal{O}_Y$-module structure on $f^{-1}\mathcal{G}$. At the level of stalks, under the canonical identifications

$$(f^{-1}\mathcal{O}_Y)_x \cong \mathcal{O}_{Y,f(x)}, \qquad (f^{-1}\mathcal{G})_x \cong \mathcal{G}_{f(x)},$$

this is precisely the natural $\mathcal{O}_{Y,f(x)}$-module structure on $\mathcal{G}_{f(x)}$.

Now we need to extend scalars. This time we use the adjoint of $f^{\#}$, namely

$$f^{-1}\mathcal{O}_Y \longrightarrow \mathcal{O}_X,$$

allowing us to form the tensor product

$$\boxed{f^*\mathcal{G} = f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X.}$$

This is, by definition, the sheaf associated to the presheaf

$$U \longmapsto (f^{-1}\mathcal{G})(U) \otimes_{(f^{-1}\mathcal{O}_Y)(U)} \mathcal{O}_X(U),$$

which in fact is canonically isomorphic to the sheafification of

$$U \longmapsto (f_{\text{pre}}^{-1}\mathcal{G})(U) \otimes_{(f_{\text{pre}}^{-1}\mathcal{O}_Y)(U)} \mathcal{O}_X(U).$$

Indeed, the canonical comparison is an isomorphism on every stalk: both sides have stalk

$$\mathcal{G}_{f(x)} \otimes_{\mathcal{O}_{Y,f(x)}} \mathcal{O}_{X,x}.$$

Summarising, we have used the two maps

$$\mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X, \qquad f^{-1}\mathcal{O}_Y \longrightarrow \mathcal{O}_X$$

for the following constructions:

- Since $f_*\mathcal{F}$ is an $f_*\mathcal{O}_X$-module, it is naturally an $\mathcal{O}_Y$-module. This yields a *pushforward functor*
$$f_*\colon \mathsf{Mod}(\mathcal{O}_X) \longrightarrow \mathsf{Mod}(\mathcal{O}_Y).$$
- Since $f^{-1}\mathcal{G}$ is an $f^{-1}\mathcal{O}_Y$-module, we can construct the $\mathcal{O}_X$-module
$$f^*\mathcal{G} = f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X.$$
This yields a *pullback functor*
$$f^*\colon \mathsf{Mod}(\mathcal{O}_Y) \longrightarrow \mathsf{Mod}(\mathcal{O}_X).$$

Note that $f^*\mathcal{O}_Y \cong \mathcal{O}_X$ canonically, for every f. In fact, for every index set I one has, canonically, $f^*(\mathcal{O}_Y^{\oplus I}) \cong \mathcal{O}_X^{\oplus I}$ and the pullback of a locally free sheaf of rank r is again locally free of rank r.

Moreover, there is a canonical isomorphism
$$\boxed{(f^*\mathcal{G})_x \cong \mathcal{G}_{f(x)} \otimes_{\mathcal{O}_{Y,f(x)}} \mathcal{O}_{X,x}}$$
for every $x \in X$, combining (1.13.2) with Lemma 1.9.2.11.

Example 1.14.0.2. For an open immersion $j\colon U \hookrightarrow X$, one has
$$j^*\mathcal{F} = j^{-1}\mathcal{F} \otimes_{j^{-1}\mathcal{O}_X} \mathcal{O}_U \cong j^{-1}\mathcal{F} \otimes_{\mathcal{O}_U} \mathcal{O}_U \cong j^{-1}\mathcal{F} \cong \mathcal{F}|_U.$$
In other words, pullback generalises ordinary restriction.

Remark 1.14.0.3. By right exactness of tensor products, it is immediate to conclude that the pullback functor $f^*\colon \mathsf{Mod}(\mathcal{O}_Y) \to \mathsf{Mod}(\mathcal{O}_X)$ is right exact for every morphism $f\colon X \to Y$ of ringed spaces.

LEMMA 1.14.0.4 (Adjunction (f^*, f_*)). *Let $f\colon X \to Y$ be a morphism of ringed spaces. Then (f^*, f_*) is an adjoint pair of functors.*

Proof. Fix $\mathcal{G} \in \mathsf{Mod}(\mathcal{O}_Y)$ and $\mathcal{F} \in \mathsf{Mod}(\mathcal{O}_X)$. We have
$$\begin{aligned}\mathrm{Hom}_{\mathcal{O}_X}(f^*\mathcal{G}, \mathcal{F}) &= \mathrm{Hom}_{\mathcal{O}_X}(f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X, \mathcal{F}) \\ &\cong \mathrm{Hom}_{f^{-1}\mathcal{O}_Y}(f^{-1}\mathcal{G}, \mathcal{F}) \\ &\cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{G}, f_*\mathcal{F}).\end{aligned}$$
The first identity is just the definition; the first isomorphism is extension/restriction of scalars; the second is the adjunction (f^{-1}, f_*), restricted to linear maps. By naturality of these identifications, they are functorial in both $\mathcal{F}$ and $\mathcal{G}$, yielding the desired adjunction. □

A | Basic category theory

A.1 Categories and functors

Definition A.1.0.1 (Category)**.** A *category* $\mathscr{C}$ is the datum of

(i) a class $\mathrm{Ob}(\mathscr{C})$ of 'objects',

(ii) a class $\mathrm{Hom}(\mathscr{C})$ of 'morphisms' (or 'arrows', or 'maps') between the objects,

(iii) class functions $d\colon \mathrm{Hom}(\mathscr{C}) \to \mathrm{Ob}(\mathscr{C})$ and $t\colon \mathrm{Hom}(\mathscr{C}) \to \mathrm{Ob}(\mathscr{C})$ specifying domain and target of every morphism,

(iv) for each pair of objects x and y, a subclass

$$\mathrm{Hom}_{\mathscr{C}}(x,y) = \left\{ f \in \mathrm{Hom}(\mathscr{C}) \,\middle|\, d(f) = x,\ t(f) = y \right\} \subset \mathrm{Hom}(\mathscr{C})$$

of morphisms with domain x and target y, partitioning the whole class $\mathrm{Hom}(\mathscr{C})$,

(v) a binary operation

$$\begin{aligned} \mathrm{Hom}_{\mathscr{C}}(x,y) \times \mathrm{Hom}_{\mathscr{C}}(y,z) &\xrightarrow{\ \circ\ } \mathrm{Hom}_{\mathscr{C}}(x,z) \\ (f,g) &\longmapsto g \circ f \end{aligned}$$

called 'composition' of morphisms, for every triple of objects x, y and z.

Such data must fulfill the following axioms:

(CAT1) For every $x \in \mathrm{Ob}(\mathscr{C})$, there is an identity morphism $\mathrm{id}_x \in \mathrm{Hom}_{\mathscr{C}}(x,x)$ enjoying the properties

$$f \circ \mathrm{id}_x = f, \quad \mathrm{id}_y \circ g = g$$

for every morphism f with domain x, and for every morphism g with target y.

(CAT2) The associativity relation

$$(h \circ g) \circ f = h \circ (g \circ f)$$

holds for every triple (f,g,h) of composable morphisms.

Definition A.1.0.2 (Small and locally small)**.** A category $\mathscr{C}$ is *small* if both $\mathrm{Ob}(\mathscr{C})$ and $\mathrm{Hom}(\mathscr{C})$ are sets, and not proper classes. We say that $\mathscr{C}$ is *locally small* if $\mathrm{Hom}_{\mathscr{C}}(x,y)$ is a set, and not a proper class, for every pair of objects x and y. For a locally small category $\mathscr{C}$, the sets $\mathrm{Hom}_{\mathscr{C}}(x,y)$ are called *Hom-sets*.

Example A.1.0.3. The following are familiar examples of categories:

- Sets, the category of sets whose morphisms are the functions between sets,
- Grp, the category of groups with morphisms the group homomorphisms,
- Ab, the category of abelian groups with morphisms the group homomorphisms,
- Rings, the category of rings with morphisms the ring homomorphisms,
- Fields, the category of fields with morphisms the field homomorphisms,
- $\mathsf{Vec}_{\mathbb{F}}$, the category of vector spaces over a field $\mathbb{F}$ with morphisms the $\mathbb{F}$-linear maps,
- Alg_R, the category of algebras over a ring R, with morphisms the R-algebra homomorphisms,
- Top, the category of topological spaces, with morphisms the continuous maps,
- Mod_R, the category of modules over a ring R, with morphisms the R-linear maps,
- Mfd, the category of smooth manifolds, with morphisms the C^∞ maps.

Remark A.1.0.4. The category Sets is locally small, but not small. This is related to Russell's Paradox, which roughly says that there is no set of all sets; large categories are handled using proper classes or universes. In fact, for the same reasons as for Sets, all of the categories in Example A.1.0.3 are locally small, but not small.

From now on, the reader is encouraged to think that all categories considered are locally small. Often, this will actually be necessary.

The following notion will be important when discussing abelian categories in Appendix A.6.

Definition A.1.0.5 (Preadditive category)**.** Let $\mathscr{C}$ be a locally small category. We say that $\mathscr{C}$ is *preadditive* if all Hom-sets $\mathrm{Hom}_{\mathscr{C}}(x,y)$ are abelian groups, and composition of morphisms is $\mathbb{Z}$-bilinear.

Example A.1.0.6. Let R be a ring. The category $\mathscr{C}$ with one object, $\mathrm{Ob}(\mathscr{C})=\{\bullet\}$, and morphisms indexed by R, i.e. $\mathrm{Hom}(\mathscr{C})=\mathrm{Hom}_{\mathscr{C}}(\bullet,\bullet)=R$. Composition is multiplication in R and the addition in R realises $\mathrm{Hom}_{\mathscr{C}}(\bullet,\bullet)$ as the underlying abelian group of R.

The categories Ab, $\mathsf{Vec}_{\mathbb{F}}$ and more generally Mod_R are also preadditive.

The following are standard, important notions of morphisms in an arbitrary category.

Definition A.1.0.7 (Mono, epi, iso)**.** Let $\mathscr{C}$ be a category, and fix a morphism $f: x \to y$. We have the following notions.

(i) We say that f is a *monomorphism* if it is left-cancellable, i.e. for every pair of morphisms $g_1, g_2: z \to x$ one has that $f \circ g_1 = f \circ g_2$ implies $g_1 = g_2$.

(ii) We say that f is an *epimorphism* if it is right-cancellable, i.e. for every pair of morphisms $g_1, g_2: y \to z$ one has that $g_1 \circ f = g_2 \circ f$ implies $g_1 = g_2$.

(iii) We say that f is an *isomorphism* if it is an invertible morphism, i.e. if there exists a morphism $g: y \to x$ satisfying $f \circ g = \mathrm{id}_y$ and $g \circ f = \mathrm{id}_x$.

Two objects $x, y \in \mathrm{Ob}(\mathscr{C})$ are said to be *isomorphic* when there is an isomorphism $x \to y$ (often denoted '$x \xrightarrow{\sim} y$').

Remark A.1.0.8. Every isomorphism is in particular both an epimorphism and a monomorphism, but the converse need not hold in general. Consider for instance the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ in the category of rings: it is a monomorphism and an epimorphism, but not an isomorphism.

Definition A.1.0.9 (Functor)**.** Let $\mathscr{C}$ and $\mathscr{C}'$ be two categories. A functor from $\mathscr{C}$ to $\mathscr{C}'$, denoted $\mathsf{F}: \mathscr{C} \to \mathscr{C}'$, is the assignment of

- an object $\mathsf{F}(x) \in \mathrm{Ob}(\mathscr{C}')$ for every $x \in \mathrm{Ob}(\mathscr{C})$, and
- a morphism $\mathsf{F}(f) \in \mathrm{Hom}_{\mathscr{C}'}(\mathsf{F}(x), \mathsf{F}(y))$ for every morphism $f \in \mathrm{Hom}_{\mathscr{C}}(x, y)$,

subject to the following axioms:

(1) $\mathsf{F}(\mathrm{id}_x) = \mathrm{id}_{\mathsf{F}(x)}$ for every $x \in \mathrm{Ob}(\mathscr{C})$,

(2) $\mathsf{F}(g \circ f) = \mathsf{F}(g) \circ \mathsf{F}(f)$ for every pair (f, g) of composable arrows.

Remark A.1.0.10. By the axioms, functors preserve isomorphisms.

A functor as in Definition A.1.0.9 is said to be *covariant*. On the other hand, a *contravariant* functor $\mathsf{F}: \mathscr{C} \to \mathscr{C}'$ assigns a morphism $\mathsf{F}(f) \in \mathrm{Hom}_{\mathscr{C}'}(\mathsf{F}(y), \mathsf{F}(x))$ for every morphism $f \in \mathrm{Hom}_{\mathscr{C}}(x, y)$, and condition (2) becomes $\mathsf{F}(g \circ f) = \mathsf{F}(f) \circ \mathsf{F}(g)$. For instance, taking an $\mathbb{F}$-vector space V to its dual $V^* = \mathrm{Hom}_{\mathbb{F}}(V, \mathbb{F})$ is a contravariant functor from $\mathsf{Vec}_{\mathbb{F}}$ to itself.

Example A.1.0.11. Every category $\mathscr{C}$ admits an *identity functor* $\mathrm{Id}_{\mathscr{C}}: \mathscr{C} \to \mathscr{C}$, sending every object and every morphism to itself. This will be essential in Definition A.3.0.4.

Define $\mathscr{C}^{\mathrm{op}}$ the *opposite category* of $\mathscr{C}$ to be the category with objects $\mathrm{Ob}(\mathscr{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathscr{C})$ and with

$$\mathrm{Hom}_{\mathscr{C}^{\mathrm{op}}}(x,y) = \mathrm{Hom}_{\mathscr{C}}(y,x) \tag{A.1.1}$$

for every $x, y \in \mathrm{Ob}(\mathscr{C})$. Then a contravariant functor $\mathscr{C} \to \mathscr{C}'$ is the same as a covariant functor $\mathscr{C}^{\mathrm{op}} \to \mathscr{C}'$.

Notation A.1.0.12. From now on, we shall simply write $x \in \mathscr{C}$ to denote that x is an object of a category $\mathscr{C}$.

A.2 Final objects and universal properties

Definition A.2.0.1 (Initial, final, zero)**.** Let $\mathscr{C}$ be a category. An *initial object* is an object $\alpha \in \mathscr{C}$ such that for every object $x \in \mathscr{C}$ there is exactly one morphism $\alpha \to x$. A *final object* (also known as a *terminal object*) is an object $\omega \in \mathscr{C}$ such that for every object $x \in \mathscr{C}$ there is exactly one morphism $x \to \omega$. A *zero object* (also called a *null object*) is an object $0 \in \mathscr{C}$ that is simultaneously initial and final.

Remark A.2.0.2 (Dualisation of initial and final objects)**.** A final, resp. initial object in a category $\mathscr{C}$ is an initial, resp. final object in $\mathscr{C}^{\mathrm{op}}$.

Example A.2.0.3. The empty set $\emptyset$, resp. the singleton $\{*\}$, is the unique initial, resp. final, object in Sets. In categories such as Grp, Ab, Mod_R, the zero object is the trivial object (trivial group, trivial R-module). In Rings, the initial object is $\mathbb{Z}$, whereas the final object is the trivial ring (the one in which $1 = 0$). In Fields, there are no initial or final objects.

When an initial, final or zero object exists, it is unique up to a unique isomorphism. This is quite a useful fact, so we spend a few words on it. How do we use it? The point is that many important structures are defined by a universal property; one way to prove that an object defined by a universal property, when it exists, is unique up to unique isomorphism (compatible with the universal property), is to realise it as a final (or initial object) in some auxiliary category $\mathscr{U}$. To be able to apply this in the future, we need to convince ourselves that if a final object exists, it is unique up to a unique isomorphism. Say $\omega, \omega' \in \mathscr{C}$ are two final objects. There exists a unique morphism $f : \omega \to \omega'$ and a unique morphism $g : \omega' \to \omega$. For the same reason, $g \circ f : \omega \to \omega' \to \omega$ and $f \circ g : \omega' \to \omega \to \omega'$ are the identity, i.e. $\omega \cong \omega'$. But again, the isomorphism is necessarily unique!

Let us make an explicit example of the principle: if an object is defined by a universal property where the word "unique" is involved in a suitable way, then it will be unique up to a unique isomorphism. By 'involved in a suitable way' we mean: allowing us to see the universal object as a final object in an appropriate category $\mathscr{U}$. Let us consider *fibre*

products, since they are so important in algebraic geometry. These are special cases of limits (cf. Definition A.5.0.1 and Example A.5.0.4) but let us ignore this for now.

Definition A.2.0.4 (Fibre product)**.** The *fibre product* of a diagram

$$\begin{array}{ccc} & & x \\ & & \downarrow{\scriptstyle f} \\ y & \xrightarrow{g} & z \end{array}$$

in a category $\mathscr{C}$ is a triple $(x \times_z y, \mathrm{pr}_1, \mathrm{pr}_2)$, where

$$x \times_z y \xrightarrow{\mathrm{pr}_1} x, \qquad x \times_z y \xrightarrow{\mathrm{pr}_2} y$$

are morphisms in $\mathscr{C}$ such that $f \circ \mathrm{pr}_1 = g \circ \mathrm{pr}_2$, with the following universal property: for any other triple (a, p_1, p_2) where $p_1 : a \to x$ and $p_2 : a \to y$ are morphisms such that $f \circ p_1 = g \circ p_2$, there exists a **unique** morphism $\phi : a \to x \times_z y$ such that $p_i = \mathrm{pr}_i \circ \phi$ for $i = 1, 2$.

The relevant category here is the category $\mathscr{U}_{f,g}$ whose objects are

$$\mathrm{Ob}(\mathscr{U}_{f,g}) = \left\{ (a, p_1, p_2) \,\middle|\, f \circ p_1 = g \circ p_2 \right\},$$

and whose morphisms $(a, p_1, p_2) \to (a', p_1', p_2')$ are given by morphisms $\psi : a \to a'$ in $\mathscr{C}$ such that $p_i' \circ \psi = p_i$ for $i = 1, 2$. Thus we see that a fibre product of f and g exists if and only if $\mathscr{U}_{f,g}$ has a final object. By the reasoning above, it will be *automatically* unique up to unique isomorphism.

Definition A.2.0.5 (Pushout)**.** The *pushout* of a diagram

$$\begin{array}{ccc} z & \longrightarrow & x \\ \downarrow & & \\ y & & \end{array}$$

in a category $\mathscr{C}$ is the fibre product of the diagram

$$\begin{array}{ccc} & & x \\ & & \downarrow \\ y & \longrightarrow & z \end{array}$$

viewed in $\mathscr{C}^{\mathrm{op}}$ via (A.1.1).



Exercise A.2.0.6. Formulate the universal property of pushout in a category $\mathscr{C}$ and identify the auxiliary category $\mathscr{U}$ analogously to the case of fibre product. Prove that the pushout in category Alg_A of algebras over a ring A coincides with the tensor product over A. That is, given A-algebras

$$\begin{array}{ccc} A & \longrightarrow & R \\ \downarrow & & \\ S & & \end{array}$$

prove that their pushout is given by

$$\begin{array}{ccc} A & \longrightarrow & R \\ \downarrow & & \downarrow \\ S & \longrightarrow & R \otimes_A S \end{array}$$

where $R \to R \otimes_A S$ sends $r \mapsto r \otimes 1_S$ and similarly for $S \to R \otimes_A S$. If you have not seen tensor products, consult Appendix B.2 first.

A.3 Equivalences

Definition A.3.0.1 (Natural transformation)**.** Let $\mathscr{C}$, $\mathscr{C}'$ be two categories. A *natural transformation* $\eta\colon \mathsf{F} \Rightarrow \mathsf{G}$ between two functors $\mathsf{F}, \mathsf{G}\colon \mathscr{C} \to \mathscr{C}'$ is the datum, for every $x \in \mathscr{C}$, of a morphism $\eta_x\colon \mathsf{F}(x) \to \mathsf{G}(x)$ in $\mathscr{C}'$, such that for every $f \in \mathrm{Hom}_{\mathscr{C}}(x_1, x_2)$ the diagram

$$\begin{array}{ccc} \mathsf{F}(x_1) & \xrightarrow{\eta_{x_1}} & \mathsf{G}(x_1) \\ {\scriptstyle \mathsf{F}(f)}\downarrow & & \downarrow{\scriptstyle \mathsf{G}(f)} \\ \mathsf{F}(x_2) & \xrightarrow{\eta_{x_2}} & \mathsf{G}(x_2) \end{array}$$

is commutative in $\mathscr{C}'$.

Definition A.3.0.2 (Natural isomorphism)**.** Let $\mathscr{C}$, $\mathscr{C}'$ be two categories. Let $\mathsf{Fun}(\mathscr{C}, \mathscr{C}')$ be the category whose objects are functors $\mathscr{C} \to \mathscr{C}'$ and whose morphisms are the natural transformations. An isomorphism in the category $\mathsf{Fun}(\mathscr{C}, \mathscr{C}')$ is called a *natural isomorphism*.



Exercise A.3.0.3. Let $\mathbb{F}$ be a field, and $\mathsf{Vec}_{\mathbb{F}}$ the category of $\mathbb{F}$-vector spaces. Consider the identity functor $\mathrm{Id}_{\mathsf{Vec}_{\mathbb{F}}}$ and the double dual functor $(-)^{**}\colon \mathsf{Vec}_{\mathbb{F}} \to \mathsf{Vec}_{\mathbb{F}}$. Show that

(i) there is a natural transformation $\mathrm{Id}_{\mathsf{Vec}_{\mathbb{F}}} \Rightarrow (-)^{**}$, and in particular a natural morphism $V \to V^{**}$ for every $V \in \mathsf{Vec}_{\mathbb{F}}$, and

(ii) if $\mathsf{Vec}_{\mathbb{F}}^{\mathsf{f}}$ denotes the category of finite-dimensional $\mathbb{F}$-vector spaces, then the construction in (i) yields a natural isomorphism.

Definition A.3.0.4 (Equivalence of categories)**.** Let $\mathscr{C}$ and $\mathscr{C}'$ be categories. An *equivalence* between them is a pair of functors

$$\mathscr{C} \xrightarrow{\mathsf{F}} \mathscr{C}', \qquad \mathscr{C}' \xrightarrow{\mathsf{G}} \mathscr{C}$$

along with a pair of natural isomorphisms

$$\mathsf{F} \circ \mathsf{G} \xrightarrow{\sim} \mathrm{Id}_{\mathscr{C}'}, \qquad \mathsf{G} \circ \mathsf{F} \xrightarrow{\sim} \mathrm{Id}_{\mathscr{C}}.$$

Terminology A.3.0.5. One often says that a functor $\mathsf{F}\colon \mathscr{C} \to \mathscr{C}'$ is an equivalence when there exists a functor $\mathsf{G}\colon \mathscr{C}' \to \mathscr{C}$ along with a pair of natural isomorphisms as in Definition A.3.0.4.

Assumption A.3.0.6. From now on, all categories are tacitly assumed locally small.

Definition A.3.0.7 (Fully faithful, essentially surjective)**.** A (covariant) functor $\mathsf{F}\colon \mathscr{C} \to \mathscr{C}'$ between locally small categories is called:

(i) *full* (resp. *faithful*, resp. *fully faithful*) if for any two objects $x, y \in \mathscr{C}$ the map of sets

$$\mathrm{Hom}_{\mathscr{C}}(x,y) \longrightarrow \mathrm{Hom}_{\mathscr{C}'}(\mathsf{F}(x),\mathsf{F}(y))$$

is surjective (resp. injective, resp. bijective).

(ii) *essentially surjective* if every object of $\mathscr{C}'$ is isomorphic to an object of the form $\mathsf{F}(x)$ for some $x \in \mathscr{C}$.

Definition A.3.0.8 (Full subcategory)**.** Let $\mathscr{C}$ be a category. A *subcategory* $\mathscr{D} \subset \mathscr{C}$ is specified by the following data: a subclass of objects $\mathrm{Ob}(\mathscr{D}) \subset \mathrm{Ob}(\mathscr{C})$ and a subclass of morphisms $\mathrm{Hom}(\mathscr{D}) \subset \mathrm{Hom}(\mathscr{C})$ such that

- $\mathrm{id}_x \in \mathrm{Hom}(\mathscr{D})$ for every $x \in \mathrm{Ob}(\mathscr{D})$,
- for every $f \in \mathrm{Hom}(\mathscr{D})$ one has $d(f) \in \mathrm{Ob}(\mathscr{D})$ and $t(f) \in \mathrm{Ob}(\mathscr{D})$, and
- the $\mathscr{C}$-composition of every pair of (composable) morphisms in $\mathrm{Hom}(\mathscr{D})$ lies again in $\mathrm{Hom}(\mathscr{D})$.

We say that $\mathscr{D}$ is a *full subcategory* of $\mathscr{C}$ if $\mathrm{Hom}_{\mathscr{C}}(x,y) = \mathrm{Hom}_{\mathscr{D}}(x,y)$ for every pair of objects $x, y \in \mathrm{Ob}(\mathscr{D})$.

Note that the conditions ensure that $\mathscr{D}$ is itself a category (in particular $\mathrm{Hom}_{\mathscr{D}}(-,-)$ is well-defined in the definition of full subcategory), and moreover it comes with a faithful functor $j\colon \mathscr{D} \to \mathscr{C}$, injective on objects — therefore we will denote such functor by $\mathscr{D} \hookrightarrow \mathscr{C}$. Moreover, $\mathscr{D}$ is a full subcategory of $\mathscr{C}$ if and only if j is full.

For instance, $\mathsf{Ab} \hookrightarrow \mathsf{Grp}$ is a full subcategory.

The following property, left as an exercise, is quite useful in practice.



Exercise A.3.0.9. Prove that a fully faithful functor $\mathsf{F}\colon \mathscr{C} \to \mathscr{C}'$ induces an equivalence of $\mathscr{C}$ with the essential image of F, namely the full subcategory of $\mathscr{C}'$ consisting of objects isomorphic to objects of the form $\mathsf{F}(x)$ for some $x \in \mathscr{C}$. In particular, a functor $\mathsf{F}\colon \mathscr{C} \to \mathscr{C}'$ induces an equivalence of categories if and only if it is fully faithful and essentially surjective. (During the proof, you will have to construct a so-called 'quasi-inverse' — the functor back from the essential image of F to $\mathscr{C}$, and this requires choosing one preimage object for each isomorphism class in the target; luckily we assumed the axiom of choice from the very start!).

Definition A.3.0.10 (Concrete category)**.** A *concrete category* is a category $\mathscr{C}$ that is equipped with a faithful functor $\mathsf{F}\colon \mathscr{C} \to \mathsf{Sets}$ to the category of sets.

The existence of a faithful functor to $\mathscr{C} \to \mathsf{Sets}$ is a property of a category; specifying a particular such functor, thus realising concreteness, is additional structure.

A.4 Adjunctions

Another notion that is rather important in category theory is that of an adjoint pair of functors.

Definition A.4.0.1 (Adjoint pair)**.** Let $\mathscr{C}$ and $\mathscr{D}$ be (locally small) categories. Let $\mathsf{F}\colon \mathscr{C} \to \mathscr{D}$ and $\mathsf{G}\colon \mathscr{D} \to \mathscr{C}$ be functors. We say that (F,G) is an *adjoint pair* of functors if for every pair of objects $(c,d) \in \mathrm{Ob}(\mathscr{C}) \times \mathrm{Ob}(\mathscr{D})$ one has a bijection of sets

$$\mathrm{Hom}_{\mathscr{D}}(\mathsf{F}(c),d) \cong \mathrm{Hom}_{\mathscr{C}}(c,\mathsf{G}(d)),$$

natural in both c and d. We say, more precisely, that F is a *left adjoint* to G and that G is a *right adjoint* to F.

Sometimes, one uses the pictorial description

$$\mathscr{C} \underset{\mathsf{G}}{\overset{\mathsf{F}}{\rightleftarrows}} \mathscr{D}$$

to say that (F,G) is an adjoint pair.

Example A.4.0.2. Here are some examples of adjunctions.

(a) Let $\mathsf{F}\colon \mathsf{Sets} \to \mathsf{Grp}$ be the functor sending a set S to the free group generated by the elements of S. Let $\Phi\colon \mathsf{Grp} \to \mathsf{Sets}$ be the forgetful functor. Then (F,Φ) is an adjoint pair.

$$\mathsf{Sets} \underset{\Phi}{\overset{\mathsf{F}}{\rightleftarrows}} \mathsf{Grp}$$

(b) Let $\mathsf{j}\colon \mathsf{Ab} \hookrightarrow \mathsf{Grp}$ be the inclusion. It is right adjoint to the abelianisation functor $\mathsf{ab}\colon \mathsf{Grp} \to \mathsf{Ab}$ sending a group G to $G^{\mathrm{ab}} = G/[G,G]$. That is, (ab,j) is an adjoint pair.

$$\mathsf{Grp} \underset{\mathsf{j}}{\overset{\mathsf{ab}}{\rightleftarrows}} \mathsf{Ab}$$

(c) Let R be a ring. Consider the functor $\mathsf{Sym}_R\colon \mathsf{Mod}_R \to \mathsf{Alg}_R$ sending $M \mapsto \mathsf{Sym}_R(M)$. Consider the forgetful functor $\Phi_R\colon \mathsf{Alg}_R \to \mathsf{Mod}_R$ sending an R-algebra to its underlying R-module. Then (Sym_R,Φ_R) is an adjoint pair.

$$\mathsf{Mod}_R \underset{\Phi_R}{\overset{\mathsf{Sym}_R}{\rightleftarrows}} \mathsf{Alg}_R$$

(d) Let $\alpha\colon R \to S$ be a ring homomorphism. Then every S-module is naturally an R-module, thus we have a forgetful functor $\Phi_\alpha\colon \mathsf{Mod}_S \to \mathsf{Mod}_R$. On the other hand, we have a functor (called extension of scalars) $-\otimes_R S\colon \mathsf{Mod}_R \to \mathsf{Mod}_S$ sending an R-module M to the S-module $M \otimes_R S$. Then $(-\otimes_R S, \Phi_\alpha)$ is an adjoint pair.

$$\mathsf{Mod}_R \underset{\Phi_\alpha}{\overset{-\otimes_R S}{\rightleftarrows}} \mathsf{Mod}_S$$

(e) Let ID be the category of integral domains (with morphisms the injective ring homomorphisms), Fields the category of fields. We have a functor $\mathrm{Frac}\colon \mathsf{ID} \to \mathsf{Fields}$ sending a domain to its fraction field, and an inclusion functor $\mathsf{j}\colon \mathsf{Fields} \hookrightarrow \mathsf{ID}$. Then $(\mathrm{Frac}, \mathsf{j})$ is an adjoint pair.

$$\mathsf{ID} \underset{\mathsf{j}}{\overset{\mathrm{Frac}}{\rightleftarrows}} \mathsf{Fields}$$

(f) Let R be a ring, $M \in \mathsf{Mod}_R$ an R-module. Consider the endofunctors on the category Mod_R given by $-\otimes_R M$ and $\mathsf{h}^M = \mathrm{Hom}_{\mathsf{Mod}_R}(M,-)$. Then $(-\otimes_R M, \mathsf{h}^M)$ is an adjoint pair. The (natural) bijections

$$\mathrm{Hom}_{\mathsf{Mod}_R}(N \otimes_R M, P) \cong \mathrm{Hom}_{\mathsf{Mod}_R}(N, \mathrm{Hom}_{\mathsf{Mod}_R}(M,P))$$

induced by the adjunction

$$\mathsf{Mod}_R \underset{\mathsf{h}^M}{\overset{-\otimes_R M}{\rightleftarrows}} \mathsf{Mod}_R$$

are in fact isomorphisms of abelian groups (recall that Mod_R is preadditive, cf. Example A.1.0.6).

It is important to remember the following properties:

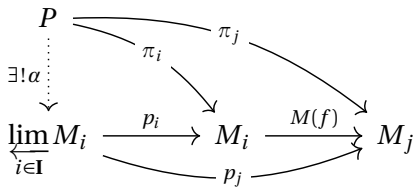
- every equivalence can be promoted to an adjoint equivalence,
- if a functor has two left (or right) adjoints, then they are naturally isomorphic.

A.5 Limits and colimits

Let $\mathscr{C}$ be a category, $\mathbf{I}$ a small category. Define an $\mathbf{I}$-diagram to be just a functor $M\colon \mathbf{I} \to \mathscr{C}$. Denote by $M_i = M(i)$ the object of $\mathscr{C}$ image of the object $i \in \mathbf{I}$ via M. If $f\colon i \to j$ is an arrow in $\mathbf{I}$, the induced arrow in $\mathscr{C}$ is denoted $M(f)\colon M_i \to M_j$.

Definition A.5.0.1 (Limit)**.** A *limit* of an $\mathbf{I}$-diagram $M\colon \mathbf{I} \to \mathscr{C}$ is an object $\varprojlim_{i\in\mathbf{I}} M_i$ of $\mathscr{C}$ along with an arrow $p_i\colon \varprojlim_{i\in\mathbf{I}} M_i \to M_i$ for every $i \in \mathbf{I}$, such that for every arrow $f\colon i \to j$ in $\mathbf{I}$ one has $p_j = M(f) \circ p_i$, and satisfying the following universal property: given an

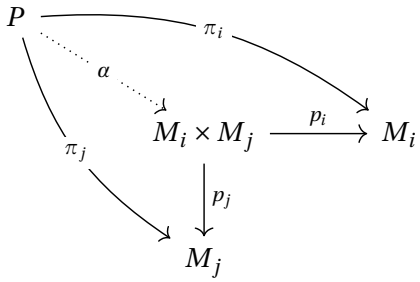
object P along with morphisms $\pi_i : P \to M_i$ such that $\pi_j = M(f) \circ \pi_i$ for every $f : i \to j$ in $\mathbf{I}$, there exists a unique arrow $\alpha : P \to \varprojlim_{i \in \mathbf{I}} M_i$ such that $\pi_i = p_i \circ \alpha$ for all $i \in \mathbf{I}$.





Exercise A.5.0.2. The limit over the empty diagram satisfies the universal property of a final object of $\mathscr{C}$.

Example A.5.0.3 (Products are limits)**.** Let $\mathbf{I}$ be the category with two objects i, j and no morphisms between them. Then an $\mathbf{I}$-diagram $M : \mathbf{I} \to \mathscr{C}$ is just the choice of two objects M_i, M_j of $\mathscr{C}$. The limit of M satisfies the universal property of the product $M_i \times M_j$.



Example A.5.0.4 (Fibre products are limits)**.** Let $\mathbf{I}$ be the category with three objects i, j, k and only two morphisms $i \to k$ and $j \to k$. Then an $\mathbf{I}$-diagram $M : \mathbf{I} \to \mathscr{C}$ is just the choice of three objects M_i, M_j and M_k of $\mathscr{C}$, along with a diagram

$$\begin{array}{ccc} & & M_i \\ & & \downarrow f \\ M_j & \xrightarrow{g} & M_k \end{array}$$

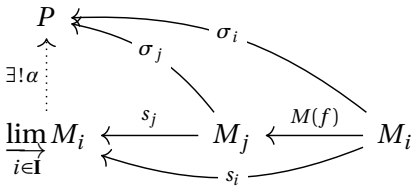
and it is clear that the limit of M satisfies the universal property of the product $(M_i \times_{M_k} M_j, \mathrm{pr}_1, \mathrm{pr}_2)$, see Definition A.2.0.4.

Example A.5.0.5 (Equalisers are limits)**.** Let $\mathbf{I}$ be the category with two objects i, j and two arrows $i \rightrightarrows j$. Then an $\mathbf{I}$-diagram $M : \mathbf{I} \to \mathscr{C}$ is just the choice of two parallel arrows $\phi, \psi : M_i \rightrightarrows M_j$ in $\mathscr{C}$. The limit of M satisfies the universal property of the equaliser of (ϕ, ψ).

Example A.5.0.6 (Kernels are limits)**.** This is because kernels are equalisers (in the previous example take $\psi = 0$ — for this to make sense of course you need to assume $\mathscr{C}$ to be preadditive).

Definition A.5.0.7 (Colimit)**.** A *colimit* of an $\mathbf{I}$-diagram $M : \mathbf{I} \to \mathscr{C}$ is an object $\varinjlim_{i \in \mathbf{I}} M_i$ of $\mathscr{C}$ along with an arrow $s_i : M_i \to \varinjlim_{i \in \mathbf{I}} M_i$ for every $i \in \mathbf{I}$, such that for every arrow

$f: i \to j$ in $\mathbf{I}$ one has $s_i = s_j \circ M(f)$, and satisfying the following universal property: given an object P along with morphisms $\sigma_i: M_i \to P$ such that $\sigma_i = \sigma_j \circ M(f)$ for every $f: i \to j$ in $\mathbf{I}$, there exists a unique arrow $\alpha: \varinjlim_{i\in\mathbf{I}} M_i \to P$ such that $\sigma_i = \alpha \circ s_i$ for all $i \in \mathbf{I}$.



Exercise A.5.0.8. The colimit over the empty diagram satisfies the universal property of an initial object of $\mathscr{C}$ (cf. Exercise A.5.0.2).

Exercise A.5.0.9. Convince yourself that coproducts, coequalisers, cokernels and pushouts are examples of colimits, along the same lines of Examples A.5.0.3, A.5.0.5, A.5.0.6 and A.5.0.4. (**Hint**: work with $\mathscr{C}^{\mathrm{op}}$ using Remark A.2.0.2).

Definition A.5.0.10 (Filtered category)**.** A nonempty category $\mathbf{I}$ is *filtered* if for every two objects $i, j \in \mathbf{I}$ the following are true:

- there exists an object $k \in \mathbf{I}$ along with two morphisms $i \to k$ and $j \to k$, and
- for any two morphisms $f, g \in \mathrm{Hom}_{\mathbf{I}}(i, j)$ there exists an object $k \in \mathbf{I}$ along with a morphism $h: j \to k$ such that $h \circ f = h \circ g$ in $\mathrm{Hom}_{\mathbf{I}}(i, k)$.

The colimit of an $\mathbf{I}$-diagram $M: \mathbf{I} \to \mathscr{C}$ where $\mathbf{I}$ is a filtered category is a *filtered colimit*.

Example A.5.0.11. In the definition of stalk of a presheaf $\mathcal{F} \in \mathsf{pSh}(X, \mathscr{C})$ at a point $x \in X$, we have been taking

$$\mathbf{I} = \{\, U \in \tau_X \mid x \in U \,\}^{\mathrm{op}}, \qquad M(U) = \mathcal{F}(U).$$

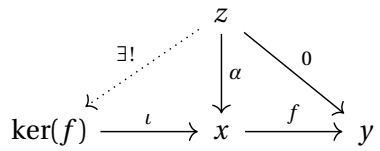
Exercise A.5.0.12. Prove that limits and colimits are unique up to unique isomorphisms, when they exist. (**Hint**: realise them as final objects in an appropriate category).

A.6 Abelian categories

Let $f: x \to y$ be an arrow in a preadditive category $\mathscr{C}$. The *kernel* of f is the final object $(\ker(f), \iota)$, if it exists, in the category[1] of pairs (z, α) where z is an object of $\mathscr{C}$ and $\alpha: z \to x$ is a morphism such that $f \circ \alpha = 0 \in \mathrm{Hom}_{\mathscr{C}}(z, y)$. The universal property of the kernel,

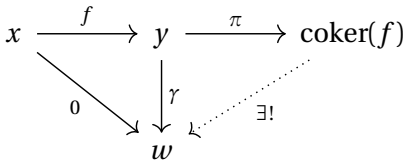
[1] As one might guess, a morphism $(z, \alpha) \to (z', \alpha')$ in this category is just a morphism $\phi: z \to z'$ such that $\alpha = \alpha' \circ \phi$.

translating the final object condition, can be represented by the diagram



where the two induced triangles commute.

The *cokernel* of f is the dual notion: it is the initial object $(\mathrm{coker}(f),\pi)$ in the category of pairs (w,γ) where w is an object of $\mathscr{C}$ and $\gamma\colon y\to w$ is a morphism such that $\gamma\circ f = 0\in\mathrm{Hom}_{\mathscr{C}}(x,w)$. The universal property of the cokernel, translating the initial object condition, can be represented by the diagram



where the two induced triangles commute.

Remark A.6.0.1. Let $f\colon x\to y$ be an arrow in a preadditive category $\mathscr{C}$. If $(\ker(f),\iota)$ is a kernel, then ι is a monomorphism, and if $(\mathrm{coker}(f),\pi)$ is a cokernel, then π is an epimorphism.

Terminology A.6.0.2. If $f\colon x\to y$ is a monomorphism in a preadditive category $\mathscr{C}$, resp. an epimorphism, we say that x is a *subobject* of y, resp. y is a *quotient object* of x.

Assume $\mathscr{C}$ has all kernels and cokernels, for simplicity. Define *image* and *coimage* of a morphism $f\colon x\to y$ to be

$$\mathrm{im}(f)=\ker(y\to\mathrm{coker}(f)),$$
$$\mathrm{coim}(f)=\mathrm{coker}(\ker(f)\to x).$$



Exercise A.6.0.3. Let $f\colon x\to y$ be an arrow in a preadditive category $\mathscr{C}$ admitting all kernels and cokernels. Show that f has a unique factorisation

$$x\longrightarrow\mathrm{coim}(f)\xrightarrow{\rho_f}\mathrm{im}(f)\longrightarrow y.$$

The following notion is extremely useful and of crucial importance in commutative algebra and algebraic geometry, as well as in the theory of sheaves.

Definition A.6.0.4 (Abelian category)**.** Let $\mathscr{C}$ be a preadditive category. We say that $\mathscr{C}$ is

- *additive*, if it admits all finite coproducts and a zero object $0\in\mathscr{C}$,
- *abelian*, if it is additive, has all kernels and cokernels and for every arrow $f\colon x\to y$ the natural map

  $$\mathrm{coim}(f)\xrightarrow{\rho_f}\mathrm{im}(f)$$

  is an isomorphism.

In an abelian category one can define a sequence

$$x \xrightarrow{f} y \xrightarrow{g} z$$

to be a complex if $g \circ f = 0$, and a complex to be *exact* (or exact *in the middle*) if the canonical map $\mathrm{im}(f) \to \ker(g)$ is an isomorphism (equivalently, if $\mathrm{im}(f)$ and $\ker(g)$ define the same subobject of y). This allows us to define $\mathbb{Z}$-indexed complexes

$$\cdots \longrightarrow x_{i-1} \xrightarrow{f_{i-1}} x_i \xrightarrow{f_i} x_{i+1} \xrightarrow{f_{i+1}} x_{i+2} \longrightarrow \cdots$$

of arbitrary length and their exactness (at each $i \in \mathbb{Z}$). A special case is given by *short exact sequences*, namely exact sequences of the form

$$0 \longrightarrow x \xrightarrow{f} y \xrightarrow{g} z \longrightarrow 0.$$

Note that, in such a sequence, $(z, g) = \mathrm{coker}(f)$ and $(x, f) = \ker(g)$.

Note also that every morphism $f : x \to y$ in an abelian category defines an exact sequence

$$0 \longrightarrow \ker(f) \longrightarrow x \xrightarrow{f} y \longrightarrow \mathrm{coker}(f) \longrightarrow 0,$$



Exercise A.6.0.5. Let $\mathscr{C}$ be an abelian category. Declare a morphism $f : x \to y$ in $\mathscr{C}$ to be injective, resp. surjective, if $\ker(f) = 0$, resp. $\mathrm{coker}(f) = 0$. Show that

(i) a morphism $f : x \to y$ is injective if and only if it is a monomorphism,

(ii) a morphism $f : x \to y$ is surjective if and only if it is an epimorphism,

(iii) a morphism $f : x \to y$ is an isomorphism if and only if it is both injective and surjective.



Exercise A.6.0.6. Let $\mathscr{C}$ be an abelian category. Prove that finite products and coproducts are the same, and deduce that $\mathscr{C}$ has all finite limits and finite colimits.

The following result is of crucial importance in the theory of abelian categories.

LEMMA A.6.0.7 (Snake lemma [7, Tag 010H]). *Given a commutative diagram*

$$\begin{array}{ccccccc} x & \xrightarrow{f} & y & \longrightarrow & z & \longrightarrow & 0 \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \\ 0 & \longrightarrow & u & \longrightarrow & v & \xrightarrow{h} & w \end{array}$$

with exact rows, there is an induced exact sequence

$$\ker(\alpha) \xrightarrow{f'} \ker(\beta) \longrightarrow \ker(\gamma) \longrightarrow \mathrm{coker}(\alpha) \longrightarrow \mathrm{coker}(\beta) \xrightarrow{h'} \mathrm{coker}(\gamma).$$

Furthermore, f' is injective if f is, and h' is surjective if h is. This property is known as the snake lemma.

Using the snake lemma (Lemma A.6.0.7), one may prove the following important result.

LEMMA A.6.0.8 (Four Lemma [7, Tag 05QA]). *Let $\mathscr{C}$ be an abelian category. Given a commutative diagram*

$$\begin{array}{ccccccc} w & \longrightarrow & x & \longrightarrow & y & \longrightarrow & z \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle \delta} \\ w' & \longrightarrow & x' & \longrightarrow & y' & \longrightarrow & z' \end{array}$$

with exact rows, one has:

(i) if α, γ are surjective and δ is injective, then β is surjective.

(ii) if β, δ are injective and α is surjective, then γ is injective.

The Four Lemma, in turn, implies the following.

LEMMA A.6.0.9 (Five Lemma). *Given a commutative diagram*

$$\begin{array}{ccccccccc} v & \longrightarrow & w & \longrightarrow & x & \longrightarrow & y & \longrightarrow & z \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \downarrow{\scriptstyle \delta} & & \downarrow{\scriptstyle \epsilon} \\ v' & \longrightarrow & w' & \longrightarrow & x' & \longrightarrow & y' & \longrightarrow & z' \end{array}$$

with exact rows, one has that if β, δ are isomorphisms, ϵ is injective, and α is surjective then γ is an isomorphism.

Definition A.6.0.10 (Additive functor)**.** Let $\mathsf{F}\colon \mathscr{C} \to \mathscr{C}'$ be a covariant functor between abelian categories. We say that F is *additive* if the natural map

$$\mathrm{Hom}_{\mathscr{C}}(x, y) \longrightarrow \mathrm{Hom}_{\mathscr{C}'}(\mathsf{F}(x), \mathsf{F}(y))$$

is a group homomorphism for every $x, y \in \mathscr{C}$.

Definition A.6.0.11 (Exactness)**.** Let $\mathsf{F}\colon \mathscr{C} \to \mathscr{C}'$ be an additive functor between abelian categories. We say that F is *left exact*, resp. *right exact*, if a short exact sequence

$$0 \longrightarrow x \longrightarrow y \longrightarrow z \longrightarrow 0$$

gets sent to an exact sequence

$$0 \longrightarrow \mathsf{F}(x) \longrightarrow \mathsf{F}(y) \longrightarrow \mathsf{F}(z),$$

resp. to an exact sequence

$$\mathsf{F}(x) \longrightarrow \mathsf{F}(y) \longrightarrow \mathsf{F}(z) \longrightarrow 0.$$

We say that F is *exact* if it is both left and right exact.



Exercise A.6.0.12 (Contravariant exactness). Formulate the notions of exactness introduced in Definition A.6.0.11 for a contravariant functor. Confirm that a covariant additive functor is left exact, resp. right exact, if and only if it turns kernels into kernels, resp. cokernels into cokernels. What about an exact contravariant functor?



Exercise A.6.0.13 (Exactness of adjoints). Let

$$\mathscr{C} \underset{\mathsf{G}}{\overset{\mathsf{F}}{\rightleftarrows}} \mathscr{D}$$

be an adjoint pair of additive functors between abelian categories, i.e. F is left adjoint to G. Prove that F (the left adjoint) preserves all colimits that exist and G (the right adjoint) preserves all limits that exist. Deduce that every right adjoint (resp. left adjoint) additive functor between two abelian categories is left exact (resp. right exact).

A.7 Yoneda Lemma

In this section we study representable functors and recall the statement of the Yoneda Lemma. This all connects again to our discussion on universal properties and final objects, cf. Appendix A.2. More details and examples can be found, for instance, in [9].

All categories are assumed to be *locally small* throughout. We ignore size issues, or equivalently work inside fixed Grothendieck universes.

Let $\mathscr{C}$ be a category. Consider the category of contravariant functors $\mathscr{C} \to \mathsf{Sets}$, i.e. the *functor category* (cf. Definition A.3.0.2)

$$\mathsf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathsf{Sets}).$$

We denote its Hom-sets, consisting of natural transformations, by

$$\mathrm{Nat}(\mathsf{F}, \mathsf{G}) = \mathrm{Hom}_{\mathsf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathsf{Sets})}(\mathsf{F}, \mathsf{G}).$$

For every object x of $\mathscr{C}$ there is a functor $\mathsf{h}_x \colon \mathscr{C}^{\mathrm{op}} \to \mathsf{Sets}$ defined by

$$u \longmapsto \mathsf{h}_x(u) = \mathrm{Hom}_{\mathscr{C}}(u, x), \quad u \in \mathscr{C}.$$

A morphism $\phi \in \mathrm{Hom}_{\mathscr{C}^{\mathrm{op}}}(u, v) = \mathrm{Hom}_{\mathscr{C}}(v, u)$ gets sent to the map of sets

$$\mathsf{h}_x(\phi) \colon \mathsf{h}_x(u) \longrightarrow \mathsf{h}_x(v), \quad \alpha \mapsto \alpha \circ \phi.$$

Consider the functor

$$\mathsf{h}_{\mathscr{C}} \colon \mathscr{C} \longrightarrow \mathsf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathsf{Sets}), \quad x \mapsto \mathsf{h}_x. \tag{A.7.1}$$

This is, indeed, a functor: for every arrow $f \colon x \to y$ in $\mathscr{C}$ and object u of $\mathscr{C}$ we can define a map of sets

$$\mathsf{h}_f u \colon \mathsf{h}_x(u) \longrightarrow \mathsf{h}_y(u), \quad \alpha \mapsto f \circ \alpha,$$

with the property that for every morphism $\phi\colon v \to u$ in $\mathscr{C}$ there is a commutative diagram

$$\begin{array}{ccc} \mathsf{h}_x(u) & \xrightarrow{\mathsf{h}_f u} & \mathsf{h}_y(u) \\ {\scriptstyle \mathsf{h}_x(\phi)}\downarrow & & \downarrow{\scriptstyle \mathsf{h}_y(\phi)} \\ \mathsf{h}_x(v) & \xrightarrow{\mathsf{h}_f v} & \mathsf{h}_y(v) \end{array} \qquad \begin{array}{ccc} u \xrightarrow{\alpha} x & \longmapsto & u \xrightarrow{f\circ\alpha} y \\ \downarrow & & \downarrow \\ v \xrightarrow{\alpha\circ\phi} x & \longmapsto & v \xrightarrow{f\circ\alpha\circ\phi} y \end{array}$$

defining a natural transformation

$$\mathsf{h}_f\colon \mathsf{h}_x \Longrightarrow \mathsf{h}_y.$$

LEMMA A.7.0.1 (Strong Yoneda). *Let* $\mathsf{F} \in \mathsf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathsf{Sets})$ *be a functor,* $x \in \mathscr{C}$ *an object. Then, there is a bijection*

$$\mathrm{Nat}(\mathsf{h}_x, \mathsf{F}) \xrightarrow{\sim} \mathsf{F}(x),$$

natural both in the object $x \in \mathscr{C}$ *and in the functor* F*.*

Proof. One can construct a map of sets

$$\mathrm{Nat}(\mathsf{h}_x, \mathsf{F}) \xrightarrow{g_x} \mathsf{F}(x)$$

by sending a natural transformation $\eta\colon \mathsf{h}_x \Rightarrow \mathsf{F}$ to the element

$$g_x(\eta) = \eta_x(\mathrm{id}_x) \in \mathsf{F}(x),$$

the image of $\mathrm{id}_x \in \mathsf{h}_x(x)$ via the map $\eta_x\colon \mathsf{h}_x(x) \to \mathsf{F}(x)$. The inverse of g_x is the map that assigns to an element $\xi \in \mathsf{F}(x)$ the natural transformation $\eta(x,\xi)\colon \mathsf{h}_x \Rightarrow \mathsf{F}$ defined as follows: for a given object $u \in \mathscr{C}$, we define

$$\mathsf{h}_x(u) \xrightarrow{\eta(x,\xi)_u} \mathsf{F}(u)$$

by sending a morphism $\alpha\colon u \to x$ to the image of ξ under $\mathsf{F}(\alpha)\colon \mathsf{F}(x) \to \mathsf{F}(u)$. This is indeed a natural transformation: we need to confirm that, given $\phi\colon v \to u$ in $\mathscr{C}$, the diagram

$$\begin{array}{ccc} \mathsf{h}_x(u) & \xrightarrow{\eta(x,\xi)_u} & \mathsf{F}(u) \\ {\scriptstyle \mathsf{h}_x(\phi)}\downarrow & & \downarrow{\scriptstyle \mathsf{F}(\phi)} \\ \mathsf{h}_x(v) & \xrightarrow{\eta(x,\xi)_v} & \mathsf{F}(v) \end{array}$$

commutes. If $\alpha \in \mathsf{h}_x(u)$, we have

$$\big(\mathsf{F}(\phi) \circ \eta(x,\xi)_u\big)(\alpha) = \mathsf{F}(\phi)(\mathsf{F}(\alpha)(\xi)) = \mathsf{F}(\alpha\circ\phi)(\xi),$$
$$\big(\eta(x,\xi)_v \circ \mathsf{h}_x(\phi)\big)(\alpha) = \eta(x,\xi)_v(\alpha\circ\phi) = \mathsf{F}(\alpha\circ\phi)(\xi),$$

as we wanted.

Finally to check that g_x is bijective with inverse $\xi \mapsto \eta(x,\xi)$, note that for every $\xi \in \mathsf{F}(x)$ and for every natural transformation $\eta\colon \mathsf{h}_x \Rightarrow \mathsf{F}$ one has

$$g_x(\eta(x,\xi)) = \eta(x,\xi)_x(\mathrm{id}_x) = \mathsf{F}(\mathrm{id}_x)(\xi) = \mathrm{id}_{\mathsf{F}(x)}(\xi) = \xi,$$
$$\eta(x, g_x(\eta)) = \eta.$$

For the second identity, fix $u \in \mathscr{C}$ and $\alpha \in \mathsf{h}_x(u)$, and compute

$$\begin{aligned}\eta(x,g_x(\eta))_u(\alpha) &= \eta(x,\eta_x(\mathrm{id}_x))_u(\alpha)\\ &= \mathsf{F}(\alpha)(\eta_x(\mathrm{id}_x))\\ &= \eta_u(\alpha).\end{aligned}$$

The last identity comes from naturality of $\eta\colon \mathsf{h}_x \Rightarrow \mathsf{F}$. Thus we have proved $\eta(x,g_x(\eta))_u = \eta_u$ for all $u \in \mathscr{C}$, which proves $\eta(x,g_x(\eta)) = \eta$ and confirms the bijectivity of g_x. We leave the naturality of this bijection as an exercise. □

The weak form of Yoneda's Lemma is the following statement about the functor $\mathsf{h}_{\mathscr{C}}$ defined above.

LEMMA A.7.0.2 (Weak Yoneda). *The functor* $\mathsf{h}_{\mathscr{C}}$ *defined in* (A.7.1) *is fully faithful.*



Exercise A.7.0.3. Show that Strong Yoneda, that we proved (Lemma A.7.0.1), implies Weak Yoneda (Lemma A.7.0.2).

Definition A.7.0.4 (Representable functor)**.** A functor $\mathsf{F} \in \mathsf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathsf{Sets})$ is *representable* if it lies in the essential image of $\mathsf{h}_{\mathscr{C}}$, i.e. if it is isomorphic to a functor h_x for some $x \in \mathscr{C}$. In this case, we say that the object $x \in \mathscr{C}$ represents F.

Note that, combining Lemma A.7.0.2 and Exercise A.3.0.9, we have that $\mathscr{C}$ is equivalent to its essential image $\mathsf{h}_{\mathscr{C}}(\mathscr{C}) \subset \mathsf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathsf{Sets})$.

Remark A.7.0.5. By Lemma A.7.0.2, if $x \in \mathscr{C}$ represents F, then x is unique up to isomorphism. Indeed, suppose we have isomorphisms

$$a\colon \mathsf{h}_x \xrightarrow{\sim} \mathsf{F}, \qquad b\colon \mathsf{h}_y \xrightarrow{\sim} \mathsf{F}$$

in the category $\mathsf{Fun}(\mathscr{C}^{\mathrm{op}}, \mathsf{Sets})$. By full faithfulness of Yoneda there is a unique isomorphism $f\colon x \xrightarrow{\sim} y$ inducing $\mathsf{h}_f = b^{-1} \circ a\colon \mathsf{h}_x \xrightarrow{\sim} \mathsf{h}_y$. Thus chosen representations, or universal pairs (x,a), are unique up to a unique compatible isomorphism. In particular, the underlying representing object (alone, without the datum of $a\colon \mathsf{h}_x \xrightarrow{\sim} \mathsf{F}$) is unique up to isomorphism.

Definition A.7.0.6 (Universal object)**.** Let $\mathsf{F}\colon \mathscr{C}^{\mathrm{op}} \to \mathsf{Sets}$ be a functor. A *universal object* for F is a pair (x,ξ) where $\xi \in \mathsf{F}(x)$, such that for every pair (u,σ) with $\sigma \in \mathsf{F}(u)$, there

exists a unique morphism $\alpha\colon u \to x$ with the property that $\mathsf{F}(\alpha)\colon \mathsf{F}(x) \to \mathsf{F}(u)$ sends ξ to σ.



Exercise A.7.0.7. Show that a pair (x, ξ) is a universal object for a functor $\mathsf{F}\colon \mathscr{C}^{\mathrm{op}} \to \mathsf{Sets}$ if and only if the natural transformation $\eta(x, \xi)$ defined in the proof of Lemma A.7.0.1 is a natural isomorphism. In particular, F is representable if and only if it has a universal object.

B | Commutative algebra

B.1 Frequently used theorems

B.1.1 Nakayama's Lemma(s)

LEMMA B.1.1.1 (Nakayama). *Let A be a ring, $I \subset A$ an ideal, M a finitely generated A-module. If $M = IM$, then there exists an element $r \in A$, with $1 - r \in I$, such that $rM = 0$. Equivalently, there exists $i \in I$ such that $im = m$ for every $m \in M$.*

The following weaker statements are also referred to as 'Nakayama's Lemma'.

LEMMA B.1.1.2. *Let A be a ring, $J_A \subset A$ its Jacobson radical (the intersection of all maximal ideals), M a finitely generated A-module. If $J_A M = M$, then $M = 0$.*

In particular, if M is a finitely generated A-module and $N \subset M$ is a submodule such that $M = N + J_A M$, then $M = N$.

LEMMA B.1.1.3. *Let $(A, \mathfrak{m}, \kappa)$ be a local ring, M a finitely generated A-module. Let $m_1, \dots, m_n \in M$ be elements such that their classes $\overline{m}_1, \dots, \overline{m}_n \in M/\mathfrak{m}M$ generate the κ-vector space $M/\mathfrak{m}M$. Then $m_1, \dots, m_n$ generate M as an A-module.*

B.1.2 Noether's Lemma

The original proof of the following fundamental result in commutative algebra can be found in Emmy Noether's work [6], published in 1926, exactly a century before this edition of the notes.

THEOREM B.1.2.1 (Noether's normalisation). *Let A be a nonzero finitely generated $\mathbb{F}$-algebra. Then, there exists an integer $d \geqslant 0$ and a finite injective ring homomorphism*

$$\mathbb{F}[x_1, \dots, x_d] \hookrightarrow A.$$

In particular, $\dim A = d$.

Proof. Write $A = \mathbb{F}[a_1, \dots, a_n]$. If $a_1, \dots, a_n$ are algebraically independent over $\mathbb{F}$, there is nothing to prove. So we assume there is a polynomial relation and we argue by induction

on n. There exists a nonzero polynomial

$$f = \sum_{\alpha \in \Lambda} c_\alpha T^\alpha \in \mathbb{F}[T_1, \ldots, T_n], \qquad T^\alpha = T_1^{\alpha_1} \cdots T_n^{\alpha_n}, \qquad c_\alpha \in \mathbb{F},$$

such that $f(a_1, \ldots, a_n) = 0$. Choose positive integers $N_1, \ldots, N_{n-1}$ such that the (finite) set

$$\left\{ \alpha_n + \sum_{i=1}^{n-1} N_i \alpha_i \;\middle|\; \alpha \in \Lambda \right\} \subset \mathbb{Z}$$

contains pairwise distinct integers. For instance, one may take the N_i to be sufficiently rapidly growing powers of a sufficiently large integer.

Set

$$b_i = a_i - a_n^{N_i}, \qquad i = 1, \ldots, n-1,$$

and put

$$B = \mathbb{F}[b_1, \ldots, b_{n-1}].$$

Consider the polynomial

$$f\big(b_1 + T^{N_1}, \ldots, b_{n-1} + T^{N_{n-1}}, T\big) \in B[T].$$

By the choice of the N_i, this polynomial has a unique term of highest degree in T, and its leading coefficient is a nonzero element of $\mathbb{F}$. After multiplying by a scalar in $\mathbb{F}^\times$, we may therefore assume that it is monic.

Since

$$f\big(b_1 + a_n^{N_1}, \ldots, b_{n-1} + a_n^{N_{n-1}}, a_n\big) = f(a_1, \ldots, a_n) = 0,$$

it follows that a_n is integral over B. Moreover,

$$a_i = b_i + a_n^{N_i} \qquad (i = 1, \ldots, n-1),$$

so

$$A = B[a_n].$$

Thus A is finite as a B-module; equivalently, the inclusion

$$B \hookrightarrow A$$

is finite.

Now B is generated as a $\mathbb{F}$-algebra by $n-1$ elements. By the induction hypothesis, there are algebraically independent elements $x_1, \ldots, x_d \in B$ such that

$$\mathbb{F}[x_1, \ldots, x_d] \hookrightarrow B$$

is finite. Since finiteness is transitive, the composite

$$\mathbb{F}[x_1, \ldots, x_d] \hookrightarrow B \hookrightarrow A$$

is finite, as required. □

Example B.1.2.2. For $A = F[x, y]/(xy - 1)$, and setting $t = x + y$, we have that A is finite over $\mathbb{F}[t]$: the element y satisfies $y^2 - ty + 1 = 0$ and $x = t - y$. This confirms $\dim A = 1$, which we also know from Krull's theorem.

B.1.3 Zariski's Lemma

LEMMA B.1.3.1 (Zariski). *Let $\mathbb{F}$ be a field and $\mathbb{E} \supset \mathbb{F}$ a field extension such that $\mathbb{E}$ is a finitely generated $\mathbb{F}$-algebra. Then $\mathbb{E}$ is finite dimensional over $\mathbb{F}$. In particular, $\mathbb{E} \supset \mathbb{F}$ is algebraic.*

Proof. This is a consequence of Noether's Normalisation Lemma. Since $\mathbb{E}$ is finitely generated as an $\mathbb{F}$-algebra, we have a finite injective homomorphism $B = \mathbb{F}[t_1, \dots, t_d] \hookrightarrow \mathbb{E}$. We claim that $d = 0$. If d were positive, we could take t_1 and note that $\alpha = 1/t_1 \in \mathbb{E}$ since $\mathbb{E}$ is a field. Then α is integral over B (since finite implies integral), which means there is a monic polynomial $q(y) \in B[y]$ such that $q(\alpha) = 0$. If we write $q(y) = y^m + b_{m-1}y^{m-1} + \cdots + b_1 y + b_0$, then we have

$$0 = \alpha^m + b_{m-1}\alpha^{m-1} + \cdots + b_1\alpha + b_0$$

in $\mathbb{E}$. But after multiplication by t_1^m, this becomes

$$0 = 1 + b_{m-1}t_1 + \cdots + b_1 t_1^{m-1} + b_0 t_1^m,$$

which says that t_1 is invertible in B, contradicting $d > 0$. □

Remark B.1.3.2. Note that the proof of Lemma B.1.3.1 actually solves **??**.

COROLLARY B.1.3.3. *If $\mathbb{F}$ is a field, A is a finitely generated $\mathbb{F}$-algebra and $\mathfrak{m} \subset A$ is a maximal ideal, the quotient $A/\mathfrak{m}$ is a finite extension of $\mathbb{F}$.*

Proof. Apply Zariski's Lemma to $\mathbb{E} = A/\mathfrak{m}$. □

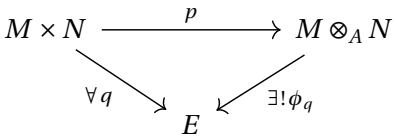
B.2 Tensor products

Definition B.2.0.1 (Tensor product of modules)**.** Let A be a ring, M and N two A-modules. The *tensor product* of M and N over A is defined to be a pair $(M \otimes_A N, p)$ where

- $M \otimes_A N$ is an A-module,
- $p : M \times N \to M \otimes_A N$ is a bilinear map,

such that the following universal property is satisfied: for every pair (E, q) where E is an A-module and $q : M \times N \to E$ is a bilinear map, there is exactly one A-linear homomorphism $\phi_q : M \otimes_A N \to E$ such that $q = \phi_q \circ p$.

The universal property of Definition B.2.0.1 can be depicted in the diagram



and, more importantly, can be rephrased by saying that there is a bijection

$$\mathsf{Bil}_A(M\times N,E)\xrightarrow{\sim}\mathrm{Hom}_{\mathsf{Mod}_A}(M\otimes_A N,E),\quad q\mapsto\phi_q,$$

functorial in E.

Regarding existence of an object $(M\otimes_A N,p)$ with the required universal property, one first considers the standard basis $\{\,e_{m,n}\mid m\in M,\ n\in N\,\}$ of the direct sum $A^{\oplus M\times N}$. One then constructs the quotient module

$$M\otimes_A N=A^{\oplus M\times N}/T$$

where $T\subset A^{\oplus M\times N}$ is the submodule generated by elements of the form

$$\begin{aligned}
&e_{m_1+m_2,n}-e_{m_1,n}-e_{m_2,n},\\
&e_{m,n_1+n_2}-e_{m,n_1}-e_{m,n_2},\\
&e_{am,n}-e_{m,an},\\
&a\,e_{m,n}-e_{am,n}.
\end{aligned}$$

The map $p\colon M\times N\to M\otimes_A N$ is defined by sending $(m,n)\mapsto[e_{m,n}]$, where the square bracket means equivalence class. One sets

$$m\otimes n=[e_{m,n}].$$

This is standard notation. Note that not all elements of $M\otimes_A N$ are of the form $m\otimes n$ for elements $m\in M$ and $n\in N$. However, every element $u\in M\otimes_A N$ can be written (non-uniquely) as a finite sum

$$u=\sum_{k=1}^{r}m_k\otimes n_k,\quad r>0.$$



Exercise B.2.0.2. Show that the pair $(M\otimes_A N,p)$ satisfies the universal property of Definition B.2.0.1.

Thanks to the universal property, $(M\otimes_A N,p)$ is unique up to a unique isomorphism. It is easy to check, using the universal property, that one has isomorphisms

$$\begin{aligned}
M\otimes_A A&\cong M,\\
M\otimes_A N&\cong N\otimes_A M,\\
(M\otimes_A N)\otimes_A P&\cong M\otimes_A(N\otimes_A P).
\end{aligned}$$

for A-modules M, N and P.



Exercise B.2.0.3. Let $m,n\in\mathbb{Z}$ be integers with $d=\gcd(m,n)$. Show that $\mathbb{Z}/m\mathbb{Z}\otimes_{\mathbb{Z}}\mathbb{Z}/n\mathbb{Z}\cong\mathbb{Z}/d\mathbb{Z}$. In particular, $\mathbb{Z}/m\mathbb{Z}\otimes_{\mathbb{Z}}\mathbb{Z}/n\mathbb{Z}=0$ when m and n are coprime.



Exercise B.2.0.4. Prove that tensor products of fields need not be fields or domains, by showing that

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \oplus \mathbb{C}.$$



Exercise B.2.0.5 ($\otimes$ commutes with $\oplus$)**.** If $(M_i)_{i\in I}$ is a family of A-modules, one has a canonical isomorphism

$$\bigoplus_{i\in I}(M_i \otimes_A N) \xrightarrow{\sim} \left(\bigoplus_{i\in I} M_i\right) \otimes_A N$$

for any A-module N.



Exercise B.2.0.6 (Tensoring is right exact)**.** Let A be a ring, M an A-module. Prove that the functor

$$\begin{aligned} \mathsf{Mod}_A &\xrightarrow{M\otimes_A -} \mathsf{Mod}_A \\ N &\longmapsto M \otimes_A N \end{aligned}$$

is right exact, i.e. that an exact sequence $N_0 \to N_1 \to N_2 \to 0$ gets sent to an exact sequence $M \otimes_A N_0 \to M \otimes_A N_1 \to M \otimes_A N_2 \to 0$.



Exercise B.2.0.7 (Restriction to submodules)**.** Let M and N be A-modules, $i : N' \hookrightarrow N$ a submodule. Denote by $i_M : M \otimes_A N' \to M \otimes_A N$ the canonical map. Show that there is a canonical isomorphism of A-modules

$$(M \otimes_A N)/\mathrm{im}(i_M) \xrightarrow{\sim} M \otimes_A (N/N').$$

In particular, if $N = A$ and $N' = I \subset A$ is an ideal, then there is a canonical isomorphism

$$M/IM \xrightarrow{\sim} M \otimes_A (A/I).$$

(**Hint**: use the universal property, or tensor the right exact sequence $N' \to N \to N/N' \to 0$).

B.3 Localisation

B.3.1 General construction for modules

Let A be a ring, M an A-module. Fix a *multiplicative subset* $S \subset A$, i.e. a subset containing the identity $1 \in A$ and such that $s_1 s_2 \in S$ whenever $s_1, s_2 \in S$.

Example B.3.1.1. The following are key examples of multiplicative subsets:

(i) $S = \{ f^n \mid n \geqslant 0 \}$ for some $f \in A$.

(ii) $S = A \setminus \mathfrak{p}$, where $\mathfrak{p} \subset A$ is a prime ideal.

(iii) $S = A \setminus 0$, if A is an integral domain.

(iv) $S = A \setminus \mathcal{Z}$, where $\mathcal{Z}$ is the set of all zero-divisors in A.

Consider the equivalence relation on $M \times S$ defined by

$$\boxed{(m,s) \sim (m',s') \Longleftrightarrow \text{there exists } u \in S \text{ such that } u(s'm - sm') = 0 \in M.}$$

We denote by m/s, or by $\frac{m}{s}$, the equivalence class of (m,s). The set of such equivalence classes

$$S^{-1}M = (M \times S)/\sim \tag{B.3.1}$$

is an abelian group via

$$\frac{m}{s} + \frac{m'}{s'} = \frac{sm' + s'm}{ss'},$$

and if $M = A$ then $S^{-1}A$ becomes a ring via

$$\frac{a}{s} \cdot \frac{a'}{s'} = \frac{aa'}{ss'}.$$

The $\mathbb{Z}$-module $S^{-1}M$ is an $S^{-1}A$-module via

$$\frac{a}{s} \cdot \frac{m}{s'} = \frac{am}{ss'}. \tag{B.3.2}$$

Here 'am' refers to the A-module structure on M.

Definition B.3.1.2 (Localisation of a module)**.** Let A be a ring, M an A-module. The localisation of M with respect to a multiplicative subset $S \subset A$ is the $S^{-1}A$-module $S^{-1}M$, where the linear structure is given by Equation (B.3.2).

Notation B.3.1.3. If $S = \{f^n \mid n \geqslant 0\}$ as in Example B.3.1.1(i) above, then we write M_f for the localisation. If $S = A \setminus \mathfrak{p}$ as in Example B.3.1.1(ii) above, then we write $M_{\mathfrak{p}}$ for the localisation. Do not confuse M_f and $M_{(f)}$ when $(f) = fA \subset A$ is a prime ideal!

Remark B.3.1.4. If $0 \in S$, then $S^{-1}M = 0$.

Remark B.3.1.5. The localisations of the form M_f and $M_{\mathfrak{p}}$ are crucial in algebraic geometry. In M_f, the equivalence relation defining the localisation reads

$$M_f \ni \frac{m}{f^d} = \frac{n}{f^e} \iff \text{there exists } k \geqslant 0 \text{ such that } f^k(f^e m - f^d n) = 0 \in M.$$

In particular, $m/f^d = 0 \in M_f$ if and only if there exists $k \geqslant 0$ such that $f^k m = 0 \in M$. In particular,

$$M_f = 0 \iff \text{for every } m \in M \text{ there exists } k \geqslant 0 \text{ such that } f^k m = 0 \in M.$$

If M is finitely generated, this is equivalent to the existence of a uniform $r \geqslant 0$ such that $f^r M = 0$. If $M = A$, one actually has that $A_f = 0$ if and only if f is nilpotent.[1]

[1] It is not true, for an arbitrary A-module, that $M_f = 0$ implies that f is nilpotent. For example, with $A = \mathbb{Z}$, $M = \mathbb{Z}/2\mathbb{Z}$, and $f = 2$, one has $M_f = 0$ although 2 is not nilpotent.

Remark B.3.1.6. We make the following observations.

- Note that $S^{-1}M$ comes with an A-linear map

$$\begin{aligned} M &\xrightarrow{\ell} S^{-1}M \\ m &\longmapsto \frac{m}{1}. \end{aligned}$$

  To see that ℓ is A-linear, simply note that the A-linear structure on $S^{-1}M$ is given by $a \cdot \frac{m}{s} = \frac{a}{1} \cdot \frac{m}{s} = \frac{am}{s}$, where the second identity uses (B.3.2), so

$$\ell(a \cdot m) = \frac{am}{1} = \frac{a}{1} \cdot \frac{m}{1} = a \cdot \ell(m).$$

- Localisation is functorial: if $\phi : N \to M$ is an A-linear map, there is an induced $S^{-1}A$-linear map

$$\begin{aligned} S^{-1}N &\xrightarrow{S^{-1}\phi} S^{-1}M, \\ \frac{n}{s} &\longmapsto \frac{\phi(n)}{s}. \end{aligned}$$

  To see that $S^{-1}\phi$ is $S^{-1}A$-linear, note that if $a/t \in S^{-1}A$ then

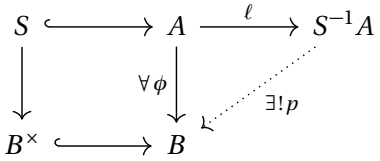
$$S^{-1}\phi\left(\frac{a}{t} \cdot \frac{n}{s}\right) = S^{-1}\phi\left(\frac{an}{ts}\right) = \frac{\phi(an)}{ts} = \frac{a \cdot \phi(n)}{ts} = \frac{a}{t} \cdot \frac{\phi(n)}{s} = \frac{a}{t} \cdot S^{-1}\phi\left(\frac{n}{s}\right).$$

B.3.2 Universal property of localisation

Set $M = A$. There is a canonical ring homomorphism

$$\ell : A \to S^{-1}A, \quad a \mapsto \frac{a}{1}$$

sending S inside the group of invertible elements of $S^{-1}A$ (the inverse of $s/1$ being $1/s$), and making the pair $(S^{-1}A, \ell)$ universal with this property: whenever one has a ring homomorphism $\phi : A \to B$ such that $\phi(S) \subset B^\times$, there is exactly one ring homomorphism $p : S^{-1}A \to B$ such that $\phi = p \circ \ell$.



Explicitly, the map p is defined by $p(a/s) = \phi(a)\phi(s)^{-1}$.

In fact, there is a universal property also for localised modules.

LEMMA B.3.2.1 ([7, Tag 07K0]). *Let A be a ring, $S \subset A$ a multiplicative subset, and fix an A-module M and an $S^{-1}A$-module N. Then, the canonical map*

$$\mathrm{Hom}_{S^{-1}A}(S^{-1}M, N) \longrightarrow \mathrm{Hom}_A(M, N)$$

sending $p : S^{-1}M \to N$ to the map $p \circ \ell : M \to S^{-1}M \to N$, where $\ell : M \to S^{-1}M$ is the localisation map, is an isomorphism.

In other words, for any A-linear map $\phi: M \to N$, there exists exactly one $S^{-1}A$-linear map $p: S^{-1}M \to N$ such that

$$\begin{array}{ccc} M & \xrightarrow{\ell} & S^{-1}M \\ {\scriptstyle \forall\, \phi}\big\downarrow & \swarrow_{\exists!\, p} & \\ N & & \end{array}$$

commutes. Explicitly, the map p is given by $p(m/s) = (1/s)\cdot\phi(m)$.

B.3.3 Ring Localisations

The following lemma is of key importance to us.

LEMMA B.3.3.1. *Let A be a ring, and $\ell: A \to S^{-1}A$ a localisation. Sending $\mathfrak{r} \mapsto \ell^{-1}(\mathfrak{r})$ establishes a bijection*

$$\begin{array}{ccc} \{\text{prime ideals } \mathfrak{r} \subset S^{-1}A\} & \xrightarrow{\simeq} & \{\text{prime ideals } \mathfrak{q} \subset A \text{ such that } \mathfrak{q} \cap S = \emptyset\} \\ \| & & \big\downarrow \\ \operatorname{Spec} S^{-1}A & & \operatorname{Spec} A \end{array}$$

having as inverse the extension operation, sending

$$\mathfrak{q} \longmapsto \mathfrak{q}S^{-1}A = \left\{ \frac{a}{f} \,\middle|\, a \in \mathfrak{q},\, f \in S \right\} \subset S^{-1}A.$$

COROLLARY B.3.3.2. *For any prime ideal $\mathfrak{p} \subset A$ the ring*

$$A_{\mathfrak{p}} = \left\{ \frac{a}{f} \,\middle|\, a \in A,\, f \notin \mathfrak{p} \right\}$$

is local, with maximal ideal

$$\mathfrak{p}A_{\mathfrak{p}} = \left\{ \frac{a}{f} \,\middle|\, a \in \mathfrak{p},\, f \notin \mathfrak{p} \right\} \subset A_{\mathfrak{p}}.$$

Proof. Indeed, the correspondence of Lemma B.3.3.1 becomes, in the case $S = A \setminus \mathfrak{p}$,

$$\{\text{prime ideals } \mathfrak{r} \subset A_{\mathfrak{p}}\} \xrightarrow{\simeq} \{\text{prime ideals } \mathfrak{q} \subset A \text{ such that } \mathfrak{q} \subset \mathfrak{p}\}$$

and since its inverse (extension along $A \to A_{\mathfrak{p}}$) is inclusion-preserving it follows that every prime ideal $\mathfrak{r} \subset A_{\mathfrak{p}}$ must be contained in $\mathfrak{p}A_{\mathfrak{p}}$. This means that $\mathfrak{p}A_{\mathfrak{p}}$ is the unique maximal ideal. □



Exercise B.3.3.3. If $(A, \mathfrak{m})$ is a local ring, then the canonical map $A \to A_{\mathfrak{m}}$ is an isomorphism.



Warning B.3.3.4. In the case when B is a graded ring and $\mathfrak{p}$ is a homogeneous prime ideal, we use the notation $B_{\mathfrak{p}}$ for the localisation of B at the multiplicative subset consisting of *homogeneous* elements that are not in $\mathfrak{p}$.

PROPOSITION B.3.3.5 ([5, Prop. 5.8]). *If $\mathfrak{m} \subset A$ is a maximal ideal and $k > 0$ is an integer, there is a natural ring isomorphism*

$$A/\mathfrak{m}^k \xrightarrow{\sim} A_{\mathfrak{m}}/(\mathfrak{m}A_{\mathfrak{m}})^k.$$

It induces isomorphisms

$$\mathfrak{m}^h/\mathfrak{m}^k \xrightarrow{\sim} (\mathfrak{m}A_{\mathfrak{m}})^h/(\mathfrak{m}A_{\mathfrak{m}})^k$$

for every $h \leqslant k$.

LEMMA B.3.3.6. *Let A be a ring, $S \subset A$ a multiplicative subset. Then $\ell : A \to S^{-1}A$ is injective if and only if S contains no zero divisors.*

Proof. Suppose $a/1 = 0/1$ in $S^{-1}A$. Then there is $u \in S$ such that $au = 0$. But u is not a zero divisor, thus $a = 0$ and ℓ is injective. Conversely, if $s \in S$ and $sa = 0$ with $a \neq 0$, then $a/1 = 0 \in S^{-1}A$, contradicting injectivity. □

Example B.3.3.7. Let A be an integral domain, which means that $(0) \subset A$ is prime. Then the localisation

$$A_{(0)} = \left\{ \frac{a}{b} \;\middle|\; a \in A,\ b \in A \setminus 0 \right\}$$

is a field, called the *fraction field* of A, that we denote by $\operatorname{Frac} A$. The canonical map $\ell : A \to \operatorname{Frac} A$ is injective by Lemma B.3.3.6.

Example B.3.3.8. Let A be a ring. Consider $S = A \setminus \mathcal{Z}$ as in Example B.3.1.1(iv). The localisation $S^{-1}A$ is called the *total ring of fractions* of A. By Lemma B.3.3.6, $S = A \setminus \mathcal{Z}$ is the largest multiplicative set such that $\ell : A \to S^{-1}A$ is injective.

Example B.3.3.9. Let $A = \mathbb{Z}$. Fix a prime number $p \in \mathbb{Z}$. Then the localisation map

$$\mathbb{Z} \hookrightarrow \mathbb{Z}_{(p)} = \left\{ \tfrac{n}{m} \;\middle|\; n \in \mathbb{Z},\ p \nmid m \right\}$$

is injective, and so is the localisation map

$$\mathbb{Z} \hookrightarrow \mathbb{Z}[1/p] = \left\{ \tfrac{n}{p^k} \;\middle|\; n \in \mathbb{Z},\ k \geqslant 0 \right\}.$$

LEMMA B.3.3.10. *If A is reduced and $S \subset A$ is a multiplicative subset, then $S^{-1}A$ is also reduced.*

Proof. Assume there exists $a \in A$, $s \in S$ and $r \in \mathbb{Z}_{>0}$ such that $0/1 = (a/s)^r = a^r/s^r \in S^{-1}A$. Then there exists $u \in S$ such that $ua^r = 0 \in A$, thus $(ua)^r = 0$, and hence $ua = 0$ by assumption. But this means $0/1 = a/1 \in S^{-1}A$, and thus $0/1 = (a/1)(1/s) = a/s \in S^{-1}A$. □

B.3.4 Exactness of localisation

LEMMA B.3.4.1. *Let A be a ring, $S \subset A$ a multiplicative subset, M an A-module. Then, there is a canonical isomorphism of $S^{-1}A$-modules*

$$\phi : S^{-1}M \xrightarrow{\sim} M \otimes_A S^{-1}A.$$

Proof. First of all, the $S^{-1}A$-module structure on $M \otimes_A S^{-1}A$ is defined (on generators) by

$$\frac{a}{t} \cdot \left(m \otimes \frac{b}{s} \right) = m \otimes \frac{ab}{ts}.$$

The map ϕ is defined by

$$\phi\left(\frac{m}{s}\right) = m \otimes \frac{1}{s}.$$

It is $S^{-1}A$-linear, since

$$\begin{aligned}
\phi\left(\frac{a}{t} \cdot \frac{m}{s}\right) &= \phi\left(\frac{am}{ts}\right) \\
&= am \otimes \frac{1}{ts} \\
&= m \otimes \frac{a}{ts} \\
&= \frac{a}{t} \cdot \left(m \otimes \frac{1}{s} \right) \\
&= \frac{a}{t} \cdot \phi\left(\frac{m}{s}\right).
\end{aligned}$$

Its inverse is given by $m \otimes (a/s) \mapsto (am)/s$. □

PROPOSITION B.3.4.2. *Let A be a ring, $S \subset A$ a multiplicative subset. Then, sending $M \mapsto S^{-1}M$ defines an exact functor from A-modules to $S^{-1}A$-modules.*

Proof. Fix a short exact sequence

$$0 \longrightarrow M \xrightarrow{\iota} N \xrightarrow{\pi} P \longrightarrow 0$$

of A-modules. We already know that

$$S^{-1}M \xrightarrow{S^{-1}\iota} S^{-1}N \xrightarrow{S^{-1}\pi} S^{-1}P \longrightarrow 0$$

is exact, since this sequence is isomorphic to

$$M \otimes_A S^{-1}A \longrightarrow N \otimes_A S^{-1}A \longrightarrow P \otimes_A S^{-1}A \longrightarrow 0$$

by Lemma B.3.4.1, and tensor product (by any A-module, e.g. $S^{-1}A$) is a right exact functor, cf. Exercise B.2.0.6. So we only need to show that

$$S^{-1}M \xrightarrow{S^{-1}\iota} S^{-1}N$$

is injective. Assume there is an element $m/s \in S^{-1}M$ such that $0 = 0/1 = S^{-1}\iota(m/s) = \iota(m)/s \in S^{-1}N$. Then there exists $u \in S$ such that $0 = u\iota(m) = \iota(um)$ in N. This implies $um = 0 \in M$, hence $m/s = um/us = 0/us = 0$. □

B.4 Length of modules

We recall some notions and facts from [7, Tag 00IU]. The theory of lengths is very important in the study of 0-dimensional coherent sheaves on varieties, and it will appear multiple times in our discussion on associated primes.

Let A be a ring, M an A-module. A chain of submodules of length n is a filtration

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M.$$

Definition B.4.0.1 (Length of a module)**.** Let A be a ring, M an A-module. Define the *length* of M to be

$$\mathsf{length}_A(M) = \sup_n \left\{ n \;\middle|\; \text{there exists a chain of submodules of length } n \right\}.$$

Remark B.4.0.2. If M has finite length, it is not obvious from the definition that all maximal chains have the same length. This is, however, true (cf. Proposition B.4.0.7).

The length is related to the notion of *support* of a module, which will also be central in later sections of this appendix.

Definition B.4.0.3 (Support of a module)**.** Let A be a ring, M an A-module. Define the *support* of M to be

$$\mathsf{Supp}_A(M) = \left\{ \mathfrak{p} \in \operatorname{Spec} A \;\middle|\; M_{\mathfrak{p}} \neq 0 \right\} \subset \operatorname{Spec} A.$$

Here are a series of easy, useful facts about length and support. Most proofs can be found in [7, Tag 00IU].

- If $\mathsf{length}_A(M) < \infty$, then M is finitely generated.
- If $0 \to M' \to M \to M'' \to 0$ is a short exact sequence of A-modules, then

$$\begin{aligned} \mathsf{length}_A(M) &= \mathsf{length}_A(M') + \mathsf{length}_A(M''), \\ \mathsf{Supp}_A(M) &= \mathsf{Supp}_A(M') \cup \mathsf{Supp}_A(M''). \end{aligned} \tag{B.4.1}$$

  The second identity is proved, for instance, in [7, Tag 00L3].

- If $(A, \mathfrak{m})$ is local and M has finite length, then $\mathfrak{m}^n M = 0$ for some $n \geqslant 0$. Conversely, if $\mathfrak{m}$ and M are finitely generated and $\mathfrak{m}^n M = 0$ for some n, then M has finite length.
- For a noetherian local ring $(A, \mathfrak{m})$ and a finitely generated module $M \neq 0$, this gives the useful equivalence

$$\mathsf{length}_A(M) < \infty \iff \mathfrak{m}^n M = 0 \text{ for some } n \iff \mathsf{Supp}_A(M) = \{\mathfrak{m}\}.$$

- If $A \to B$ is a ring homomorphism and M is a B-module, then $\mathsf{length}_A(M) \geqslant \mathsf{length}_B(M)$, with equality if $A \to B$ is surjective.
- In particular, if $B = S^{-1}A$ for some multiplicative subset S, since every B-submodule $N' \subset S^{-1}M$ is of the form $S^{-1}N$ for some A-submodule $N \subset M$, we deduce that $\mathsf{length}_A(M) \geqslant \mathsf{length}_{S^{-1}A}(S^{-1}M)$.
- Let $(A, \mathfrak{m})$ be a local ring, M an A-module with $\mathfrak{m}M = 0$. Then

$$\mathsf{length}_A(M) = \dim_{A/\mathfrak{m}} M,$$

and $\mathsf{length}_A(M) < \infty$ if and only if M is finitely generated.

Remark B.4.0.4. The implication

$$(A, \mathfrak{m}) \text{ local, } M \text{ finitely generated} \quad \Longrightarrow \quad \mathfrak{m}^n M = 0 \text{ for some } n.$$

is false. For example, take $M = A = \mathbb{F}[x]_{(x)}$. Then M is finitely generated over the local ring A, but $x^n A \neq 0$ for every n. One really needs finite length for the conclusion to hold.



Exercise B.4.0.5. Recall from **??** the characterisations of 0-dimensional noetherian local rings A. We propose now the following further characterisation: show that 0-dimensionality, for a noetherian local ring A, is equivalent to $\mathsf{length}_A(A) < \infty$.

Recall that a module $M \neq 0$ is *simple* if it has no proper nonzero submodules.

LEMMA B.4.0.6 (Length and simple modules [7, Tag 00J2]). *Let A be a ring, $M \neq 0$ an A-module. The following are equivalent:*

(i) *M is simple,*

(ii) *$\mathsf{length}_A(M) = 1$, and*

(iii) *$M \cong A/\mathfrak{m}$ for some maximal ideal $\mathfrak{m} \subset A$.*

The following proposition, in particular the last statement about the "local length", is tightly linked to Proposition B.5.1.13.

PROPOSITION B.4.0.7 ([7, Tag 00J3]). *Let A be a ring, M an A-module of finite length. Fix a maximal chain of submodules*

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M.$$

Then $n = \mathsf{length}_A(M)$, each subquotient M_i/M_{i-1} is simple, hence isomorphic to $A/\mathfrak{m}_i$ for some maximal ideal $\mathfrak{m}_i \subset A$, and for every maximal ideal $\mathfrak{m} \subset A$ one has that

$$\#\{\, i \mid \mathfrak{m}_i = \mathfrak{m} \,\} = \mathsf{length}_{A_\mathfrak{m}}(M_\mathfrak{m}).$$



Exercise B.4.0.8. Let M be a finitely generated module over a local ring $(A, \mathfrak{m}, \kappa)$. Assume $\mathfrak{m}^n M = 0$ for some $n \geqslant 0$. Prove that

$$\mathsf{length}_A(M) = \sum_{i=0}^{n-1} \dim_\kappa \left(\frac{\mathfrak{m}^i M}{\mathfrak{m}^{i+1} M} \right).$$

B.5 Associated points, and embedded components

For the theory of associated points we mainly follow Vakil's exposition [8, Ch. 6] and Eisenbud's [1, Ch. 3], whereas for Cohen–Macaulay schemes we follow [7].

What are associated points? Just like the generic points, they are very specific points of a (locally noetherian) scheme X. They are more relevant than all other points, in that they reveal part of the behaviour of the structure sheaf.

The generic points of the irreducible components of X are always associated points. There may also be nonminimal associated points, called embedded points. The closure $\overline{\{x\}} \subset X$ of an embedded point $x \in X$ is called an *embedded component*, and is always contained, set-theoretically, in an irreducible component of X. A reduced locally noetherian scheme never has embedded points. The presence of embedded points records additional scheme-theoretic structure concentrated on proper closed subsets. However, a nonreduced scheme need not have embedded points: its nilpotent structure may just be supported along an irreducible component.

In this section, the "best" results are obtained for *finitely generated* modules over a *noetherian* ring. We will treat the general theory without making these assumptions right away, but the reader can safely make them from the start.

B.5.1 Associated points: local case

Let A be a commutative ring, and let M be an A-module. If $m \in M$, we set

$$\begin{aligned}
\mathsf{Ann}_A(m) &= \{\, a \in A \mid am = 0 \,\} = \ker(A \xrightarrow{\cdot m} M) \subset A,\\
\mathsf{Ann}_A(M) &= \{\, a \in A \mid am = 0 \text{ for all } m \in M \,\} \subset A,\\
\mathsf{Supp}_A(m) &= \left\{\, \mathfrak{p} \in \operatorname{Spec} A \;\middle|\; m_{\mathfrak{p}} \neq 0 \,\right\} \subset \operatorname{Spec} A,\\
\mathsf{Supp}_A(M) &= \left\{\, \mathfrak{p} \in \operatorname{Spec} A \;\middle|\; M_{\mathfrak{p}} \neq 0 \,\right\} \subset \operatorname{Spec} A.
\end{aligned}$$

In the third identity, $m_{\mathfrak{p}}$ denotes the image of $m \in M$ in the localisation $M_{\mathfrak{p}}$. The support $\mathsf{Supp}_A(M)$ had already been introduced in Definition B.4.0.3.

We have the identity

$$\mathsf{Supp}_A(m) = \mathrm{V}(\mathsf{Ann}_A(m)),$$

and if $M = mA$ is principal, then

$$\mathsf{Supp}_A(M) = \mathsf{Supp}_A(m) = \mathrm{V}(\mathsf{Ann}_A(m)).$$

In general, it is clear that

$$\mathsf{Supp}_A(M) \subset \mathrm{V}(\mathsf{Ann}_A(M)) = \{\, \mathfrak{p} \in \operatorname{Spec} A \mid \mathfrak{p} \supset \mathsf{Ann}_A(M) \,\}. \tag{B.5.1}$$

Indeed, fix $\mathfrak{p}$ such that $M_{\mathfrak{p}} \neq 0$. If $\mathfrak{p} \notin \mathrm{V}(\mathsf{Ann}_A(M))$, then there exists $a \in \mathsf{Ann}_A(M)$ such that $a \notin \mathfrak{p}$. But $a \notin \mathfrak{p}$ implies $\overline{a} = a/1 \in A_{\mathfrak{p}}^{\times} \subset A_{\mathfrak{p}}$ is a unit, whereas $a \in \mathsf{Ann}_A(M)$ implies

that $\overline{a}M_{\mathfrak{p}} = 0$. But a unit cannot annihilate $M_{\mathfrak{p}}$ unless $M_{\mathfrak{p}} = 0$, contradiction. To obtain the reverse inclusion $\mathsf{V}(\mathsf{Ann}_A(M)) \subset \mathsf{Supp}_A(M)$, on the other hand, one needs to assume M to be finitely generated, as the next exercise proposes to confirm.



Exercise B.5.1.1 ($\mathsf{Supp}_A(M) = \mathsf{V}(\mathsf{Ann}_A(M))$ if M is finite)**.** Assume M is a finitely generated A-module, and there exists $a \in A$ such that $a \in \mathfrak{p}$ for all $\mathfrak{p} \in \mathsf{Supp}_A(M)$. Then there exists $r > 0$ such that $a^r \in \mathsf{Ann}_A(M)$. In other words,

$$\bigcap_{\mathfrak{p} \in \mathsf{Supp}_A(M)} \subset \sqrt{\mathsf{Ann}_A(M)}.$$

Coupled with (B.5.1), we obtain, when M is finitely generated, the identity

$$\mathsf{Supp}_A(M) = \mathsf{V}(\mathsf{Ann}_A(M)).$$

Definition B.5.1.2 (Associated primes)**.** Let A be a commutative ring, and let M be an A-module. A prime ideal $\mathfrak{p} \in \operatorname{Spec} A$ is said to be *associated to* M if $\mathfrak{p} = \mathsf{Ann}_A(m)$ for some $m \in M$. The set of all primes associated to M is denoted

$$\mathsf{Ass}_A(M) = \{\, \mathfrak{p} \in \operatorname{Spec} A \mid \mathfrak{p} \text{ is associated to } M \,\} \subset \operatorname{Spec} A.$$

We have the following characterisation of associated primes.

LEMMA B.5.1.3. *Let A be a ring, and let M be an A-module. Let $\mathfrak{p} \subset A$ be a prime ideal. Then $\mathfrak{p} \in \mathsf{Ass}_A(M)$ if and only if $A/\mathfrak{p}$ is an A-submodule of M.*

Proof. If $\mathfrak{p} = \mathsf{Ann}_A(m)$ for some $m \in M$, consider the map $\phi_m \colon A \to M$ defined by $\phi_m(a) = a \cdot m$. Since its kernel is by definition $\mathsf{Ann}_A(m)$, the quotient $A/\mathfrak{p}$ is an A-submodule of M. Conversely, given an A-linear inclusion $i \colon A/\mathfrak{p} \hookrightarrow M$, consider the composition $\phi \colon A \twoheadrightarrow A/\mathfrak{p} \hookrightarrow M$. Then $\phi_{i(1)}(a) = a \cdot i(1) = i(a + \mathfrak{p}) = \phi(a)$ for all $a \in A$, i.e. $\phi = \phi_{i(1)}$. □

Example B.5.1.4. If A is an integral domain, the only associated prime of $M = A$ is the trivial ideal $(0) \subset A$. If A is a UFD and $f \in A$ is a nonzero nonunit, the only associated primes in $M = A/fA$ are the prime factors of f. See [8, Ex. 6.6.E and 6.6.F].

Remark B.5.1.5. Let A be a ring, and let M be an A-module. If $\mathfrak{p} \in \mathsf{Ass}_A(M)$ then $\mathfrak{p} \supset \mathsf{Ann}_A(M)$. Thus, $\mathsf{Ass}_A(M) \subset \mathsf{V}(\mathsf{Ann}_A(M))$. In fact, something finer is true: we always have

$$\boxed{\mathsf{Ass}_A(M) \subset \mathsf{Supp}_A(M) \subset \mathsf{V}(\mathsf{Ann}_A(M)).}$$

Indeed, if $\mathfrak{p} = \mathsf{Ann}_A(m) \in \mathsf{Ass}_A(M)$, then $am \neq 0$ for every $a \in A \setminus \mathfrak{p}$, and this tells us that $0 \neq m/1 \in M_{\mathfrak{p}}$, in particular $\mathfrak{p} \in \mathsf{Supp}_A(M)$. The inclusion $\mathsf{Supp}_A(M) \subset \mathsf{V}(\mathsf{Ann}_A(M))$, on the other hand, was confirmed in (B.5.1).



Exercise B.5.1.6. An element $a \in A$ is said to be a zero-divisor on an A-module M if $am = 0$ for some nonzero $m \in M$. Denote by $\mathsf{Z}_A(M)$ the set of zero-divisors on M. Assume A is noetherian. Prove that

$$\mathsf{Z}_A(M) = \bigcup_{\mathfrak{p} \in \mathsf{Ass}_A(M)} \mathfrak{p}.$$

Note that this generalises **??**.



Exercise B.5.1.7. Let A be a noetherian ring, M an A-module. Show that

(i) $\mathsf{Ass}_A(M) = \emptyset$ if and only if $M = 0$, and

(ii) the natural map $M \to \prod_{\mathfrak{p} \in \mathsf{Ass}_A(M)} M_{\mathfrak{p}}$ is injective,

For (i), you can look at [8, Prop. 6.6.10] or [1, Prop. 3.4] if you need a hint.

Notation B.5.1.8. We denote by $\mathsf{Min}(S) \subset S$ the subset of minimal elements, with respect to inclusion, in a set S.

Definition B.5.1.9 (Isolated primes)**.** Let A be a ring, M an A-module. The minimal elements (with respect to inclusion) in $\mathsf{Supp}_A(M)$ are called *isolated primes* of M. In symbols, this is the subset $\mathsf{Min}(\mathsf{Supp}_A(M)) \subset \mathsf{Supp}_A(M)$.

Note that it is not yet clear at this point that all isolated primes of M belong to $\mathsf{Ass}_A(M)$. All we have at this point is

$$\begin{array}{ccccccc} & & \mathsf{Ass}_A(M) & & & & \\ & \overset{\cdots\nearrow}{} & \downarrow & & & & \\ \mathsf{Min}(\mathsf{Supp}_A(M)) & \hookrightarrow & \mathsf{Supp}_A(M) & \hookrightarrow & \mathsf{V}(\mathsf{Ann}_A(M)) & \hookrightarrow & \operatorname{Spec} A \end{array}$$

but the inclusion $\mathsf{Min}(\mathsf{Supp}_A(M)) \subset \mathsf{Ass}_A(M)$ will be true for $M \neq 0$ finitely generated over a noetherian ring A by Theorem B.5.1.11, and the above picture will become

$$\begin{array}{ccccccc} & & \mathsf{Ass}_A(M) & \overset{\text{finite} \neq \emptyset}{\longleftarrow} & & & \\ & \overset{\text{Theorem B.5.1.11}}{\nearrow} & \downarrow & & & & \\ \mathsf{Min}(\mathsf{Supp}_A(M)) & \hookrightarrow & \mathsf{Supp}_A(M) & = & \mathsf{V}(\mathsf{Ann}_A(M)) & \hookrightarrow & \operatorname{Spec} A, \end{array}$$

at least very informally speaking.

The next result records the key steps in the general theory.

THEOREM B.5.1.10. *The following statements hold true.*

(1) *If* $0 \to M' \to M \to M'' \to 0$ *is a short exact sequence of* A*-modules, there are inclusions*

$$\mathsf{Ass}_A(M') \subset \mathsf{Ass}_A(M) \subset \mathsf{Ass}_A(M') \cup \mathsf{Ass}_A(M''). \tag{B.5.2}$$

(2) *If M is finitely generated, then $\overline{\mathrm{Ass}_A(M)} \subset \mathrm{Supp}_A(M)$, and if A is furthermore assumed to be noetherian, one has*

$$\overline{\mathrm{Ass}_A(M)} = \mathrm{Supp}_A(M).$$

(3) *If M is a finitely generated module over a noetherian ring A, and $S \subset A$ is a multiplicative subset, then*

$$\mathrm{Ass}_{S^{-1}A}(S^{-1}M) = \left\{ S^{-1}\mathfrak{p} \;\middle|\; \mathfrak{p} \in \mathrm{Ass}_A(M),\, \mathfrak{p} \cap S = \emptyset \right\} \subset \mathrm{Spec}\, S^{-1}A.$$

In particular, for $\mathfrak{p} \in \mathrm{Spec}\, A$, there is an identity

$$\mathrm{Ass}_{A_\mathfrak{p}}(M_\mathfrak{p}) = \left\{ \mathfrak{q}A_\mathfrak{p} \;\middle|\; \mathfrak{q} \in \mathrm{Ass}_A(M),\, \mathfrak{q} \subset \mathfrak{p} \right\}.$$

Proof. We provide references or give quick proofs for all the statements.

For part (1), the inclusions (B.5.2) are proved in [8, Prop. 6.6.15] (see also [1, Lemma 3.6] and [7, Tag 02M3]).

The inclusion $\overline{\mathrm{Ass}_A(M)} \subset \mathrm{Supp}_A(M)$ in part (2) follows from the inclusion $\mathrm{Ass}_A(M) \subset \mathrm{Supp}_A(M)$ of Remark B.5.1.5 combined with the closedness of $\mathrm{Supp}_A(M) = \mathrm{V}(\mathrm{Ann}_A(M))$, cf. Exercise B.5.1.1, whereas the identity $\overline{\mathrm{Ass}_A(M)} = \mathrm{Supp}_A(M)$ is proved in [8, Prop. 6.6.24] under the stated noetherian assumption. Part (3) is proved in [8, Prop. 6.6.26] or [1, Thm. 3.1]. □

From now on we assume A is noetherian and M is finitely generated. We have the following result.

THEOREM B.5.1.11. *Let A be a noetherian ring, $M \neq 0$ a finitely generated A-module. Then $\mathrm{Ass}_A(M)$ is a finite nonempty set containing all isolated primes.*

Proof. The fact that $\mathrm{Ass}_A(M) \neq \emptyset$ is just Exercise B.5.1.7(i).

Finiteness follows from the following standard fact in commutative algebra: every nonzero finitely generated module M over a noetherian ring A admits a finite 'prime filtration'

$$\boxed{0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_s = M, \qquad M_i/M_{i-1} \cong A/\mathfrak{p}_i.}$$

To obtain such a prime filtration, one proceeds as follows (as done e.g. in [1, Prop. 3.7]): Since $M \neq 0$, by Exercise B.5.1.7(i) we have at least one element $\mathfrak{p}_1 \in \mathrm{Ass}_A(M)$, so by Lemma B.5.1.3 we know $M_1 = A/\mathfrak{p}_1 \subset M$ is a submodule. Apply this reasoning again to M/M_1 to produce M_2, and use the fact that submodules of M satisfy the ascending chain condition to conclude that the process stops, i.e. $M_n = M$ for some integer n. See also [7, Tag 00L0].

Caveat: a prime filtration as above is not unique, and neither the full set of primes $\{\mathfrak{p}_i\}$ nor their multiplicities is invariant in general. What is always true is that, for any such filtration, giving rise to prime ideals $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_s\}$, one has

$$\mathrm{Ass}_A(M) \subset \{\mathfrak{p}_1, \ldots, \mathfrak{p}_s\} \subset \mathrm{Supp}_A(M).$$

The first inclusion is easily proved combining Equation (B.5.2) with an inductive argument, and implies that $\mathsf{Ass}_A(M)$ is a finite set.

It remains to confirm that $\mathsf{Ass}_A(M)$ contains all isolated primes. Let $\mathfrak{p} \in \mathsf{Min}(\mathsf{Supp}_A(M))$ be an isolated prime. Localising at $\mathfrak{p}$, we get

$$\mathsf{Supp}_{A_\mathfrak{p}}(M_\mathfrak{p}) = \{\mathfrak{p}A_\mathfrak{p}\}.$$

Since $M_\mathfrak{p} \neq 0$ is finitely generated over the noetherian ring $A_\mathfrak{p}$, we have

$$\emptyset \neq \mathsf{Ass}_{A_\mathfrak{p}}(M_\mathfrak{p}) \subset \mathsf{Supp}_{A_\mathfrak{p}}(M_\mathfrak{p}).$$

So necessarily

$$\mathfrak{p}A_\mathfrak{p} \in \mathsf{Ass}_{A_\mathfrak{p}}(M_\mathfrak{p}) = \left\{ \mathfrak{q}A_\mathfrak{p} \;\middle|\; \mathfrak{q} \in \mathsf{Ass}_A(M),\, \mathfrak{q} \subset \mathfrak{p} \right\},$$

where we used Theorem B.5.1.10(3). Thus $\mathfrak{p}A_\mathfrak{p} = \mathfrak{q}A_\mathfrak{p}$ for some $\mathfrak{q} \subset \mathfrak{p}$. Since $\mathsf{Ass}_A(M) \subset \mathsf{Supp}_A(M)$ and $\mathfrak{p}$ is minimal in $\mathsf{Supp}_A(M)$, we must have $\mathfrak{q} = \mathfrak{p}$, hence $\mathfrak{p} \in \mathsf{Ass}_A(M)$. □

Example B.5.1.12. Fix a nonzero finitely generated module M over a noetherian ring A, and consider a prime filtration

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_s = M, \qquad M_i/M_{i-1} \cong A/\mathfrak{p}_i. \tag{B.5.3}$$

Note that 'extra primes' may occur. For $A = \mathbb{F}[x] = M$, the filtration $0 \subsetneq A$ has the single factor $A/(0)$, giving rise to the single prime $\mathfrak{p}_1 = (0)$, whereas $0 \subsetneq xA \subsetneq A$ gives rise to primes $\mathfrak{p}_2 = xA$ (second inclusion) and $\mathfrak{p}_1 = (0)$ (since $xA/(0) = xA \cong A = A/(0)$). However, $\mathfrak{p}_1 = (0)$ is the only isolated prime.

There is nevertheless an invariant multiplicity at the *minimal* primes of the support. If $\mathfrak{p}$ is minimal in $\mathsf{Supp}_A(M)$, i.e. it is isolated, then the number of indices i with $\mathfrak{p}_i = \mathfrak{p}$ is independent of the prime filtration and equals

$$\mathsf{length}_{A_\mathfrak{p}} M_\mathfrak{p}.$$

The length of a module was introduced in Definition B.4.0.1.

It is worth stating this fact in a precise way.

PROPOSITION B.5.1.13. *Let $M \neq 0$ be a finitely generated module over a noetherian ring A. Then, the following sets of prime ideals in A coincide:*

(i) *the isolated primes of M, i.e.* $\mathsf{Min}(\mathsf{Supp}_A(M))$,

(ii) *the minimal primes in* $\mathsf{Ass}_A(M)$, *i.e.* $\mathsf{Min}(\mathsf{Ass}_A(M))$, *and*

(iii) *the minimal primes in $S = \{\mathfrak{p}_1, \ldots, \mathfrak{p}_s\}$, where S arises from any prime filtration as in* (B.5.3).

Furthermore, if $\mathfrak{p}$ belongs to any of these sets, then the number of times it occurs in a prime filtration as in (B.5.3) *is independent of the filtration, and it equals* $\mathsf{length}_{A_\mathfrak{p}} M_\mathfrak{p}$.

Proof. The sets in (i) and (iii) agree because $\mathrm{Supp}_A(M) = \bigcup_{1\leqslant i\leqslant s} \mathrm{V}(\mathfrak{p}_i)$. See [7, Tag 02CE] for the other equivalence.

To obtain the last statement, localise the filtration at $\mathfrak{p}$. A factor $(A/\mathfrak{p}_i)_\mathfrak{p}$ is nonzero exactly when $\mathfrak{p}_i \subset \mathfrak{p}$. Since every $\mathfrak{p}_i$ lies in $\mathrm{Supp}_A(M)$ and $\mathfrak{p}$ is minimal there, this forces $\mathfrak{p}_i = \mathfrak{p}$. Each surviving factor is the residue field $\kappa(\mathfrak{p})$, of length one. □

Definition B.5.1.14 (Embedded primes)**.** Let A be a noetherian ring, M a finitely generated A-module. The non-isolated primes in $\mathrm{Ass}_A(M)$ are called the *embedded primes* of M.

In particular, in the setup of Definition B.5.1.14, one has

$$\mathrm{Ass}_A(M) = \{\text{ embedded primes of } M\ \} \amalg \{\text{ isolated primes of } M\ \}.$$

The most boring situation is when A is an integral domain, in which case the generic point $\xi \in \operatorname{Spec} A$ is the only associated (and clearly isolated, being minimal) prime, cf. Example B.5.1.4. A similar situation occurs with reduced affine schemes, as the following exercise illustrates.



Exercise B.5.1.15 (Reduced implies no embedded points, I)**.** Let A be a reduced noetherian ring. Show that $M = A$ has no embedded associated primes. In other words, $\mathrm{Ass}_A(A) = \mathrm{Min}(\operatorname{Spec} A)$, i.e. associated primes precisely correspond to the irreducible components of $\operatorname{Spec} A$.

Remark B.5.1.16. We make the following observations. Let M be a finitely generated module over a noetherian ring A.

(i) We know that
$$\mathrm{Min}(\mathrm{Supp}_A(M)) \subset \mathrm{Ass}_A(M) \subset \mathrm{Supp}_A(M),$$
therefore $\mathrm{Min}(\mathrm{Supp}_A(M)) = \mathrm{Min}(\mathrm{Ass}_A(M))$. In other words, the isolated associated primes are precisely the minimal associated primes.

(ii) The isolated associated primes, i.e. the elements of $\mathrm{Min}(\mathrm{Supp}_A(M)) = \mathrm{Min}(\mathrm{Ass}_A(M))$, are precisely the generic points of the irreducible components of $\mathrm{Supp}_A(M)$, namely the minimal primes lying above the ideal $\mathrm{Ann}_A(M) \subset A$.

(iii) Let $\mathfrak{q} \in \mathrm{Ass}_A(M) \setminus \mathrm{Min}(\mathrm{Supp}_A(M))$ be an *embedded* prime. Then there is an isolated prime $\mathfrak{p} \in \mathrm{Min}(\mathrm{Supp}_A(M))$ such that $\mathfrak{p} \subsetneq \mathfrak{q}$. This means that $\mathrm{V}(\mathfrak{q})$ is entirely contained in some irreducible component $\mathrm{V}(\mathfrak{p}) \subset \mathrm{Supp}_A(M)$.

The important point in Remark B.5.1.16(iii) is that embedded primes are *not visible from the underlying topological space* $\mathrm{Supp}_A(M)$ alone. They record extra module-theoretic information, such as torsion or scheme-theoretic structure concentrated along a proper closed subset. We illustrate this with the following example.

Example B.5.1.17. Set $A=\mathbb{F}[x,y]$ and $M=A/(x)\oplus A/(x,y)$, so that $\mathsf{Supp}_A(M)=\mathrm{V}(x)\subset \mathbb{A}^2_{\mathbb{F}}$. However, $\mathsf{Ass}_A(M)=\{(x),(x,y)\}$. The prime (x) is isolated: it is the generic point of the unique irreducible component $\mathrm{V}(x)$ of $\mathsf{Supp}_A(M)=\mathrm{V}(\mathsf{Ann}_A(M))$. The prime (x,y) is embedded: geometrically it is the origin sitting inside the line $\mathrm{V}(x)$.

Of course, the A-module $M'=A/(x)$ has the same support as M, but the two modules are not the same! The difference is recorded by the presence, for M, of an embedded prime. Informally speaking, the support of a module tells us *where* the module lives, whereas its associated primes tell us *how* it lives there.

B.5.2 Primary decomposition and embedded components

Definition B.5.2.1 (Primary ideal)**.** Let A be a noetherian ring. A proper ideal $\mathfrak{q}\subsetneq A$ is called *primary* if whenever a product xy lies in $\mathfrak{q}$, either x or a power of y lies in $\mathfrak{q}$. Equivalently, $\mathfrak{q}\subsetneq A$ is primary if every zero-divisor in $A/\mathfrak{q}$ is nilpotent.

Let A be a noetherian ring. Every proper ideal $I\subset A$ has a (non-unique) *primary decomposition*, i.e. an expression as intersection

$$I=\mathfrak{q}_1\cap\cdots\cap\mathfrak{q}_r$$

of primary ideals. See [1, Prop. 3.10] for a proof. One verifies that the radical of a primary ideal is prime, and one says that $\mathfrak{q}$ is $\mathfrak{p}$-primary if $\sqrt{\mathfrak{q}}=\mathfrak{p}$. One can always ensure that the decomposition is irredundant, i.e. removing any $\mathfrak{q}_i$ changes the intersection, and $\sqrt{\mathfrak{q}_i}\neq\sqrt{\mathfrak{q}_j}$ for all $i\neq j$. Note that (even irredundant) primary decompositions are not unique! However, the *radicals* of the primary ideals appearing in an irredundant primary decomposition are intrinsic, as the next proposition will confirm.

PROPOSITION B.5.2.2 ([1, Prop. 3.10.b]). *Let $I\subset A$ be an ideal with irredundant primary decomposition $I=\mathfrak{q}_1\cap\cdots\cap\mathfrak{q}_r$. Then, the set*

$$\{\sqrt{\mathfrak{q}_i}\mid 1\leqslant i\leqslant r\}\subset\operatorname{Spec}A$$

is determined by I. More precisely, it coincides with $\mathsf{Ass}_A(A/I)$ and there are inclusions $\mathsf{Ass}_A(A/I)\subset\mathsf{Supp}_A(A/I)=\mathrm{V}(\mathsf{Ann}_A(A/I))=\mathrm{V}(I)$.



Exercise B.5.2.3. Set $A=\mathbb{F}[x,y]$ and $I=(xy,y^2)$. Show that $\mathsf{Ass}_A(A/I)=\{(y),(x,y)\}$.

Definition B.5.2.4 (Embedded component)**.** Let I be an ideal in a noetherian ring A. An *embedded component* of $\operatorname{Spec}A/I$ is a closed subset $Z\subset\operatorname{Spec}A/I$ of the form $Z=\mathrm{V}(\mathfrak{p})=\overline{\{\mathfrak{p}\}}$ for some embedded prime $\mathfrak{p}\in\mathsf{Ass}_A(A/I)\setminus\mathsf{Min}(\mathsf{Ass}_A(A/I))$.

Remark B.5.2.5. Let $I\subset A$ be an ideal with a chosen irredundant primary decomposition $I=\mathfrak{q}_1\cap\cdots\cap\mathfrak{q}_r$. Let

$$\mathsf{Min}(\mathsf{Ass}_A(A/I))\subset\mathsf{Ass}_A(A/I)=\{\sqrt{\mathfrak{q}_i}\mid 1\leqslant i\leqslant r\}\subset\operatorname{Spec}A$$

be the set of minimal primes, and choose an embedded prime $\sqrt{\mathfrak{q}_i} \in \mathsf{Ass}_A(A/I) \setminus \mathsf{Min}(\mathsf{Ass}_A(A/I))$. Then, as already mentioned, there exists $\mathfrak{p} \in \mathsf{Min}(\mathsf{Ass}_A(A/I))$ such that $\mathfrak{p} \subsetneq \sqrt{\mathfrak{q}_i}$. Set-theoretically, we have $\mathrm{V}(\mathfrak{q}_i) = \mathrm{V}(\sqrt{\mathfrak{q}_i}) \subsetneq \mathrm{V}(\mathfrak{p})$, but *scheme-theoretically* we do not have

$$\operatorname{Spec} A/\mathfrak{q}_i \subset \operatorname{Spec} A/\mathfrak{p}!$$

Example B.5.2.6. Consider $A = \mathbb{F}[x,y]$ and $I = (xy, y^2)$, as in Exercise B.5.2.3. A primary decomposition of I is

$$I = (x,y)^2 \cap (y).$$

However, the closed subscheme $\operatorname{Spec} A/(x,y)^2 \subset \operatorname{Spec} A = \mathbb{A}^2_{\mathbb{F}}$ is not scheme-theoretically contained in $\operatorname{Spec} A/(y)$.

One might be tempted to call a closed subscheme of the form

$$\operatorname{Spec} A/\mathfrak{q}_i \hookrightarrow \operatorname{Spec} A/I,$$

with $I \subset \mathfrak{q}_i \subset A$ as in Remark B.5.2.5, a *scheme-theoretic embedded component*.[2]

But these objects are not canonical, as they depend on the chosen (irredundant) primary decomposition; what is the canonical is the collection of embedded components in the (purely topological) sense of Definition B.5.2.4. For instance, going back to Example B.5.2.6, one has also the primary decompositions

$$(xy, y^2) = (x, y^2) \cap (y) = (x+y, y^2) \cap (y)$$

where of course both $\mathfrak{q} = (x, y^2)$ and $\mathfrak{q}' = (x+y, y^2)$ have radical (x,y). They correspond to the same embedded component, and they (incidentally) give rise to abstractly isomorphic schemes, as $\mathbb{F}[x,y]/(x,y^2) \cong \mathbb{F}[y]/y^2 \cong \mathbb{F}[x,y]/(x+y,y^2)$, but they are not isomorphic (even abstractly) to $\operatorname{Spec} \mathbb{F}[x,y]/(x,y)^2$, as the latter has length 3. To summarise, embedded components contain only topological information, whereas any 'scheme-theoretic embedded component' gives a closed *subscheme* $Y \subset \operatorname{Spec} A/I$ which remembers *how* the nonreduced structure 'causes' the presence of the (topological) embedded components.

B.5.3 Associated points: global case

Let X be a locally noetherian scheme and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module, so that $\mathsf{Supp}(\mathcal{F})$ is a closed subset of X. The set of associated points of $\mathcal{F}$ is the set

$$\mathsf{Ass}_X(\mathcal{F}) = \left\{ x \in X \;\middle|\; \mathfrak{m}_x \in \mathsf{Ass}_{\mathcal{O}_{X,x}}(\mathcal{F}_x) \right\},$$

where $\mathfrak{m}_x$ is the maximal ideal of the local ring $\mathcal{O}_{X,x}$. This set is locally finite,[3] i.e. its intersection with every affine open in X is finite. In fact, $\mathsf{Ass}_X(\mathcal{F})$ is finite if X is noetherian,

[2] This terminology is not standard.

[3] Note that it really is *not* finite in general: for instance, $X = \coprod_{n \geqslant 1} \operatorname{Spec} \mathbb{F}$ is locally noetherian and $\mathcal{O}_X$ is coherent, but $\mathsf{Ass}_X(\mathcal{O}_X) = X$ is infinite.

or more generally if $\mathsf{Supp}(\mathcal{F})$ is quasicompact. This definition of $\mathsf{Ass}_X(\mathcal{F})$ is affine-local: if $U = \operatorname{Spec} A \subset X$ is open and $\mathcal{F}|_U \cong \widetilde{M}$, with M finitely generated, then

$$\mathsf{Ass}_X(\mathcal{F}) \cap U = \mathsf{Ass}_A(M) \subset U.$$

An associated point $x \in \mathsf{Ass}_X(\mathcal{F})$ is isolated if it is minimal in the support in the affine-local sense. The isolated associated points are precisely the generic points $\{\eta_i\}_{i\in I}$ of the irreducible components $Z_i \subset \mathsf{Supp}(\mathcal{F})$. An associated point $x \in \mathsf{Ass}_X(\mathcal{F})$ is embedded if it is not isolated, i.e. it is not a generic point of an irreducible component of the support. Equivalently, in an affine neighbourhood $U = \operatorname{Spec} A$ with $\mathcal{F}|_U \cong \widetilde{M}$, the corresponding prime $\mathfrak{q}$ is an embedded associated prime of M. Geometrically, if x is embedded, then its closure $\overline{\{x\}}$ is properly contained in an irreducible component of $\mathsf{Supp}(\mathcal{F})$. The embedded points of $\mathscr{O}_X$ are simply called the embedded points of X. An *embedded component* of X is an irreducible closed subset of the form $\overline{\{x\}}$, where $x \in \mathsf{Ass}_X(\mathscr{O}_X)$ is an embedded point.

Let $Z \subset Y$ be a locally noetherian closed subscheme of a scheme Y. Then Z has an embedded associated point if and only if there exists a dense open $U \subset Y$ such that $Z \cap U$ is topologically dense in Z but not schematically dense.

Concretely, one has to keep in mind the following situation. Let X be locally noetherian and let $p \in X$ be a closed point which is not the generic point of an irreducible component. Then p is an embedded associated point of X if and only if $X \setminus \{p\}$ is topologically dense in X but is not schematically dense, equivalently if its scheme-theoretic closure in X is a proper closed subscheme.

For instance, if $X = \operatorname{Spec} \mathbb{F}[x,y]/(xy, y^2)$, the scheme-theoretic closure of $X \setminus \{0\} \subset X$ is $\operatorname{Spec} \mathbb{F}[x,y]/y \cong \mathbb{A}^1_{\mathbb{F}} \cong X_{\mathsf{red}}$, which is not equal to X as a scheme.

B.5.4 Links with finite length modules

In this section we will see (further) links between length, associated primes, and support.

THEOREM B.5.4.1. *Let A be a noetherian ring and let M be a finitely generated A-module. The following are equivalent:*

(i) $\mathsf{length}_A(M) < \infty$,

(ii) $\mathsf{Supp}_A(M)$ *is a finite set of maximal ideals, and*

(iii) $A/\mathsf{Ann}_A(M)$ *is an artinian ring.*

Proof. Set $B = A/\mathsf{Ann}_A(M)$. Notice first that, since M is finitely generated, Exercise B.5.1.1 gives $\mathsf{Supp}_A(M) = \mathrm{V}(\mathsf{Ann}_A(M))$. Therefore there is a canonical homeomorphism

$$\operatorname{Spec} B \xrightarrow{\simeq} \mathsf{Supp}_A(M).$$

We now proceed proving the equivalences.

(i) ⇒ (ii). Assume $\mathsf{length}_A(M) < \infty$. By Proposition B.4.0.7, M admits a composition series

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M$$

whose factors are simple. By Lemma B.4.0.6, for every i there is a maximal ideal $\mathfrak{m}_i \subset A$ such that

$$M_i/M_{i-1} \cong A/\mathfrak{m}_i.$$

Applying repeatedly the support identity of (B.4.1), we obtain

$$\mathsf{Supp}_A(M) = \bigcup_{i=1}^{n} \mathsf{Supp}_A(M_i/M_{i-1}) = \bigcup_{i=1}^{n} \mathrm{V}(\mathfrak{m}_i) = \{\mathfrak{m}_1, \ldots, \mathfrak{m}_n\}.$$

Thus $\mathsf{Supp}_A(M)$ is a finite set consisting of maximal ideals.

(ii) ⇒ (iii). By the homeomorphism $\operatorname{Spec} B \simeq \mathsf{Supp}_A(M)$, every prime ideal of B is maximal. Hence $\dim B = 0$. Moreover, B is noetherian because it is a quotient of the noetherian ring A. Therefore the standard characterisation of 0-dimensional noetherian rings, cf. **??**, gives that B is artinian.

(iii) ⇒ (i). Since B is a quotient of the noetherian ring A, it is noetherian. Thus B is both noetherian and Artinian. Consequently B, viewed as a module over itself, has finite length.

The A-module structure on M factors through B, and M is finitely generated as a B-module. Hence there is a surjection $B^{\oplus r} \twoheadrightarrow M$ for some $r \geqslant 0$. Since B has finite length, so does $B^{\oplus r}$, and therefore, by additivity of length,

$$\mathsf{length}_B(M) < \infty.$$

Finally, since $A \to B$ is surjective, the A-submodules of M are precisely its B-submodules. Hence $\mathsf{length}_A(M) = \mathsf{length}_B(M) < \infty$, as desired. □

COROLLARY B.5.4.2. *Let A be noetherian and let M be a finitely generated A-module of finite length. Then*

$$\mathsf{Ass}_A(M) = \mathsf{Supp}_A(M),$$

and the canonical map

$$M \longrightarrow \bigoplus_{\mathfrak{m} \in \mathsf{Supp}_A(M)} M_{\mathfrak{m}}$$

is an isomorphism. In particular,

$$\mathsf{length}_A(M) = \sum_{\mathfrak{m} \in \mathsf{Supp}_A(M)} \mathsf{length}_{A_{\mathfrak{m}}}(M_{\mathfrak{m}}).$$

Proof. By Theorem B.5.4.1, $\mathsf{Supp}_A(M)$ consists of finitely many maximal ideals. Since every minimal element of the support is associated, and every element of the support is now minimal, we obtain $\mathsf{Supp}_A(M) \subset \mathsf{Ass}_A(M)$. The reverse inclusion always holds.

Next we prove that $M \to \bigoplus_{\mathfrak{m}\in \mathsf{Supp}_A(M)} M_{\mathfrak{m}}$ is bijective arguing by induction on $n = \mathsf{length}_A(M)$.

If $n = 0$, then $M = 0$, so there is nothing to prove. Assume $n = 1$. Then M is simple, hence by Lemma B.4.0.6 we have $M \cong A/\mathfrak{m}$ for some maximal ideal $\mathfrak{m} \subset A$. Therefore

$$\mathsf{Supp}_A(M) = \{\mathfrak{m}\}.$$

Moreover, the canonical map $M \to M_{\mathfrak{m}}$ is an isomorphism since every element of $A \setminus \mathfrak{m}$ acts invertibly on $A/\mathfrak{m}$.

Suppose now that $n > 1$. By Proposition B.4.0.7, we can choose a submodule $N \subsetneq M$ such that $Q = M/N$ is simple. By Lemma B.4.0.6, there is a maximal ideal $\mathfrak{m}_0 \subset A$ such that $Q \cong A/\mathfrak{m}_0$. Thus we have a short exact sequence

$$0 \longrightarrow N \longrightarrow M \longrightarrow Q \longrightarrow 0.$$

Set $S = \mathsf{Supp}_A(M)$. By the behaviour of support in short exact sequences,

$$S = \mathsf{Supp}_A(N) \cup \mathsf{Supp}_A(Q).$$

Moreover, $\mathsf{Supp}_A(Q) = \{\mathfrak{m}_0\}$, and Q has length 1 by Lemma B.4.0.6(ii). It follows from (B.4.1) that $\mathsf{length}_A(N) = n - 1$, so the induction hypothesis can be applied to both Q and N.

For every $\mathfrak{m} \in S$, exactness of localisation gives an exact sequence

$$0 \longrightarrow N_{\mathfrak{m}} \longrightarrow M_{\mathfrak{m}} \longrightarrow Q_{\mathfrak{m}} \longrightarrow 0.$$

Since S is finite, taking the direct sum over $\mathfrak{m} \in S$ gives an exact sequence

$$0 \longrightarrow \bigoplus_{\mathfrak{m}\in S} N_{\mathfrak{m}} \longrightarrow \bigoplus_{\mathfrak{m}\in S} M_{\mathfrak{m}} \longrightarrow \bigoplus_{\mathfrak{m}\in S} Q_{\mathfrak{m}} \longrightarrow 0.$$

We therefore obtain a commutative diagram with exact rows

$$\begin{array}{ccccccccc} 0 & \longrightarrow & N & \longrightarrow & M & \longrightarrow & Q & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \theta_N} & & \downarrow{\scriptstyle \theta_M} & & \downarrow{\scriptstyle \theta_Q} & & \\ 0 & \longrightarrow & \bigoplus_{\mathfrak{m}\in S} N_{\mathfrak{m}} & \longrightarrow & \bigoplus_{\mathfrak{m}\in S} M_{\mathfrak{m}} & \longrightarrow & \bigoplus_{\mathfrak{m}\in S} Q_{\mathfrak{m}} & \longrightarrow & 0. \end{array}$$

Both outer maps are isomorphisms by induction. Induction can be applied to both N and Q because, regarding N, we have $N_{\mathfrak{m}} = 0$ whenever $\mathfrak{m} \notin \mathsf{Supp}_A(N)$, so enlarging the index set from $\mathsf{Supp}_A(N)$ to S merely adds zero summands. Regarding Q, we already observed it has only $\mathfrak{m}_0$ in its support, so the same reasoning applies. Hence the statement follows from the Five Lemma (cf. Lemma A.6.0.9). □

Definition B.5.4.3 (Length of a 0-dimensional scheme)**.** Let Z be a 0-dimensional noetherian scheme. We define the *length* of Z by

$$\mathsf{length}(Z) = \sum_{z\in Z} \mathsf{length}_{\mathcal{O}_{Z,z}}(\mathcal{O}_{Z,z}).$$

Note that the definition makes sense: the right hand side is a finite sum of positive integers: indeed, a 0-dimensional noetherian scheme has finitely many points, and each local ring $\mathscr{O}_{Z,z}$ is a 0-dimensional noetherian local ring, hence an artinian local ring.

If $Z = \operatorname{Spec} B$, then B is artinian and

$$\mathsf{length}(Z) = \mathsf{length}_B(B).$$

If Z is finite over a field $\mathbb{F}$, we have

$$\dim_{\mathbb{F}} \mathrm{H}^0(Z, \mathscr{O}_Z) = \sum_{z \in Z} [\kappa(z) : \mathbb{F}] \, \mathsf{length}_{\mathscr{O}_{Z,z}}(\mathscr{O}_{Z,z}).$$

Therefore, if Z is finite over an algebraically closed field $\mathbf{k}$, we obtain

$$\mathsf{length}(Z) = \dim_{\mathbf{k}} \mathrm{H}^0(Z, \mathscr{O}_Z).$$

B.5.5 The cycle associated to a coherent sheaf

Let X be a noetherian scheme. We denote by

$$\mathrm{Z}_*(X)$$

the free abelian group generated by the irreducible closed subsets of X (equivalently, by the integral closed subschemes of X, where an irreducible closed subset is endowed with its reduced induced scheme structure). Its elements are called *cycles* on X, and can be written as

$$\sum_{1 \leqslant i \leqslant s} m_i [Y_i], \quad m_i \in \mathbb{Z},$$

for some integer $0 \leqslant s < \infty$.

Let $\mathcal{F}$ be a coherent $\mathscr{O}_X$-module. Since X is noetherian, the closed subset $\mathsf{Supp}(\mathcal{F}) \subset X$ has finitely many irreducible components, say $Z_1, \ldots, Z_r \subset X$. Let ξ_i be the generic point of Z_i. The points ξ_i are precisely the isolated associated points of $\mathcal{F}$.

LEMMA B.5.5.1. *Let X be a noetherian scheme, let $\mathcal{F}$ be a coherent $\mathscr{O}_X$-module, and let $\xi \in \mathsf{Supp}(\mathcal{F})$ be the generic point of an irreducible component of* $\mathsf{Supp}(\mathcal{F})$*. Then*

$$\mathsf{length}_{\mathscr{O}_{X,\xi}}(\mathcal{F}_\xi) < \infty.$$

Proof. Since ξ is a minimal point of $\mathsf{Supp}(\mathcal{F})$, after localising at ξ we have

$$\mathsf{Supp}_{\mathscr{O}_{X,\xi}}(\mathcal{F}_\xi) = \{\mathfrak{m}_\xi\},$$

where $\mathfrak{m}_\xi \subset \mathscr{O}_{X,\xi}$ is the maximal ideal. Indeed, the prime ideals in the support of $\mathcal{F}_\xi$ correspond to points of $\mathsf{Supp}(\mathcal{F})$ specialising to ξ, and by minimality of ξ there are no such points other than ξ itself.

Now $\mathcal{F}_\xi$ is a finitely generated module over the noetherian local ring $\mathcal{O}_{X,\xi}$. Hence, by the finite-length criterion of Theorem B.5.4.1, we have

$$\mathsf{length}_{\mathcal{O}_{X,\xi}}(\mathcal{F}_\xi) < \infty,$$

as desired. □

Definition B.5.5.2 (Cycle associated to a coherent sheaf)**.** Let X be a noetherian scheme and let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module. For every irreducible component $Z \subset \mathsf{Supp}(\mathcal{F})$, with generic point ξ, define the *multiplicity of Z in $\mathcal{F}$* to be

$$m_Z(\mathcal{F}) = \mathsf{length}_{\mathcal{O}_{X,\xi}}(\mathcal{F}_\xi).$$

The *cycle associated to $\mathcal{F}$* is[4]

$$[\mathcal{F}] = \sum_{\substack{\xi \in \mathsf{Ass}_X(\mathcal{F}) \\ \xi \text{ isolated}}} m_Z(\mathcal{F})[Z],$$

where in the sum Z is related to ξ via $Z = \overline{\{\xi\}}$.

Example B.5.5.3. Set $A = \mathbb{F}[x,y]$ and $M = A/(x^2y^3)$. Then $(x^2y^3) = (x^2) \cap (y^3)$ is an irredundant primary decomposition, and thus all associated primes of M are isolated,

$$\mathsf{Min}(\mathsf{Supp}_A(M)) = \mathsf{Ass}_A(M) = \{(x),(y)\}.$$

At (x) the element y is invertible, so $M_{(x)} \cong A_{(x)}/x^2A_{(x)}$ and

$$\mathsf{length}_{A_{(x)}}(M_{(x)}) = 2.$$

Similarly,

$$\mathsf{length}_{A_{(y)}}(M_{(y)}) = 3.$$

Consider $\mathcal{F} = \widetilde{M}$. The two irreducible components have generic multiplicities 2 and 3, and we obtain the cycle

$$[\mathcal{F}] = 2[\mathrm{V}(x)] + 3[\mathrm{V}(y)] \in \mathrm{Z}_*(\mathbb{A}^2_{\mathbb{F}}).$$



Warning B.5.5.4. Only the isolated associated points of $\mathcal{F}$ contribute to the cycle $[\mathcal{F}]$. Embedded associated points are *not recorded* by $[\mathcal{F}]$.

Thus the cycle associated to a coherent sheaf records its generic multiplicity along the irreducible components of its support, but does not remember scheme-theoretic structure concentrated in smaller closed subsets.

[4] See also [7, Tag 02QV] for a more general construction.

Example B.5.5.5. Returning to Example B.5.1.17, set $A = \mathbb{F}[x,y]$ and $M = A/(x) \oplus A/(x,y)$. Consider $\mathcal{F} = \widetilde{M}$. Then

$$\mathsf{Ass}_A(M) = \{(x),(x,y)\},$$

but (x,y) is embedded. The support has the unique irreducible component $\mathrm{V}(x)$, and $M_{(x)} \cong \mathrm{Frac}(A/(x))$. Thus

$$\mathsf{length}_{A_{(x)}} M_{(x)} = 1$$

and we obtain the cycle

$$[\mathcal{F}] = [\mathrm{V}(x)] \in \mathrm{Z}_*(\mathbb{A}^2_{\mathbb{F}}).$$

In particular, the embedded summand $A/(x,y)$ is invisible to the cycle.



Warning B.5.5.6. Our definition of the cycle associated to a coherent sheaf does not satisfy the natural additivity property on short exact sequences. This is already clear from Example B.5.5.5. If one wants additivity on short exact sequences, a slighter finer theory has to be developed, see [7, Tag 02QV].

B.6 Depth and Cohen–Macaulay schemes

We recall some notions, mainly from [7, Tag 00LE].

Definition B.6.0.1 (Regular sequence [7, Tag 00LF])**.** Let M be a module over a ring A. Then a sequence $f_1,\dots,f_r$ of elements in A is called an *M-regular sequence* (of length r) if, for every i, multiplication by f_i is injective on $M/(f_1,\dots,f_{i-1})M$ and $M/(f_1,\dots,f_r)M \neq 0$. If $M = A$, we simply call it a *regular sequence*.

Definition B.6.0.2 (Depth of a module [7, Tag 00LI])**.** Let M be a finitely generated module over a ring A, and let $I \subset A$ be an ideal. If $IM \neq M$, define the *I-depth* of M as

$$\mathrm{depth}_I(M) = \sup_r \left\{ r \;\middle|\; \text{there exists an } M\text{-regular sequence } f_1,\dots,f_r \in I \right\}.$$

If $IM = M$, define $\mathrm{depth}_I(M) = +\infty$. In particular, $\mathrm{depth}_A(M) = +\infty$ for all M.

If $(A,\mathfrak{m})$ is a local ring, we call $\mathrm{depth}_{\mathfrak{m}}(M)$ simply the *depth* of M.

If A is noetherian and $IM \neq M$, then $\mathrm{depth}_I(M) < +\infty$, see [7, Tag 0AUJ].

The following list of facts reflects the fairly good behaviour of depth for finite modules over *local noetherian* rings. Also note that item (ii) connects to the theory of associated points. Since depth is used to define Cohen–Macaulay schemes, it will not be surprising when we will see that Cohen–Macaulay schemes can be described via embedded points, see Lemma B.6.0.9.

Fact B.6.0.3. Assume $(A,\mathfrak{m})$ is noetherian and local, and $M \neq 0$ is finitely generated. Then, the following statements hold true.

(i) By [7, Tag 00LW], one has

$$\operatorname{depth}_{\mathfrak{m}}(M) = \min_i \left\{ i \in \mathbb{N} \,\middle|\, \operatorname{Ext}^i(A/\mathfrak{m}, M) \neq 0 \right\}.$$

(ii) By [7, Tag 0BK4], one has

$$\operatorname{depth}_{\mathfrak{m}}(M) \leqslant \dim A/\mathfrak{p}, \qquad \text{for all } \mathfrak{p} \in \operatorname{Ass}_A(M).$$

(iii) For an arbitrary prime $\mathfrak{p} \in \operatorname{Spec} A$, by [7, Tag 0FCC] one has

$$\operatorname{depth}_{\mathfrak{m}}(M) \leqslant \dim A/\mathfrak{p} + \operatorname{depth}_{\mathfrak{p}A_{\mathfrak{p}}}(M_{\mathfrak{p}}).$$

(iv) By [7, Tag 00LK], one has

$$\dim \operatorname{Supp}_A(M) \geqslant \operatorname{depth}_{\mathfrak{m}}(M).$$



Exercise B.6.0.4. Let $(A, \mathfrak{m}, \kappa)$ be a local noetherian ring and $M \neq 0$ a finitely generated A-module. Prove that the following conditions are equivalent:

(i) $\mathfrak{m} \in \operatorname{Ass}_A(M)$

(ii) $\operatorname{Hom}_A(\kappa, M) \neq 0$, and

(iii) $\operatorname{depth}_{\mathfrak{m}}(M) = 0$.

The A-module $\operatorname{Hom}_A(\kappa, M)$ is called the *socle* of M.

Definition B.6.0.5 (Cohen–Macaulay module)**.** A finitely generated module $M \neq 0$ over a noetherian local ring $(A, \mathfrak{m})$ is said to be *Cohen–Macaulay* if $\dim \operatorname{Supp}_A(M) = \operatorname{depth}_{\mathfrak{m}}(M)$. A noetherian local ring $(A, \mathfrak{m})$ is said to be *Cohen–Macaulay* if it is Cohen–Macaulay as a module over itself.

Definition B.6.0.6 (Cohen–Macaulay scheme)**.** Let X be a locally noetherian scheme. We say that X is *Cohen–Macaulay* if for every $x \in X$ the local ring $\mathcal{O}_{X,x}$ is Cohen–Macaulay, i.e. if

$$\operatorname{depth}_{\mathfrak{m}_x}(\mathcal{O}_{X,x}) = \dim \mathcal{O}_{X,x}.$$

Example B.6.0.7. Consider the rings

$$A_1 = \mathbb{F}[x, y]/(xy), \qquad A_2 = \mathbb{F}[x, y]/(x^2), \qquad A_3 = \mathbb{F}[x, y]/(x^2, xy).$$

Set $X_i = \operatorname{Spec} A_i$. Then X_1 is reduced Cohen–Macaulay with two isolated associated primes, X_2 is nonreduced Cohen–Macaulay with one isolated associated prime, and finally X_3 has the embedded prime (x, y) and is not Cohen–Macaulay.

Lastly, we recall Serre's conditions and basic results from [7, Tag 033P]. Let $k \in \mathbb{Z}_{\geqslant 0}$ be a nonnegative integer. We say that a locally noetherian scheme X satisfies

(i) *Serre's condition* S_k if for every point $x \in X$, we have

$$\operatorname{depth}_{\mathfrak{m}_x}(\mathscr{O}_{X,x}) \geqslant \min(k, \dim \mathscr{O}_{X,x}).$$

(ii) condition R_k if for every point $x \in X$ we have that $\dim \mathscr{O}_{X,x} \leqslant k$ implies that $\mathscr{O}_{X,x}$ is regular.

LEMMA B.6.0.8 ([7, Tag 033P]). *Let X be a locally noetherian scheme. Then,*

(a) *X is regular if and only if it satisfies R_k for every $k \geqslant 0$.*

(b) *X is Cohen–Macaulay if and only if it satisfies S_k for every $k \geqslant 0$.*

(c) *X is reduced if and only if it satisfies S_1 and R_0.*

(d) *X is normal if and only if it satisfies S_2 and R_1.*

The notion of embedded point connects to the notion of Cohen–Macaulay scheme through the following result.

LEMMA B.6.0.9 ([7, Tag 05AJ]). *Let X be a locally noetherian scheme. Then,*

(i) *X has no embedded points if and only if it satisfies Serre's condition S_1.*

(ii) *If $\dim X \leqslant 1$, then X is Cohen–Macaulay if and only if it has no embedded points.*

(iii) *Let $U \subset X$ be an open subscheme. Then, the following are equivalent:*

- *For every open subscheme $V \subset X$, the scheme-theoretic closure of $U \cap V$ in V is equal to V,*[5]
- *U is dense in X and U contains all embedded points of X, and*
- *$\operatorname{Ass}_X(\mathscr{O}_X) \subset U$.*

We propose part of the previous lemma as an exercise.



Exercise B.6.0.10. Let X be a locally noetherian scheme and $j\colon U \hookrightarrow X$ an open immersion. Prove that $\mathscr{O}_X \to j_*\mathscr{O}_U$ is injective if and only if $\operatorname{Ass}_X(\mathscr{O}_X) \subset U$.

We close this section with a couple of useful remarks.

Remark B.6.0.11 (Reduced implies no embedded points, II)**.** Let X be a reduced locally noetherian scheme. Then X has no embedded points. This follows combining Lemma B.6.0.9(i) and Lemma B.6.0.8(c) or, more simply, from the affine-local description of $\operatorname{Ass}_X(\mathscr{O}_X)$ combined with Exercise B.5.1.15.

[5] This condition is saying that U is *schematically dense* in X, see [7, Tag 01RB].

Remark B.6.0.12 (Cohen–Macaulay does not imply reduced)**.** By Lemma B.6.0.9(ii), we know that a reduced 1-dimensional locally noetherian scheme is Cohen–Macaulay. However, there can be nonreduced Cohen–Macaulay curves: for example, curves with a nonreduced component, such as the affine plane curve $\operatorname{Spec}\mathbb{F}[x,y]/x^2 \subset \mathbb{A}^2$, to be visualised as a "double y-axis". In this example, one indeed has $\mathsf{Ass}_A(A) = \{(\overline{x})\}$, where $A = \mathbb{F}[x,y]/x^2$.

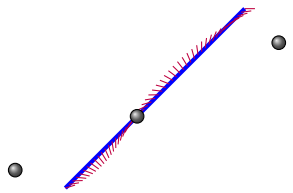


Figure B.1: A thickened (non-Cohen–Macaulay) curve with an embedded point and two isolated (possibly fat) points.

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